Training with EventWeight

ttHcc Analysis

Lenart Jerala, with the Ljubljana group

ATLAS Experiment · Jožef Stefan Institute, Ljubljana · University of Ljubljana, FMF

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Training with EventWeight — the one-line story

- A recipe for training with *signed event weights*, for any analysis
- Let me be honest about scope first: I have said before that we cannot get a better *classifier* — and I stand by that. The point is a better, *rigorous observable*

The next two slides make that precise — what I have *not* achieved, then what I *have* — and then:

- short introduction to the method;
- does it work, and results

These are my most important slides on ML so far. They go much further than just one novelty – they complete the puzzle.

Disconnected: Miha suggested I shout out Claude – apparently LLMs aren't as popular as they should be.

What I have *not*: a better classifier

- Classification is capped by the inputs, not by our training choices. Larger gains require stronger inputs, e.g. a continuous FTAG score: input quality dominates
- *Architecture does not rescue it.* A rank-3 (three-body) attention transformer ("CANDY") looked like a large gain, but a same-size ParT reproduced it; this motivates caution about auxiliary-task gains
- *Correlation-aware inductive bias buys little at full statistics.* In our and others' tests, physics-inspired bias and pretraining help mainly in the low-statistics regime ($\sim$ 100k events); against the full-sample baseline they add little, though they do not appear to hurt
- *What still moved the needle:* thinking like an ML engineer, not like a physicist – make network learn biases through better trianing conditions (EMA, PLE, ...); take from the industry!

What I *achieved*: a rigorous observable

Same classifier — but a rigorous observable

The method does not make the *classifier* better — same $q(y \mid x)$, same AUC against the generation labels. **It makes the delivered *observable* rigorous: the composition into the physical posterior π is the Neyman–Pearson-optimal discriminant for the *physical* problem**

- How, in one line: compose the *unweighted* classifier with a *learned per-event mean weight* into π — optimal for the *weighted* (physical) test, not the generation one
- Practical: baked into ONNX at export — no more manually entered class yields
- Headline: 12% lower combined stat-only expected limit

Chapter 1: The optimal observable

The classification problem — recap

- Simulator emits *labelled, weighted* events (x, y, w): observables $x \in \mathbb{R}^d$, class y, and a signed MC weight $w \in \mathbb{R}$
- Signal-like or background-like? Build a test statistic $t : \mathbb{R}^d \to \mathbb{R}$ collapsing the d dimensions to one, then cut
- The question: which t best separates the physical signal density $p_s(x)$ from background $p_b(x)$?
- *Neyman–Pearson*: the optimal t is the likelihood ratio (or any monotone map of it),

$$t(x) = \frac{p_s(x)}{p_b(x)}, \qquad \text{in practice } \log \frac{p_s(x)}{p_b(x)}.$$

- Completely general and optimal — no handcrafted combination of observables does better

No closed form $\Rightarrow$ learn it

- No analytic densities: matrix element, PDFs, shower, hadronisation, detector response and reconstruction all shape them
- Let $f_y(x) = q(x \mid y)$ denote the generation shape and $\rho_y = q(y)$ its generated class fraction. A calibrated, class-imbalanced classifier outputs $q(y \mid x)$, and in the binary case

$$\frac{f_s(x)}{f_b(x)} = \frac{\rho_b}{\rho_s} \frac{q(s \mid x)}{q(b \mid x)}.$$

  Proven thoroughly in Cranmer, Pavez & Louppe [arXiv:1506.02169]
- But: this is the generation ratio f_s / f_b; Neyman–Pearson needs the physical ratio p_s / p_b. The weights connect the two

Two laws (probability distributions), two posteriors

- Colour code throughout: blue is the generation law Q — what the network counts; green is the physical law P — what the analysis reports
- For every (X, Y)-measurable h,

$$\mathbb{E}_P[h] := \frac{\mathbb{E}_Q[Wh]}{Z}, \qquad Z := \mathbb{E}_Q[W] > 0.$$

- Their class posteriors are the generation posterior $q(y \mid x)$ and the physical posterior $\pi(y \mid x)$. The task is to recover the latter without putting signed weights in an empirical loss

The obvious weighted loss — and why it breaks

- Per-event weight w_i — obvious reflex: put it in the loss,

$$\mathcal{L} = -\frac{1}{n}\sum_i \log q_\theta(y_i \mid x_i) \quad \longrightarrow \quad \mathcal{L}_w = -\frac{1}{n}\sum_i w_i \log q_\theta(y_i \mid x_i).$$

- Plain CE is strictly proper and recovers q. The *population* weighted CE targets π
- But its empirical signed-weight version is ill-posed: a negative local weighted count drives the loss to $-\infty$; large $|w_i|$ also dominate the gradient

The constant-Z_y correction used today

- Today we train with class-weighted CE ($c_y \propto 1/\rho_y$): the softmax learns the prior-removed posterior $b(y \mid x) \propto q(y \mid x)/\rho_y \propto f_y(x)$, and the physical yields $N_y = \sum_{i:Y_i=y} w_i$ are entered by hand. Here n_y is the generated count, $\rho_y = n_y/n$, and

$$Z_y := \mathbb{E}_Q[W \mid Y = y] = \frac{N_y}{n_y}.$$

- The new method uses no class weights: plain, class-imbalanced CE gives $q(y \mid x)$, and the matching per-class scalar is Z_y. Both routes compose to the *same* scalar-corrected posterior,

$$D_y^{(Z)}(x) = \frac{N_y\, b(y \mid x)}{\sum_{y'} N_{y'}\, b(y' \mid x)} = \frac{Z_y\, q(y \mid x)}{\sum_{y'} Z_{y'}\, q(y' \mid x)} = \mathrm{softmax}_y\big(\log q(y \mid x) + \log Z_y\big).$$

- This label-shift correction equals $\pi(y \mid x)$ only if $\bar{w}_y(x) \equiv Z_y$. In general the physical weight also varies within each class [Saerens et al. 2002; Menon et al. 2021]

Punchline: everything depends on the weight through one object

CMOK is a name for this conditional-mean composition; an arXiv write-up is in preparation.

Sufficiency lemma — the most important piece of CMOK

For every (X, Y)-measurable h,

$$\mathbb{E}_Q[\, W\, h(X, Y)\,] \;=\; \mathbb{E}_Q[\, \bar{w}_Y(X)\, h(X, Y)\,].$$

Every physical posterior and expected yield/bin content sees W only through its conditional mean $\bar{w}_Y(x)$

- Write $W = \bar{w}_Y(X) + \eta$ with $\mathbb{E}_Q[\eta \mid X, Y] = 0$: the residue cancels from all *central values*
- It does *not* disappear from MC uncertainty: $\mathbb{E}[W^2]$ still controls the variance. Same $\bar{w}$ means identical expectations, not identical error bars

The factorisation and the master identity

Train two ordinary estimators on a shared representation, never weighting a loss:

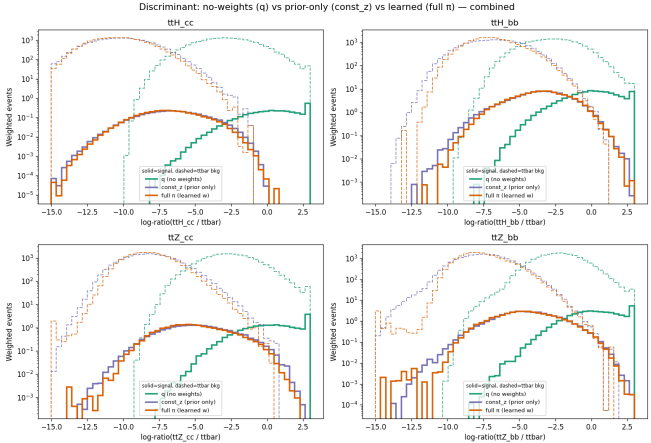
- the softmax probability $q(y \mid x)$ — *which class does this event look like?*
- the conditional mean $\bar{w}_y(x) = \mathbb{E}_Q[W \mid x, y] \geq 0$ under the physicality assumption — the weight is a regression *target*, never a loss multiplier
- $\bar{w}_y(x) = Z_y[1 + \delta_y(x)]$: per-class scale Z_y times the feature-dependent correction

At inference, upgrade the scalar Z_y to the per-event mean $\bar{w}_y(x)$:

$$\underbrace{\frac{Z_y\, q(y \mid x)}{\sum_{y'} Z_{y'}\, q(y' \mid x)}}_{D_y^{(Z)}(x)} \xrightarrow{Z_y \to \bar{w}_y(x)} \boxed{\pi(y \mid x) = \frac{\bar{w}_y(x)\, q(y \mid x)}{\sum_{y'} \bar{w}_{y'}(x)\, q(y' \mid x)}} \qquad (\star)$$

- Our life is easier: this is baked into ONNX at export — no more manually entered yields

Discriminant: q vs prior-only (Z_y) vs full π



Changes might look small but they bring HUGE differences.

Population equivalence and well-posedness

Classifier head

$$L_{\mathrm{CE}} = \mathbb{E}[-\log q_\theta(Y \mid X)] \;\geq\; 0$$

Strictly proper $\Rightarrow$ minimiser is the generation posterior $q(\cdot \mid x)$

Weight head

$$L_{\mathrm{reg}} = \mathbb{E}\big[(\mu_{\phi,Y}(X) - W)^2\big] \geq \mathbb{E}[v(X,Y)] \geq 0$$

Minimiser is the conditional mean $\bar{w}$; bounded below *even when* $W < 0$

Here $v(x,y) := \mathrm{Var}_Q(W \mid x, y)$ is the irreducible scatter. Under physicality, finite variance and $Z > 0$:

(i) unweighted CE $\to$ generation posterior $q(\cdot \mid x)$;
(ii) squared error $\to$ conditional mean $\bar{w}_y(x)$;
(iii) composing through $(\star) \to$ the physical posterior $\pi(\cdot \mid x)$;
(iv) the *population* weighted CE has the same minimiser π; its empirical signed-weight version remains ill-posed

Another payoff: a negative-weight-free template

- We want a sample reproducing the physical density with only nonnegative weights — no classifier needed
- What we do:
  1. train a flow on the *unweighted* events of class $y \Rightarrow$ a samplable count density $f_y(x) = q(x \mid y)$ — no weight enters, so signs / dynamic range never touch it;
  2. regress $\bar{w}_y(x) = \mathbb{E}[W \mid x, y]$ (in practice β-NLL; same population mean);
  3. normalised physical density: $p_y(x) = f_y(x)\,\bar{w}_y(x)/Z_y$
- Draw $x_m \sim f_y$ and attach a prediction constrained/floored nonnegative at export. At population level,

$$\frac{1}{M}\sum_m \bar{w}_y(x_m)\, g(x_m) \xrightarrow{M\to\infty} \int g\, f_y\, \bar{w}_y = Z_y \int g\, p_y, \qquad Z_y = \frac{N_y}{n_y}.$$

  This gives the physical template per generated event; finite effective statistics and model uncertainty still remain

Chapter 2: Training and model checks

How we train: era IDs as learned structure

- Run 2 and partial Run 3 are trained together, with an *era ID* embedded and added to the feature representation:
  - Categorical embedding lets $\bar{w}$ specialise per era: separate $\bar{w}_{\mathrm{R2}}(x)$ and $\bar{w}_{\mathrm{R3}}(x)$ — needed, since per-class $Z_y = \mathbb{E}[W \mid y]$ differs 0.6–2.4$\times$ between eras
- Without the ID, the head learns an MC-count-weighted mixture of the era means, tied to arbitrary generated proportions and unsuitable for per-era fits
- **Systematics next:** the same conditioning idea can be tested for systematic samples; NOT IN THIS PRESENTATION

Model and training, in brief

- Standard analysis classes, with the HF1b/1B and HF1c/1C splits this and further class splitting did not improve the classifier
- Direct from ROOT training using eventModulo; model details:

Backbone — ParT (d=128, depth 6)

- Attention: 8 heads $\times$ 16 dim-head = 128; SwiGLU FF $\times$4; RMSNorm + QK-norm; drop-path 0.1, LayerScale 10^{-4}
- Pairwise embedder: depth 2, $\times$2 SwiGLU — ΔR, k_t, z, m^2, m_T + FTAG interactions
- MET: gated injection (per-token sigmoid)
- Pooling: multi-query cross-attn

Two factorised heads (each: 2-layer MLP, 128, SwiGLU $\times$4, RMSNorm, residual, drop 0.1)

- Classifier $q(y \mid x)$: $\rightarrow$ 11 logits (unweighted CE)
- Weight head $\bar{w}_y(x)$: per-class β-NLL *regression* (β=0.5); target W/Z_y is $O(1)$
- Compose $\rightarrow \pi$ via $(\star)$, only at inference

Chapter 3: Results

Combined expected limit improves 12% (stat-only)

Same 3.2.0-NTuple test sample, channels and binning (etc) as Jan two weeks ago; the comparison only changes weight treatment

	Run 2	Partial Run 3	Run 2 + Partial Run 3
0L	3.740 ± 0.134 3.210 ± 0.149 −14%	3.122 ± 0.111 2.656 ± 0.078 −15%	2.397 ± 0.061 2.046 ± 0.053 −15%
1L	3.975 ± 0.125 3.663 ± 0.067 −8%	3.685 ± 0.108 3.236 ± 0.089 −12%	2.702 ± 0.058 2.425 ± 0.042 −10%
2L	8.457 ± 0.359 7.937 ± 0.315 −6%	7.666 ± 0.227 7.519 ± 0.254 −2%	5.680 ± 0.143 5.458 ± 0.141 −4%
Combined	2.593 ± 0.057 2.310 ± 0.059 −11%	2.275 ± 0.050 1.981 ± 0.038 −13%	1.710 ± 0.027 $\mathbf{1.504 \pm 0.023}$ −12%

Table 1: expected 95% CL upper limit on $\mu(\mathrm{ttH} \to c\bar{c})$; lower is better. Top: Jan's baseline; bottom: CMOK. Combined entries use inverse-quadrature under a stat-only Gaussian approximation; **not a joint fit**.

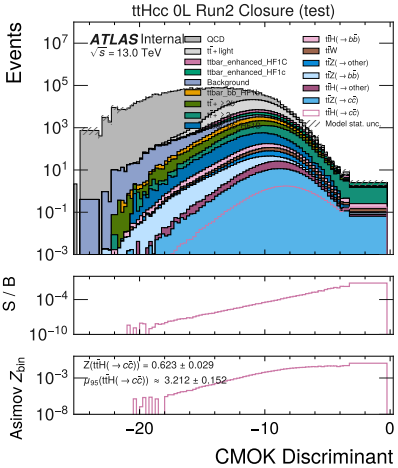
The other three signals — expected significances

	Run 2	Partial Run 3	Run 2 + Partial Run 3
ttH$\to b\bar{b}$			
0L	20.85 ± 0.45	24.95 ± 0.54	32.52 ± 0.51
1L	15.88 ± 0.13	16.67 ± 0.18	23.02 ± 0.16
2L	6.04 ± 0.11	6.08 ± 0.10	8.57 ± 0.10
Combined	26.90 ± 0.36	30.61 ± 0.45	$\mathbf{40.75 \pm 0.41}$
ttZ$\to c\bar{c}$			
0L	4.49 ± 0.18	5.20 ± 0.17	6.87 ± 0.17
1L	3.44 ± 0.08	3.62 ± 0.13	4.99 ± 0.11
2L	1.69 ± 0.12	1.67 ± 0.11	2.37 ± 0.11
Combined	5.90 ± 0.15	6.55 ± 0.15	$\mathbf{8.82 \pm 0.15}$
ttZ$\to b\bar{b}$			
0L	10.21 ± 0.47	9.61 ± 0.29	14.02 ± 0.39
1L	7.25 ± 0.11	6.96 ± 0.12	10.05 ± 0.11
2L	2.76 ± 0.08	2.63 ± 0.08	3.81 ± 0.08
Combined	12.82 ± 0.38	12.16 ± 0.24	$\mathbf{17.66 \pm 0.32}$

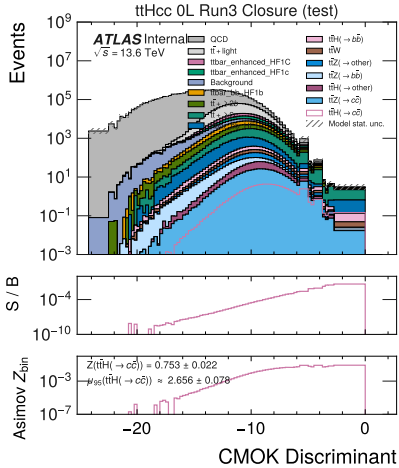
Tables 2–4: CMOK expected Asimov figures of merit, stat-only. Combined entries are quadrature combinations (Gaussian approximation), **not joint fits**.

Chapter 4: Discriminant plots

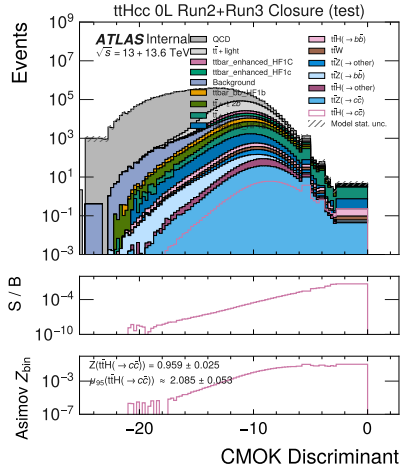
ttH→ $c\bar{c}$ discriminant — 0L, MC closure



Run 2

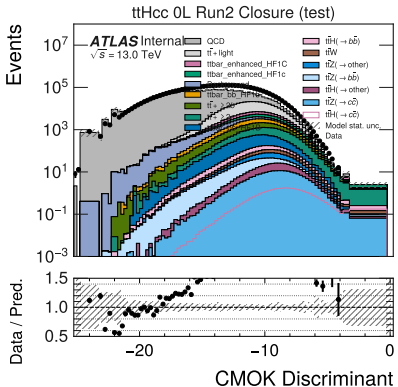


Partial Run 3

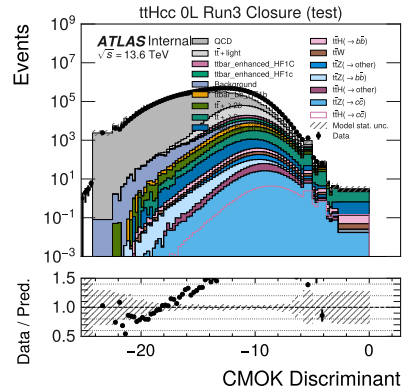


Run 2 + Partial Run 3

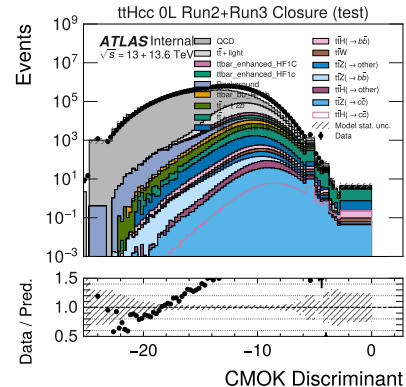
ttH→ $c\bar{c}$ discriminant — 0L, data / prediction



Run 2

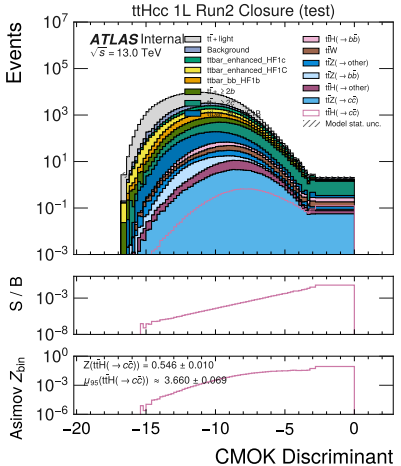


Partial Run 3

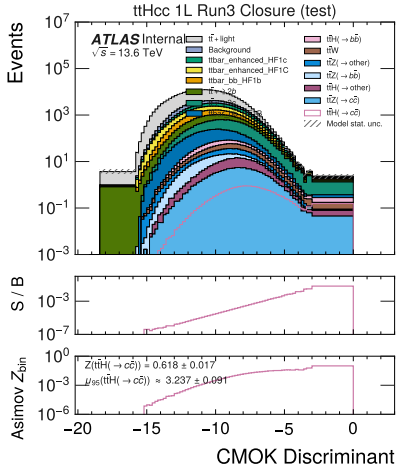


Run 2 + Partial Run 3

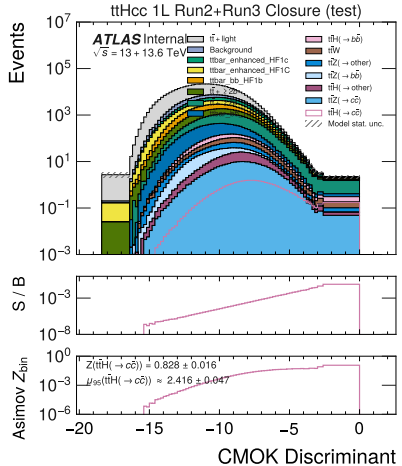
ttH→ $c\bar{c}$ discriminant — 1L, MC closure



Run 2

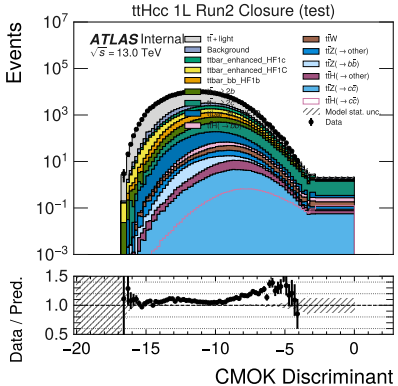


Partial Run 3

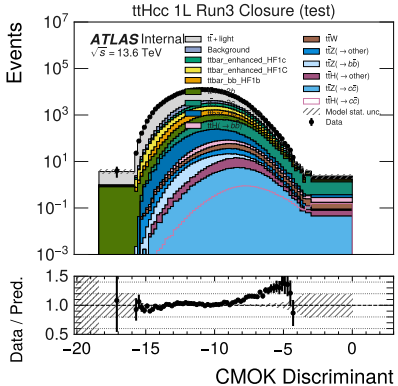


Run 2 + Partial Run 3

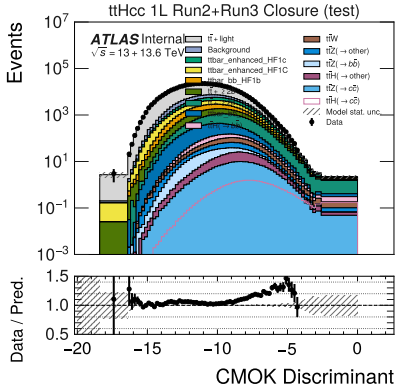
ttH→ $c\bar{c}$ discriminant — 1L, data / prediction



Run 2

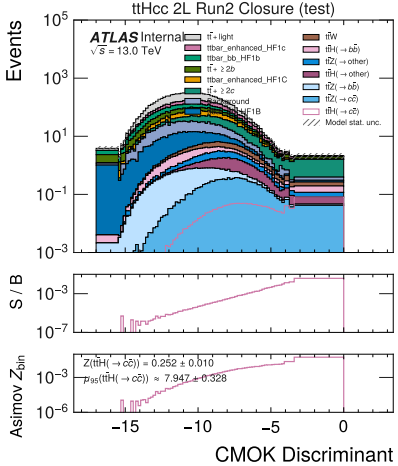


Partial Run 3

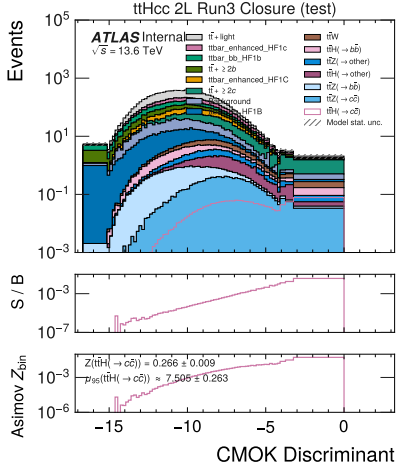


Run 2 + Partial Run 3

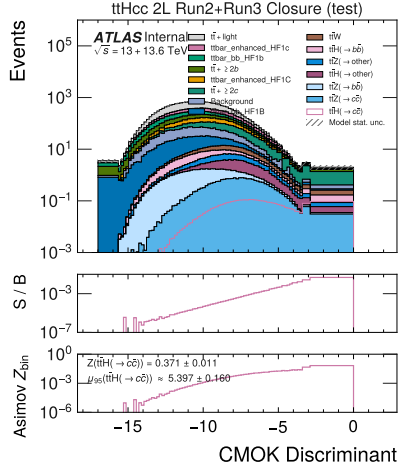
ttH→ $c\bar{c}$ discriminant — 2L, MC closure



Run 2

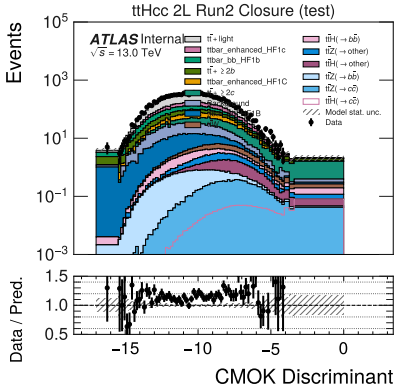


Partial Run 3

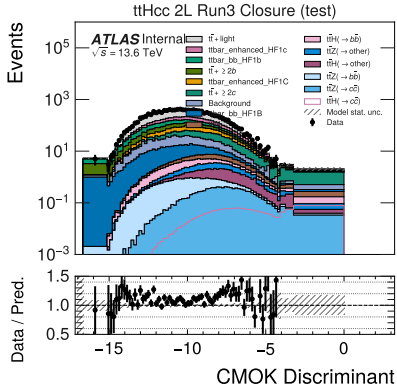


Run 2 + Partial Run 3

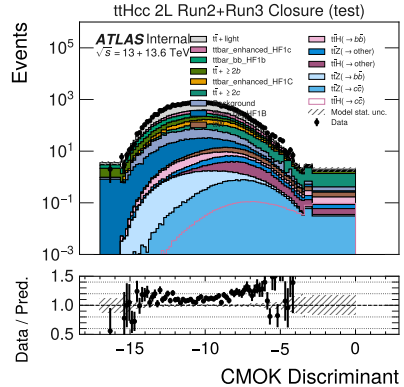
ttH→ $c\bar{c}$ discriminant — 2L, data / prediction



Run 2



Partial Run 3

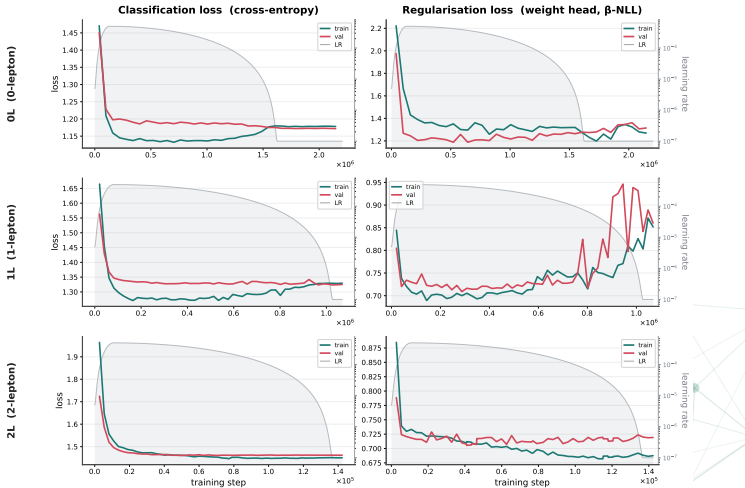


Run 2 + Partial Run 3

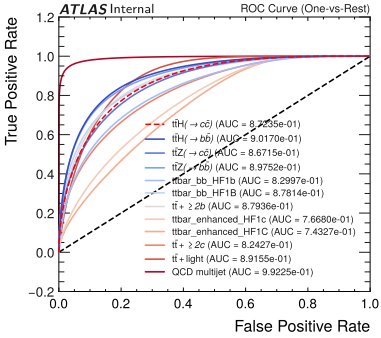
Chapter 5: Model validation

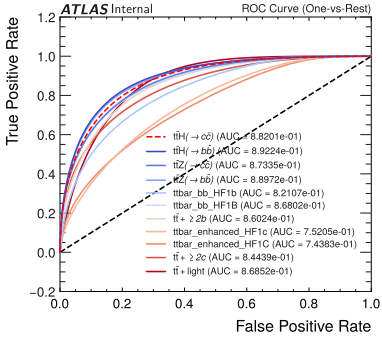
Metrics and learning the weights

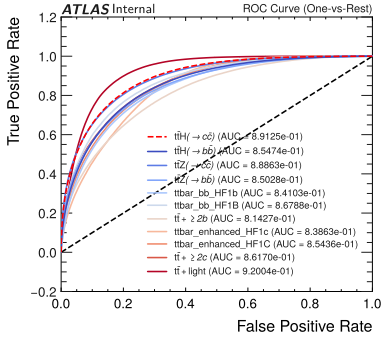
- **No confusion matrix / accuracy:** argmax metrics are swamped by the $t\bar{t}$ light majority due to no class weights (resolvable; perhaps also worth it to spend some time on balanced classes); class-resolved ROC is more informative
- β-NLL is a regression loss, not a direct closure metric; its rise seems not to impact performance
- CE and β-NLL versus the learning-rate (LR) schedule diagnose optimisation; region-resolved closure is the current physics proxy



One-vs-rest ROC (AUC per class) — 0L / 1L / 2L



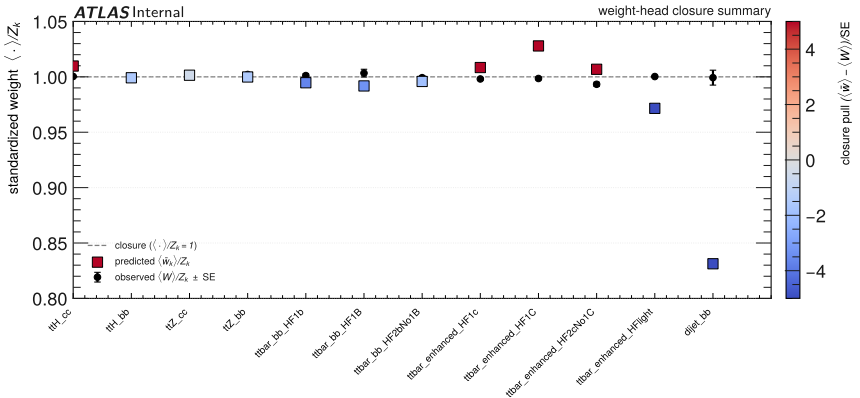




0L 1L 2L

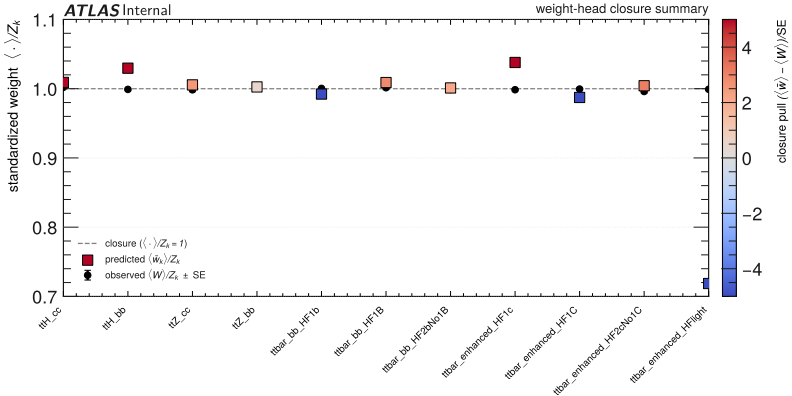
Per class: TPR vs FPR ranking that class against all others; AUC in the legend. **AUC is base-rate invariant** — it ranks by the LR f_s / f_b (no prior), so unlike accuracy/CM imbalance moves the *threshold*, not the ranking.

Weight-head closure summary (all classes) — 0L



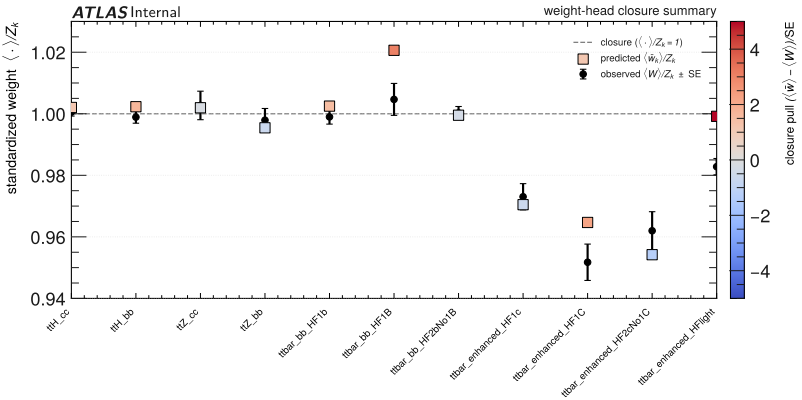
Per class: standardised mean weight — *predicted* $\langle\bar{w}_y\rangle / Z_y$ (squares) vs *observed* $\langle W\rangle / Z_y \pm$SE (dots), coloured by pull; on the dashed line = closes.

Weight-head closure summary (all classes) — 1L



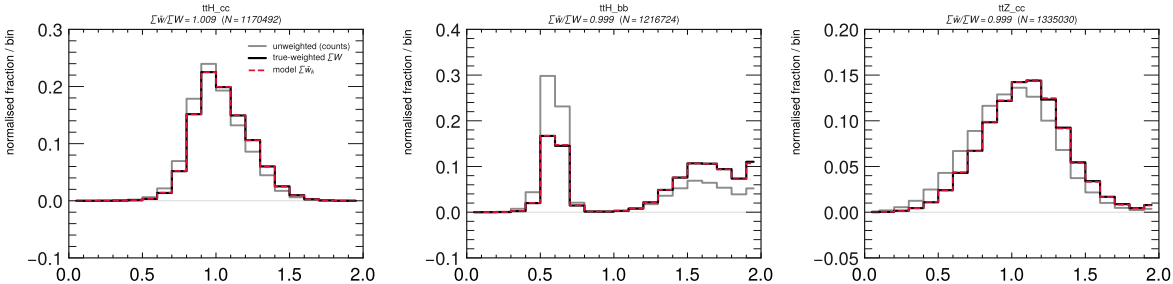
Per class: standardised mean weight — *predicted* $\langle \bar{w}_y \rangle / Z_y$ (squares) vs *observed* $\langle W \rangle / Z_y \pm$SE (dots), coloured by pull; on the dashed line = closes.

Weight-head closure summary (all classes) — 2L



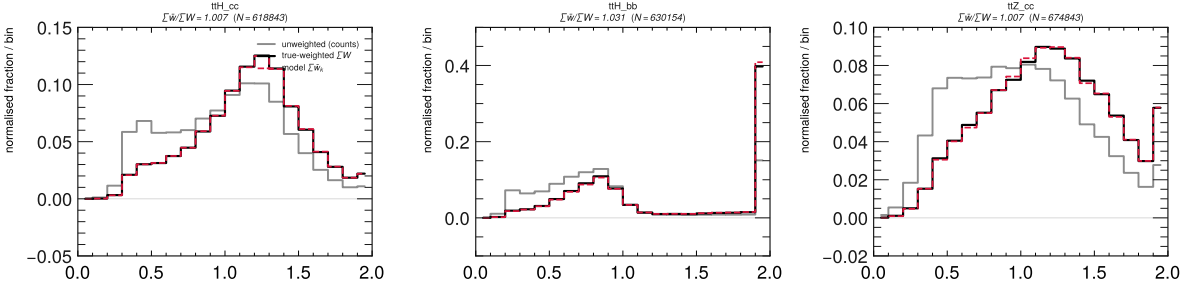
Per class: standardised mean weight — *predicted* $\langle\bar{w}_y\rangle / Z_y$ (squares) vs *observed* $\langle W\rangle / Z_y \pm$SE (dots), coloured by pull; on the dashed line = closes.

Weight-shape closure, 0L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



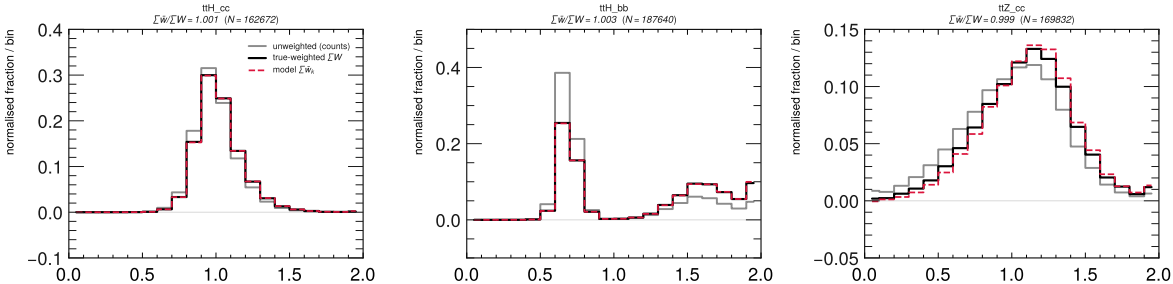
Predicted standardised weight $r_y(x) = \bar{w}_y(x)/Z_y$ per class: unweighted counts (grey), true weighted ΣW (black), model $\Sigma\bar{w}$ (red dashed); red $\approx$ black $\Rightarrow$ the head reproduces the shape. *All remaining classes are in backup.*

Weight-shape closure, 1L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



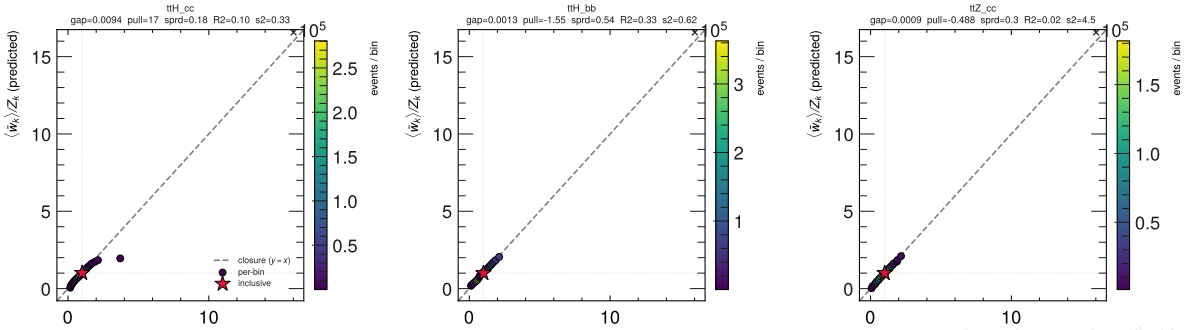
Predicted standardised weight $r_y(x) = \bar{w}_y(x)/Z_y$ per class: unweighted counts (grey), true weighted ΣW (black), model $\Sigma\bar{w}$ (red dashed); red $\approx$ black $\Rightarrow$ the head reproduces the shape. *All remaining classes are in backup.*

Weight-shape closure, 2L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



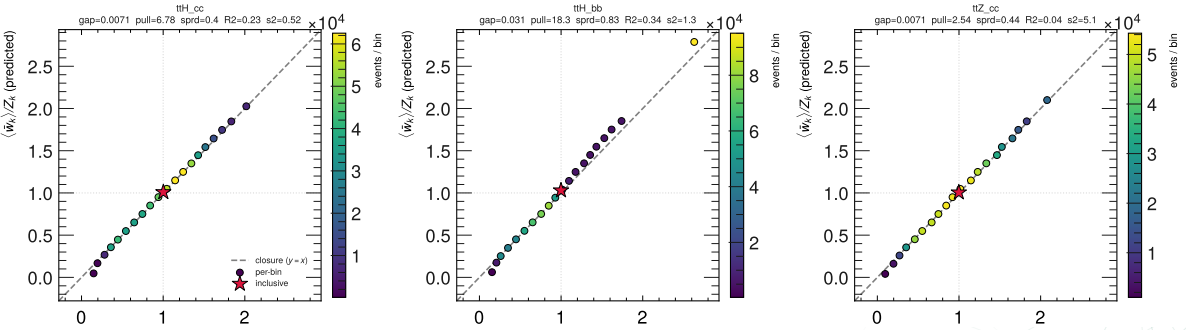
Predicted standardised weight $r_y(x) = \bar{w}_y(x)/Z_y$ per class: unweighted counts (grey), true weighted ΣW (black), model $\Sigma\bar{w}$ (red dashed); red $\approx$ black $\Rightarrow$ the head reproduces the shape. *All remaining classes are in backup.*

Weight-head closure, 0L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



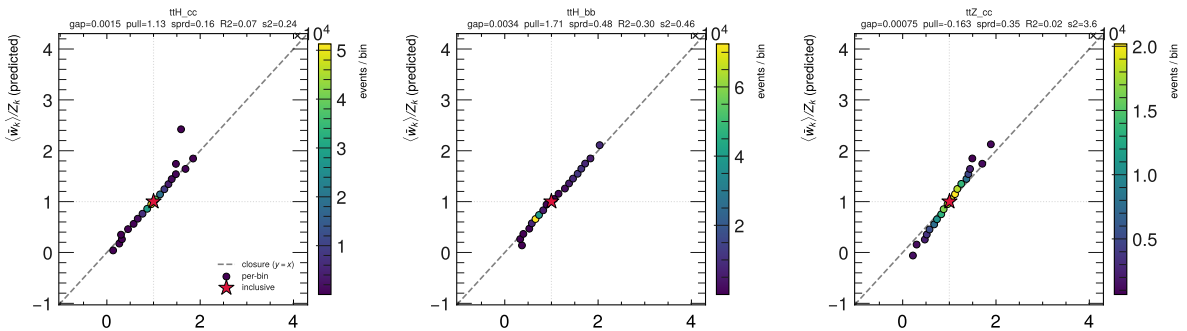
Region-resolved: *predicted* vs *observed* standardised mean *per bin* (colour = events/bin) + inclusive point (star); on $y{=}x$ = closes; box: gap / pull / spread / R^2. *All remaining classes are in backup.*

Weight-head closure, 1L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



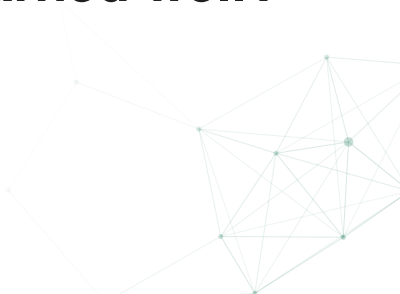
Region-resolved: *predicted* vs *observed* standardised mean *per bin* (colour = events/bin) + inclusive point (star); on $y{=}x$ = closes; box: gap / pull / spread / R^2. *All remaining classes are in backup.*

Weight-head closure, 2L — ttH $c\bar{c}$, ttH $b\bar{b}$, ttZ $c\bar{c}$



Region-resolved: *predicted* vs *observed* standardised mean *per bin* (colour = events/bin) + inclusive point (star); on $y{=}x$ = closes; box: gap / pull / spread / R^2. *All remaining classes are in backup.*

Chapter 6: Has our model learned well?



Validation I — the physical $\log(S/B)$ linearity test

The question this test asks

Is the delivered D the physical log-likelihood ratio, not merely the generation one?

Procedure:

1. bin the validation events in $D(x)$;
2. per bin, sum physical yields $S = \sum_{i \in \text{bin},\, y_i = s} w_i$, B likewise — the target carries the signed weights: it probes the weight treatment, not just the classifier;
3. plot $\log(S/B)$ against D and fit a straight line.

If $D = \log(p_S/p_B)$ exactly, every bin obeys

$$\frac{S}{B}(x) = \frac{N_S\, p_S(x)}{N_B\, p_B(x)} = \frac{N_S}{N_B}\, e^{D(x)} \quad \Longrightarrow \quad \log \frac{S}{B} = \underbrace{D}_{\text{slope 1}} + \underbrace{\log \frac{N_S}{N_B}}_{\text{intercept}}$$

Pass condition

Fitted slope one, flat residual. Slope $\neq 1$ = residual calibration or compression of the learned physical log-likelihood ratio.

Validation I — why the weights must live *inside* the discriminant

The target itself contains a *per-event* weight ratio (recall $p_y = f_y\,\bar{w}_y / Z_y$):

$$\log \frac{p_S(x)}{p_B(x)} = \log \frac{f_S(x)}{f_B(x)} + \log \frac{\bar{w}_S(x)}{\bar{w}_B(x)} + \text{const.}$$

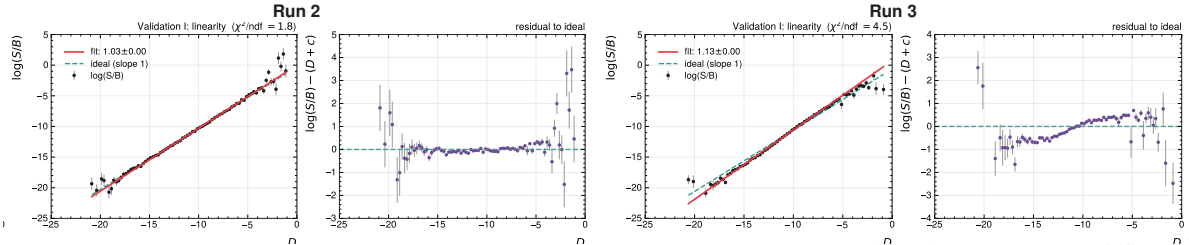
What each recipe delivers:

recipe	delivered log-odds	per-event weight info?
unweighted CE q	$\log \frac{f_S}{f_B} + \log \frac{\rho_S}{\rho_B}$	×
class-weighted CE (c_y)	$\log \frac{f_S}{f_B} + \log \frac{c_S \rho_S}{c_B \rho_B}$	×
constant-Z_y corrected $D^{(Z)}$	$\log \frac{f_S}{f_B} + \log \frac{Z_S \rho_S}{Z_B \rho_B}$	×
full π	$\log \frac{f_S}{f_B} + \log \frac{\rho_S}{\rho_B} + \log \frac{\bar{w}_S(x)}{\bar{w}_B(x)}$	✓

A scalar can move a line; it cannot straighten a curve

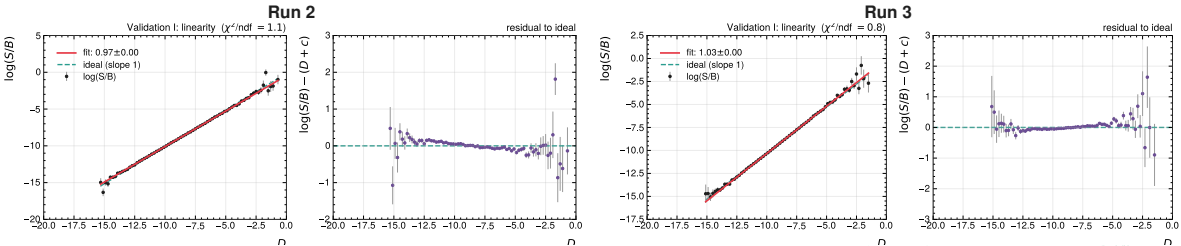
The three weight-blind rows differ from each other only by *constants*: same ranking, same shape, different intercept. All miss the *same per-event* term $\log(\bar{w}_S(x)/\bar{w}_B(x))$ — exactly what bends $\log(S/B)$, so slope $\neq 1$ and curvature survive any scalar shift. Only π can be linear with slope one.

ttH→ $c\bar{c}$ — 0L: physical $\log(S/B)$ linearity



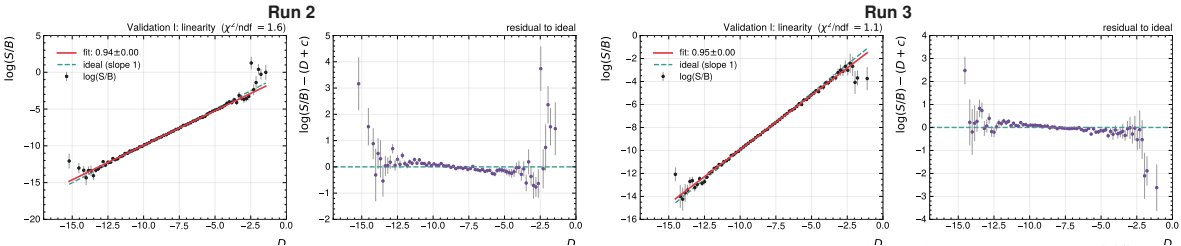
Per era: binned physical $\log(S/B)$ versus the discriminant D with the fitted line (left; ideal slope one), and the residual to the ideal line, $\log(S/B) - (D + c)$ (right; flat = passes). Other signals in **Backup 2**.

ttH→ $c\bar{c}$ — 1L: physical $\log(S/B)$ linearity



Per era: binned physical $\log(S/B)$ versus the discriminant D with the fitted line (left; ideal slope one), and the residual to the ideal line, $\log(S/B) - (D + c)$ (right; flat = passes). Other signals in **Backup 2**.

ttH→ $c\bar{c}$ — 2L: physical $\log(S/B)$ linearity



Per era: binned physical $\log(S/B)$ versus the discriminant D with the fitted line (left; ideal slope one), and the residual to the ideal line, $\log(S/B) - (D + c)$ (right; flat = passes). Other signals in **Backup 2**.

Validation II — the morphing (reweighting) closure test

The question this test asks

Is D the physical log-likelihood ratio *event by event* — strong enough to *transport* the signal density onto the background one, not merely to rank events?

The procedure, step by step:

1. per event, compute the morph factor $m(x) = e^{-D(x)}$;
2. reweight every signal event by $m(x_i)\, w_i$ — the morph factor *times* its physical weight;
3. compare the morphed signal shape with the physical background shape, observable by observable, and check that $\log(S_w / B)$ is flat.

If D is exact, $m(x) = p_B(x)/p_S(x)$, so $m(x)\, p_S(x) = p_B(x)$: the morph closes every function of the model inputs x in the population limit.

Pass condition

After reweighting, the signal and background shapes overlap and $\log(S_w / B)$ is flat. We compare the full π composition against the honest constant-Z_y baseline.

Validation II — what each weight treatment leaves behind

Apply each candidate's morph to the physical signal density $p_S = f_S\,\bar{w}_S/Z_S$:

- **weight-blind** D_q ($m_q = f_B/f_S$): morphs the unweighted count shapes exactly — but on the physical ones it leaves

$$\frac{m_q(x)\,p_S(x)}{p_B(x)} = \frac{(f_B/f_S)\;f_S\,\bar{w}_S/Z_S}{f_B\,\bar{w}_B/Z_B} = \frac{r_S(x)}{r_B(x)}, \qquad r_y(x) = \frac{\bar{w}_y(x)}{Z_y},$$

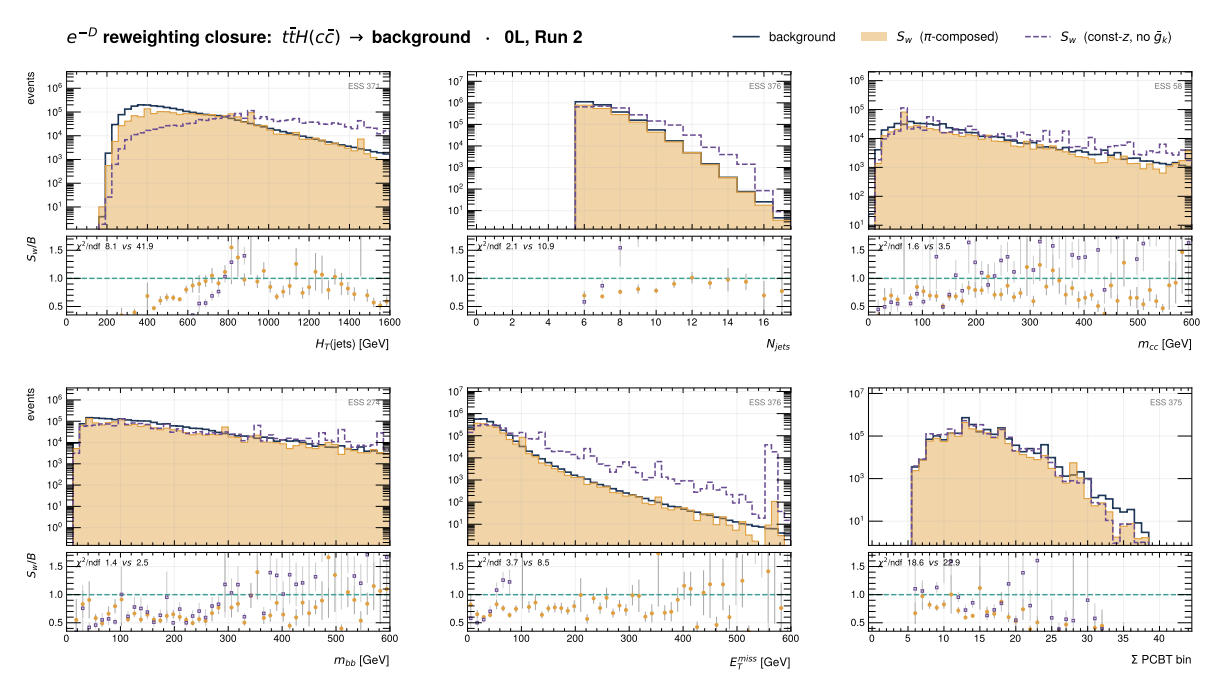
  a per-event residual, $\neq 1$ wherever the standardised weights differ between the classes;

- **constant-Z_y** multiplies m_q by the *constant* Z_S/Z_B: it can fix the overall *normalisation*, never the residual shape $r_S(x)/r_B(x)$;
- **full** π: $m_\pi = e^{-D_\pi} \propto p_B/p_S$ — the residual is identically 1; every function of x closes.

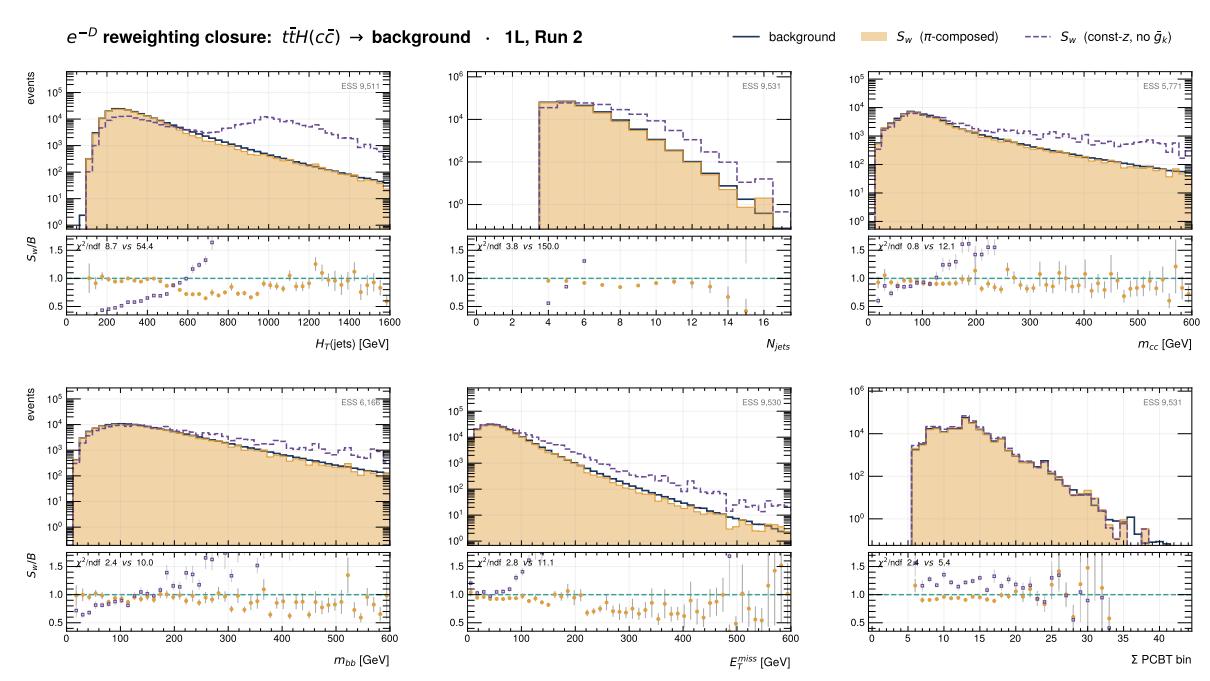
Where to see it in the plots

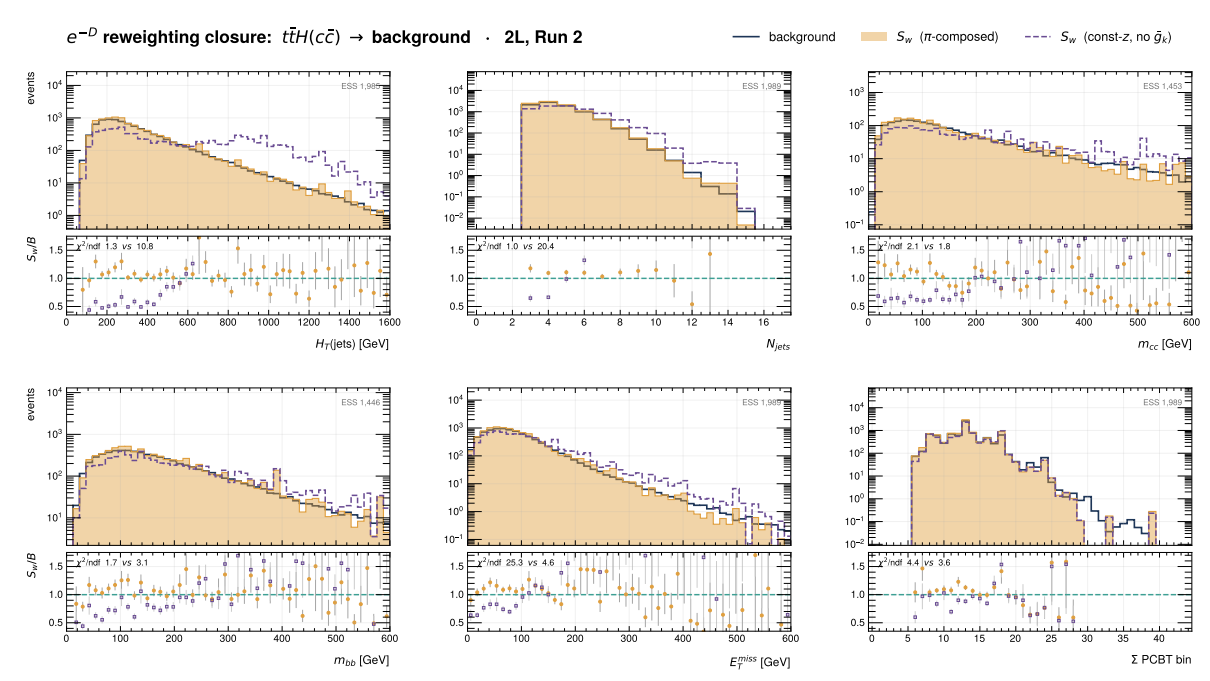
The χ^2/ndf heatmap columns (π vs constant-Z_y) quantify exactly this residual, observable by observable; the per-observable overlays show it bin by bin.

Shown in the body: ttH→ $c\bar{c}$ in 0L/1L/2L for Run 2 and Run 3. Other signals in **Backup 2**.


e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 372
χ^2/ndf 8.1 vs 41.9
H_T(jets) [GeV]
ESS 376
χ^2/ndf 2.1 vs 10.9
N_{jets}
ESS 58
χ^2/ndf 1.6 vs 3.5
m_{cc} [GeV]
ESS 275
χ^2/ndf 1.4 vs 2.5
m_{bb} [GeV]
ESS 376
χ^2/ndf 3.7 vs 8.5
E_T^{miss} [GeV]
ESS 375
χ^2/ndf 18.6 vs 22.9
Σ PCBT bin

e^{-D} **reweighting closure:** $t\bar{t}H(c\bar{c})$ **→ background · 1L, Run 2**

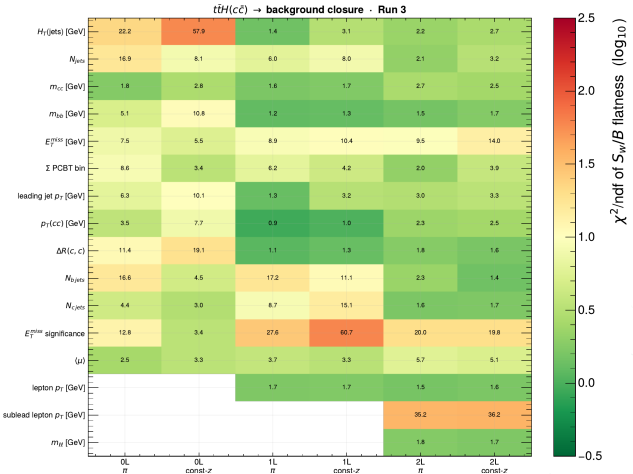



e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 1,985
χ^2/ndf 1.3 vs 10.8
H_T(jets) [GeV]
ESS 1,989
χ^2/ndf 1.0 vs 20.4
N_{jets}
ESS 1,453
χ^2/ndf 2.1 vs 1.8
m_{cc} [GeV]
ESS 1,446
χ^2/ndf 1.7 vs 3.1
m_{bb} [GeV]
ESS 1,985
χ^2/ndf 25.3 vs 4.6
E_T^{miss} [GeV]
ESS 1,989
χ^2/ndf 4.4 vs 3.6
Σ PCBT bin

All-observable morphing closure — Run 2

Each row is one observable; within 0L, 1L and 2L, the columns compare morphing with the **full physical posterior** π against the **constant-Z_y baseline**. The value and colour show χ^2/ndf for S_w/B flatness (lower is better).

All-observable morphing closure — Run 3

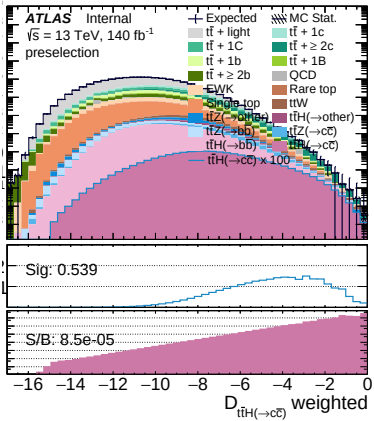


Each row is one observable; within 0L, 1L and 2L, the columns compare morphing with the **full physical posterior** π against the **constant-Z_y baseline**. The value and colour show χ^2/ndf for S_w/B flatness (lower is better).

Chapter 7: TNA inference

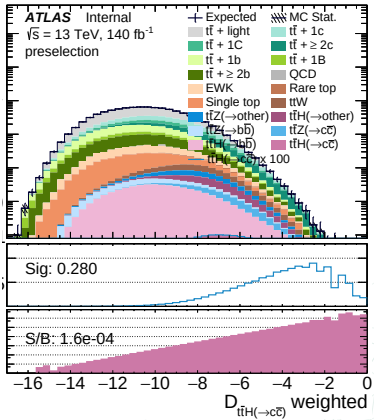
TNA inference — blinded preselection

1L



Blinded analysis-region inference at preselection.

2L



Thanks, Zoja.

Conclusion

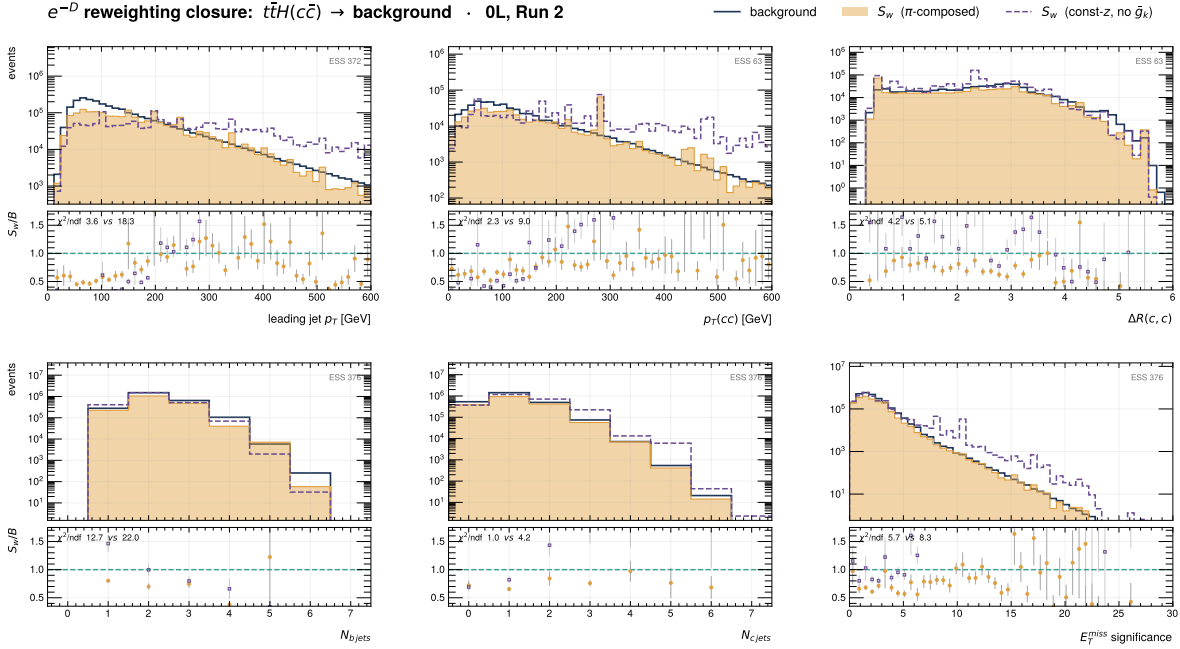
What this work delivers

A rigorous observable, $\pi(y \mid x) \propto \bar{w}_y(x)\, q(y \mid x)$, for the physical weighted problem — learned with two ordinary objectives and composed at inference

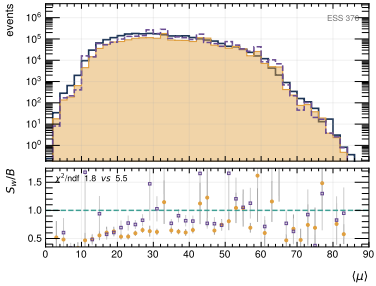
- **Result:** the combined stat-only expected limit improves by 12%
- **Validation:** physical-S/B linearity and signal-to-background morphing test the composition; model largely closes
- **Inference:** simple
- **Fits:** preliminary studies suggest smaller limit degradation; pending more investigation
- **Further plans:** train on 3.3.0 NTuples, with systematic samples, back to 9 classes, then fit

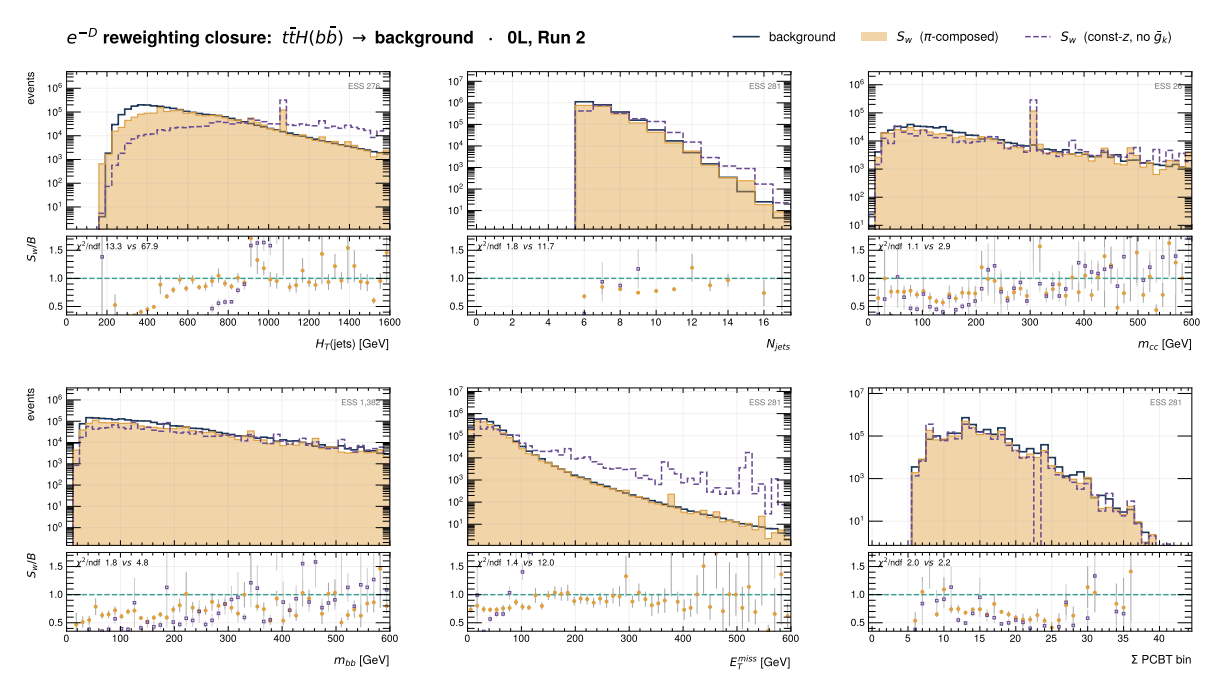
Questions?

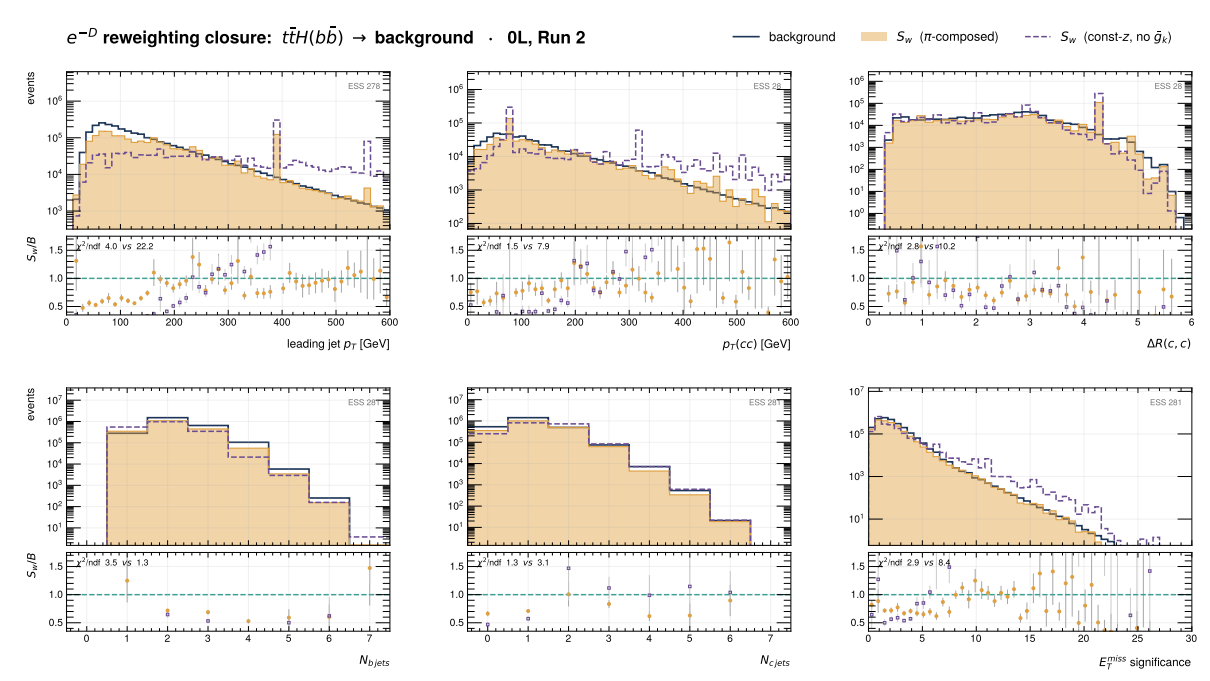
Backup 1: Multi-observable closure


e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 372
ESS 63
ESS 63
S_w/B
χ^2/ndf 3.6 vs 18.3
χ^2/ndf 2.3 vs 9.0
χ^2/ndf 4.2 vs 5.1
leading jet p_T [GeV]
$p_T(cc)$ [GeV]
$\Delta R(c,c)$
ESS 375
ESS 375
ESS 376
χ^2/ndf 12.7 vs 22.0
χ^2/ndf 1.0 vs 4.2
χ^2/ndf 5.7 vs 8.3
$N_{b\,jets}$
$N_{c\,jets}$
E_T^{miss} significance

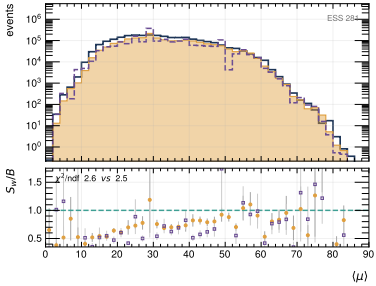
e^{-D} reweighting closure: $t\bar{t}H(c\bar{c}) \rightarrow$ background · 0L, Run 2

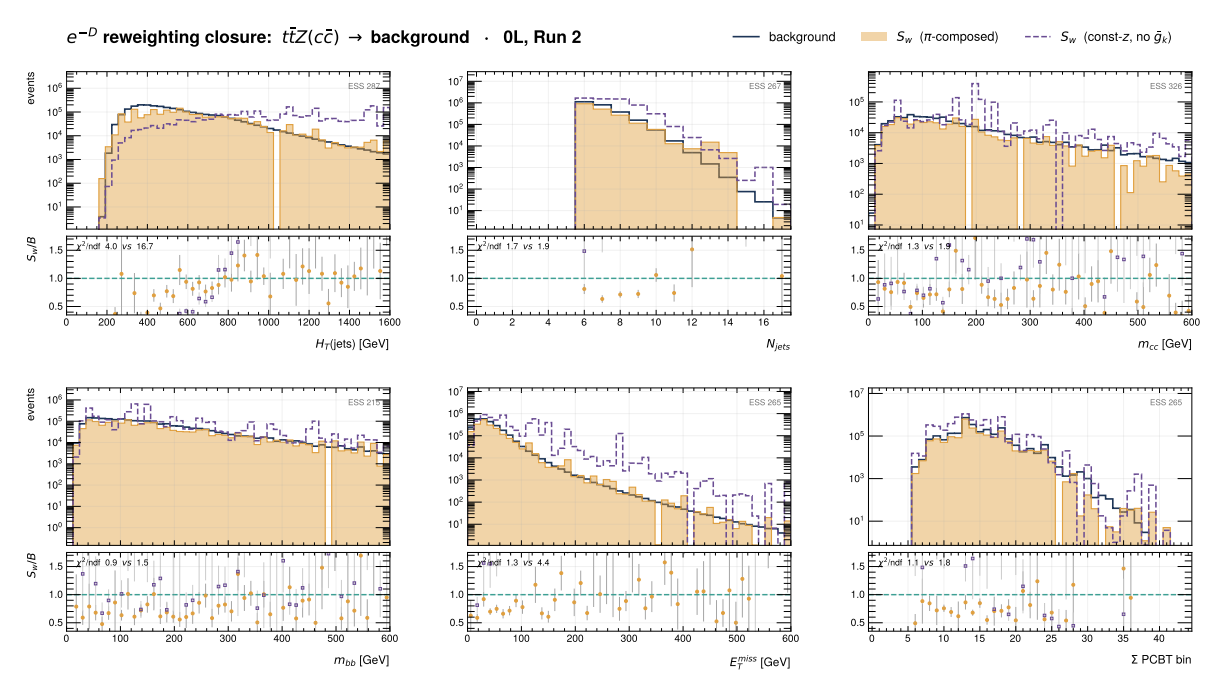


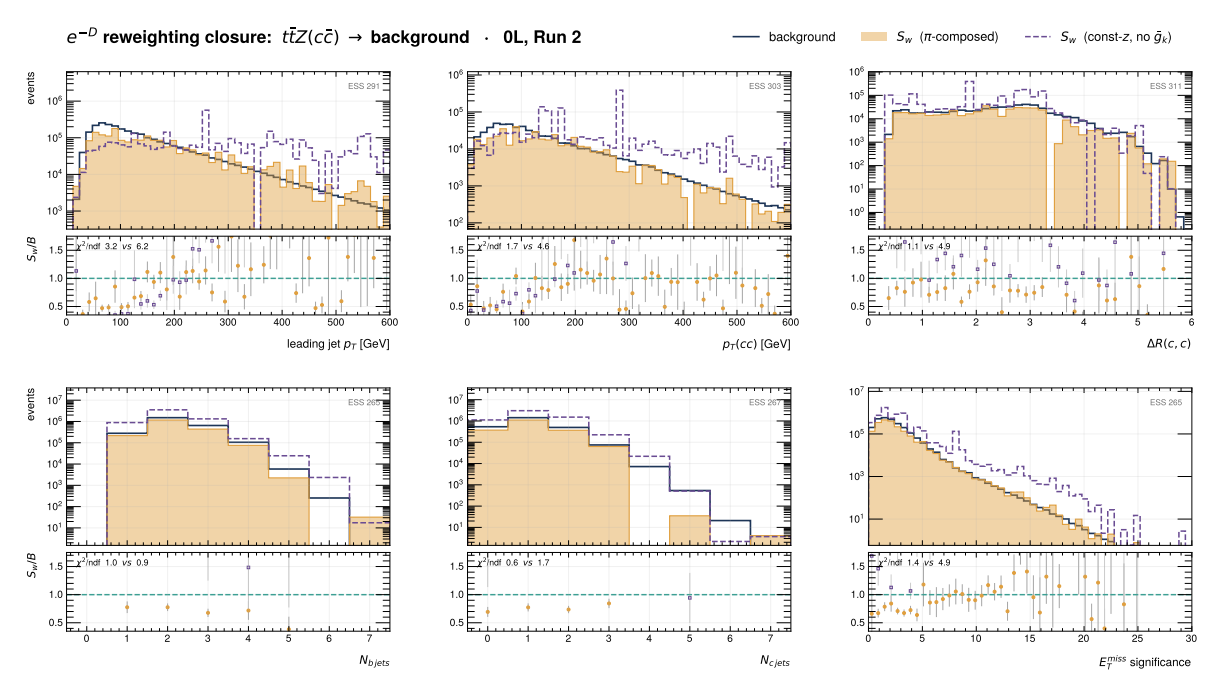

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 27k
χ^2/ndf 13.3 vs 67.9
H_T(jets) [GeV]
ESS 281
χ^2/ndf 1.8 vs 11.7
N_{jets}
ESS 29
χ^2/ndf 1.1 vs 2.9
m_{cc} [GeV]
ESS 1.38k
χ^2/ndf 1.8 vs 4.8
m_{bb} [GeV]
ESS 281
χ^2/ndf 1.4 vs 12.0
E_T^{miss} [GeV]
ESS 281
χ^2/ndf 2.0 vs 2.2
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
ESS 278
χ^2/ndf 4.0 vs 22.2
S_w/B
leading jet p_T [GeV]
ESS 28
χ^2/ndf 1.5 vs 7.9
$p_T(cc)$ [GeV]
ESS 28
χ^2/ndf 2.8 vs 10.2
$\Delta R(c,c)$
ESS 287
χ^2/ndf 3.5 vs 1.3
$N_{b\,jets}$
ESS 287
χ^2/ndf 1.3 vs 3.1
$N_{c\,jets}$
ESS 281
χ^2/ndf 2.9 vs 8.4
E_T^{miss} significance

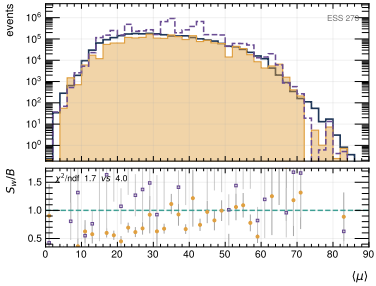
e^{-D} reweighting closure: $t\bar{t}H(b\bar{b}) \rightarrow$ background · 0L, Run 2

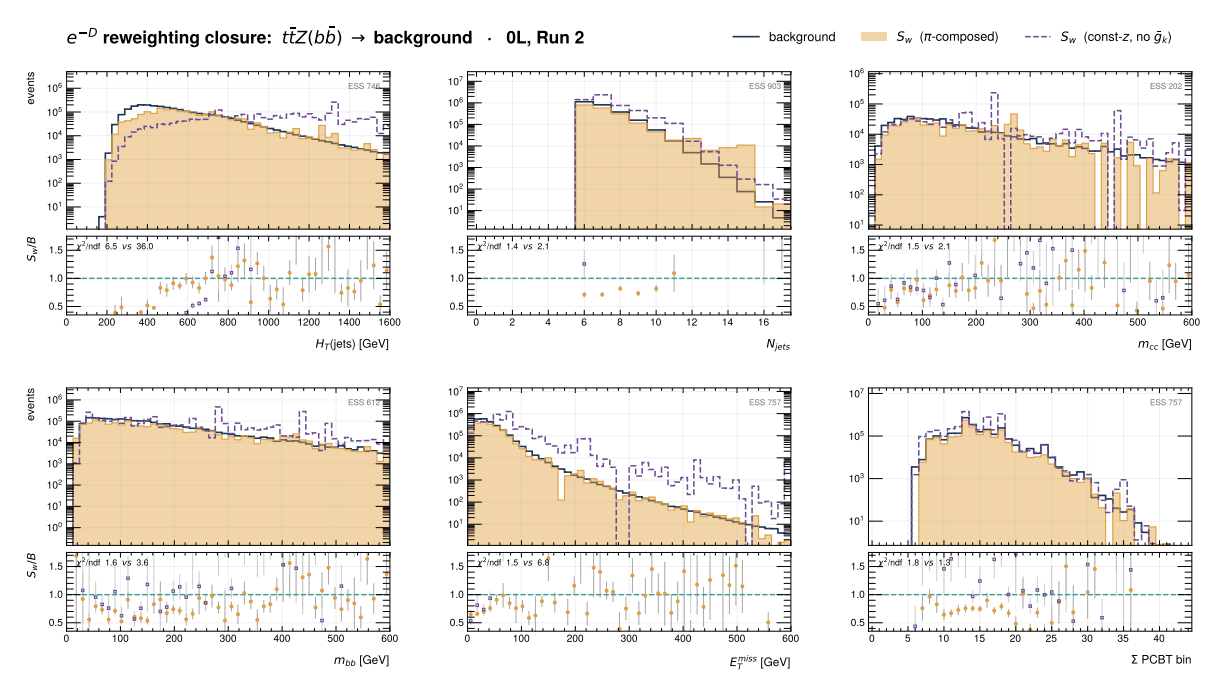


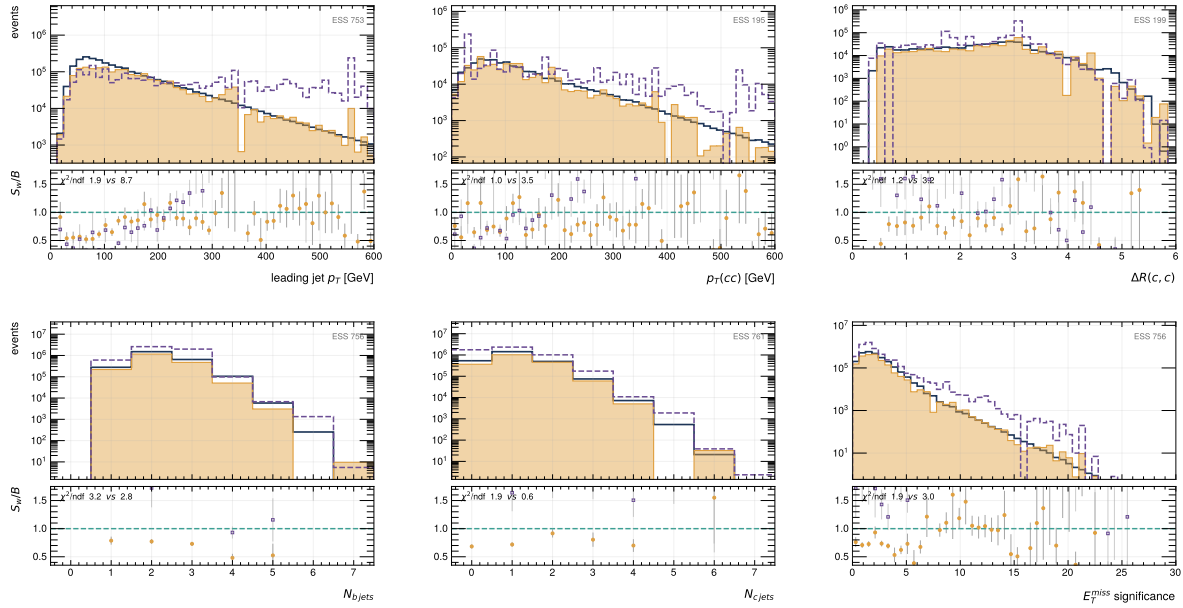

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 282
χ^2/ndf 4.0 vs 16.7
H_T(jets) [GeV]
ESS 267
χ^2/ndf 1.7 vs 1.9
N_{jets}
ESS 326
χ^2/ndf 1.3 vs 1.9
m_{cc} [GeV]
ESS 215
χ^2/ndf 0.9 vs 1.5
m_{bb} [GeV]
ESS 265
χ^2/ndf 1.3 vs 4.4
E_T^{miss} [GeV]
ESS 265
χ^2/ndf 1.1 vs 1.8
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 291
χ^2/ndf 3.2 vs 6.2
leading jet p_T [GeV]
ESS 303
χ^2/ndf 1.7 vs 4.6
$p_T(cc)$ [GeV]
ESS 311
χ^2/ndf 1.1 vs 4.9
$\Delta R(c,c)$
ESS 265
χ^2/ndf 1.0 vs 0.9
$N_{b\,jets}$
ESS 267
χ^2/ndf 0.6 vs 1.7
$N_{c\,jets}$
ESS 265
χ^2/ndf 1.4 vs 4.9
E_T^{miss} significance

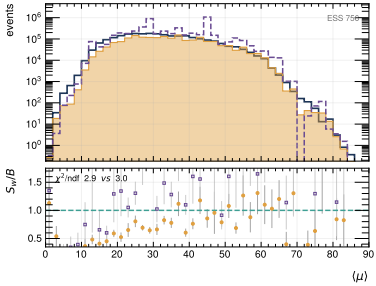
e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 0L, Run 2

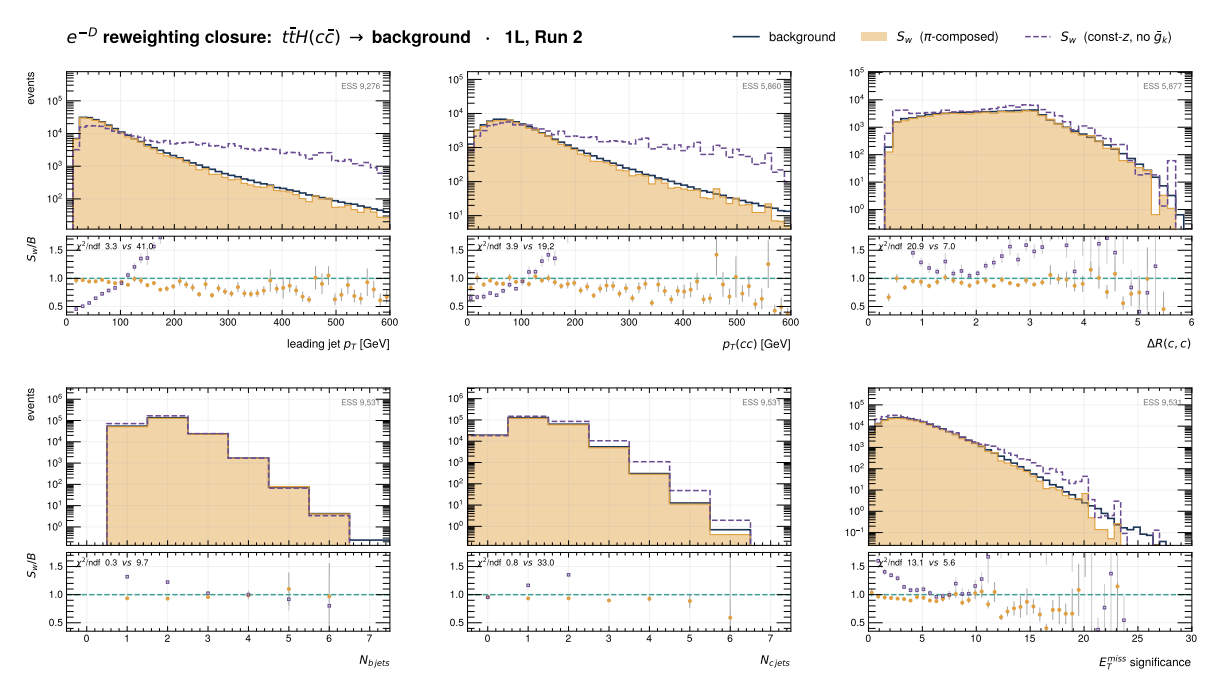



e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 748
χ^2/ndf 6.5 vs 36.0
H_T(jets) [GeV]
ESS 903
χ^2/ndf 1.4 vs 2.1
N_{jets}
ESS 202
χ^2/ndf 1.5 vs 2.1
m_{cc} [GeV]
ESS 617
χ^2/ndf 1.6 vs 3.6
m_{bb} [GeV]
ESS 757
χ^2/ndf 1.5 vs 6.8
E_T^{miss} [GeV]
ESS 757
χ^2/ndf 1.8 vs 1.3
Σ PCBT bin

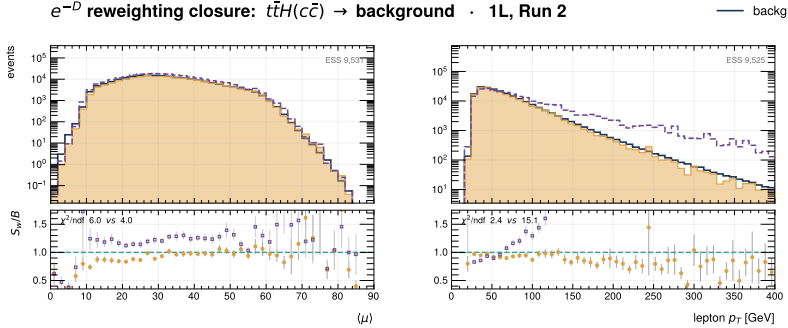

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 753
χ^2/ndf 1.9 vs 8.7
leading jet p_T [GeV]
ESS 195
χ^2/ndf 1.0 vs 3.5
$p_T(cc)$ [GeV]
ESS 199
χ^2/ndf 1.2 vs 3.2
$\Delta R(c,c)$
ESS 755
χ^2/ndf 3.2 vs 2.8
$N_{b\,jets}$
ESS 767
χ^2/ndf 1.9 vs 0.6
$N_{c\,jets}$
ESS 756
χ^2/ndf 1.9 vs 3.0
E_T^{miss} significance

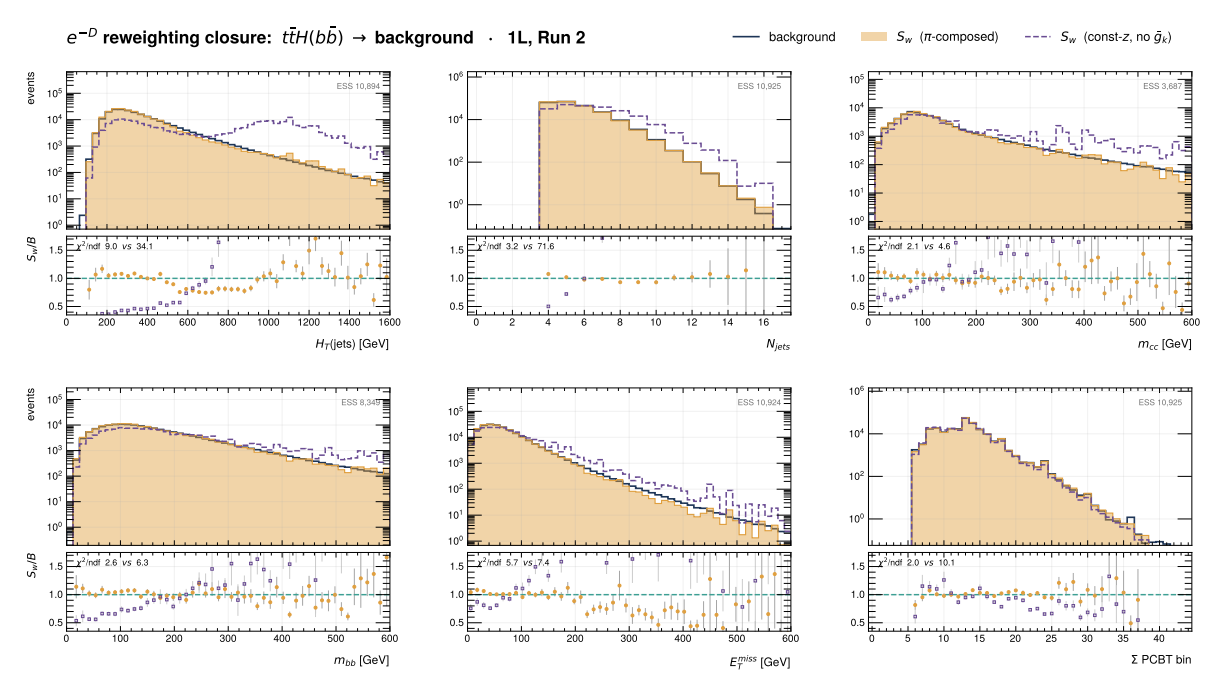
e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 2

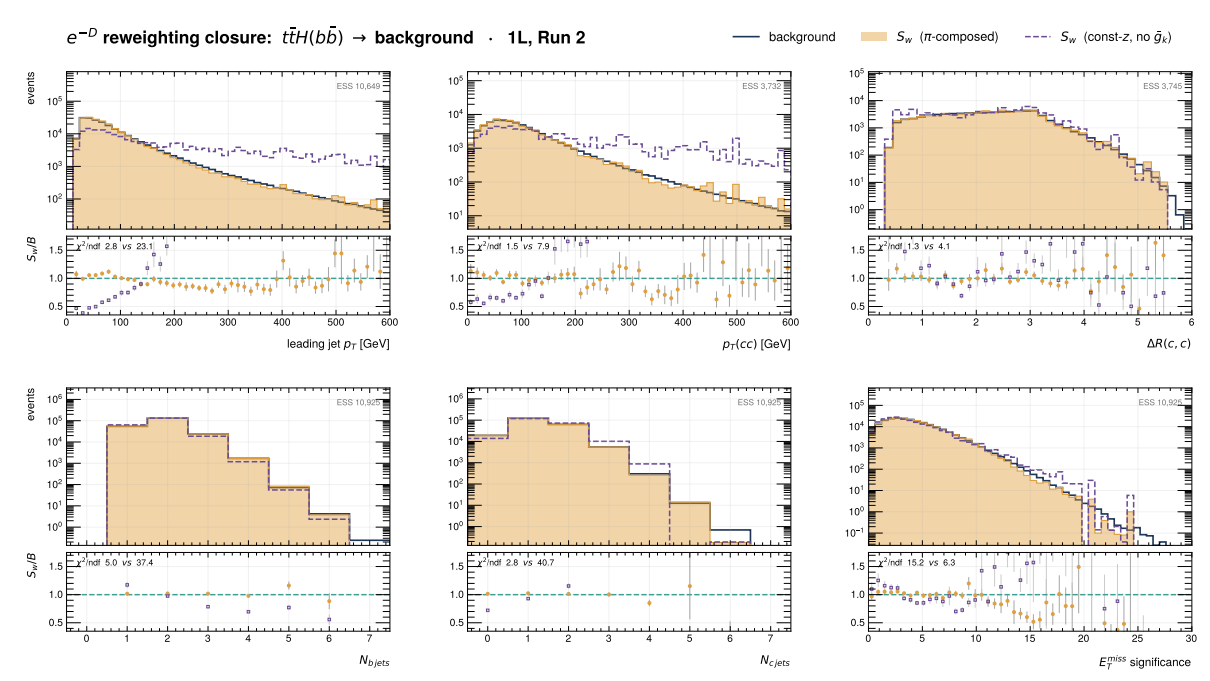



e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 9,276
χ^2/ndf 3.3 vs 41.0
leading jet p_T [GeV]
ESS 5,860
χ^2/ndf 3.9 vs 19.2
$p_T(cc)$ [GeV]
ESS 5,877
χ^2/ndf 20.9 vs 7.0
$\Delta R(c,c)$
ESS 9,537
χ^2/ndf 0.3 vs 9.7
$N_{b\,jets}$
ESS 9,537
χ^2/ndf 0.8 vs 33.0
$N_{c\,jets}$
ESS 9,537
χ^2/ndf 13.1 vs 5.6
E_T^{miss} significance

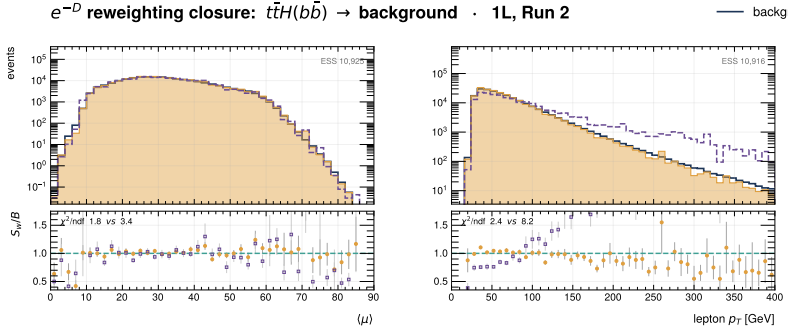
e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 1L, Run 2

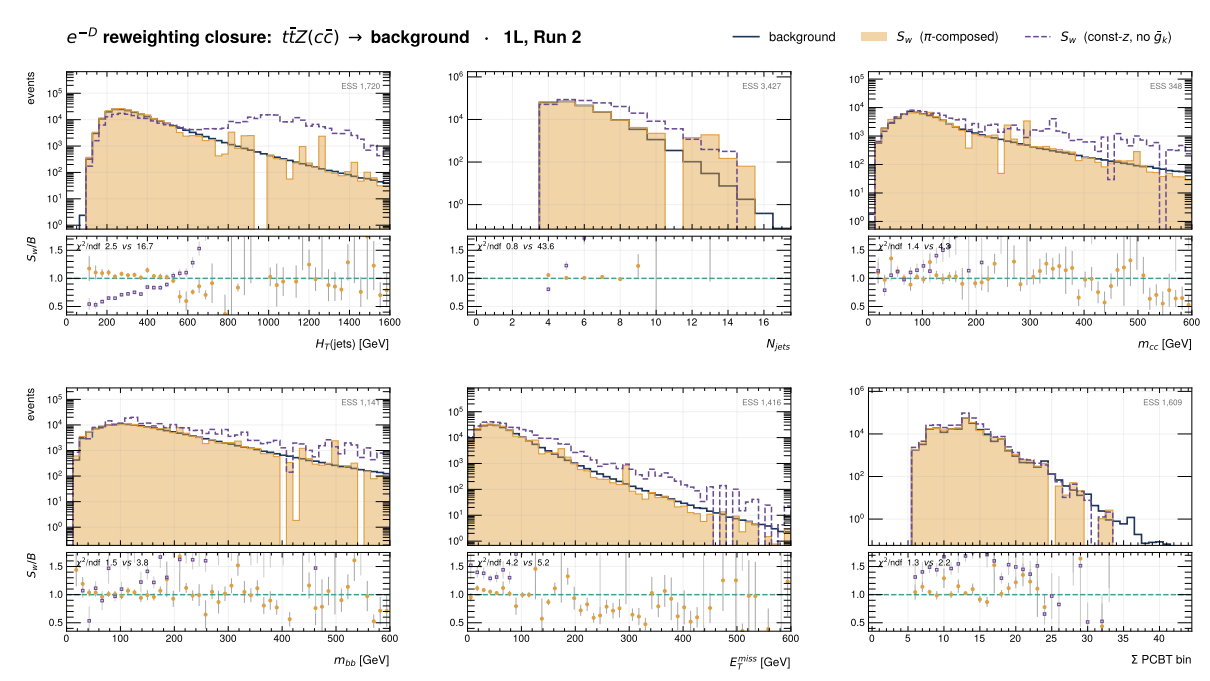


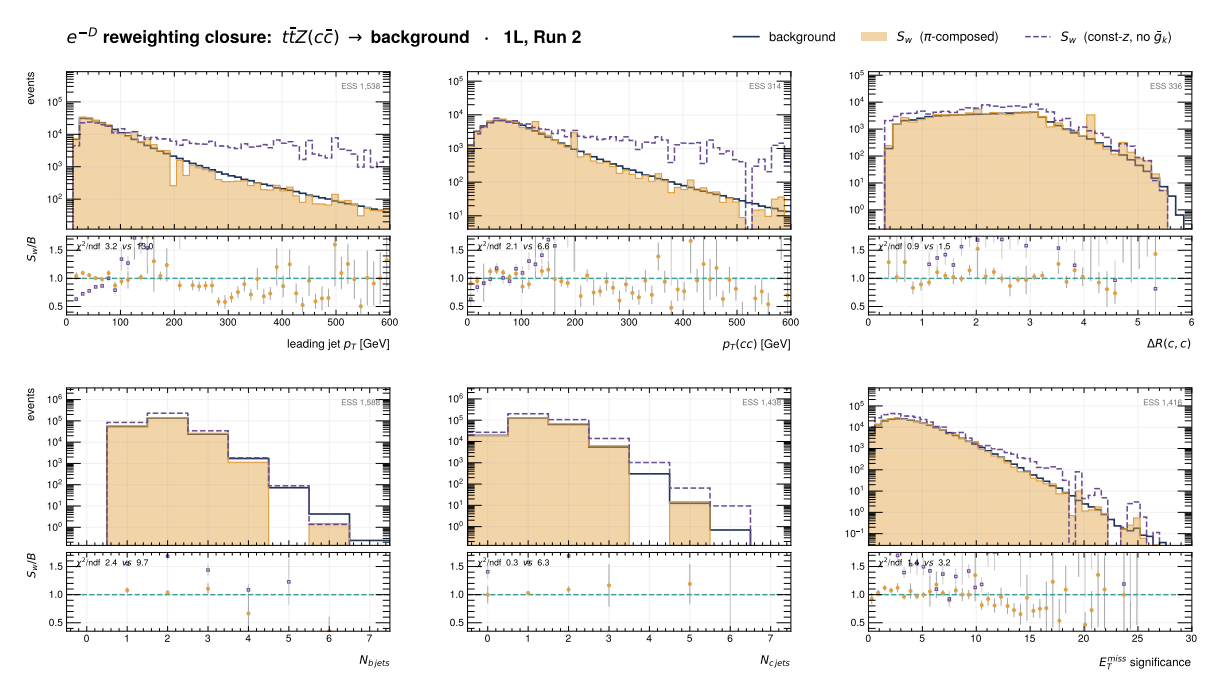

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 10,894
χ^2/ndf 9.0 vs 34.1
H_T(jets) [GeV]
ESS 10,925
χ^2/ndf 3.2 vs 71.6
N_{jets}
ESS 3,687
χ^2/ndf 2.1 vs 4.6
m_{cc} [GeV]
ESS 8,349
χ^2/ndf 2.6 vs 6.3
m_{bb} [GeV]
ESS 10,924
χ^2/ndf 5.7 vs 7.4
E_T^{miss} [GeV]
ESS 10,925
χ^2/ndf 2.0 vs 10.1
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
ESS 10,649
S_w/B
χ^2/ndf 2.8 vs 23.1
leading jet p_T [GeV]
ESS 3,732
χ^2/ndf 1.5 vs 7.9
$p_T(cc)$ [GeV]
ESS 3,745
χ^2/ndf 1.3 vs 4.1
$\Delta R(c,c)$
ESS 10,925
χ^2/ndf 5.0 vs 37.4
$N_{b\,jets}$
ESS 10,925
χ^2/ndf 2.8 vs 40.7
$N_{c\,jets}$
ESS 10,925
χ^2/ndf 15.2 vs 6.3
E_T^{miss} significance

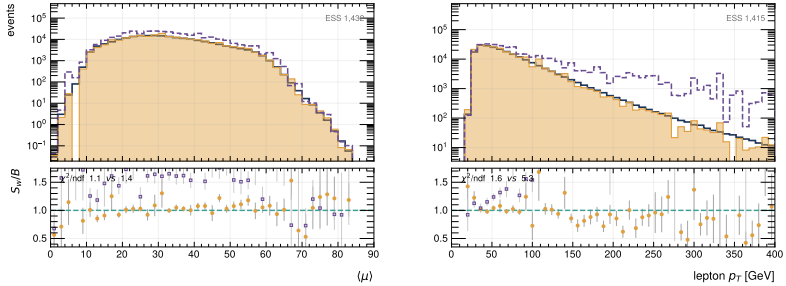
e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 1L, Run 2

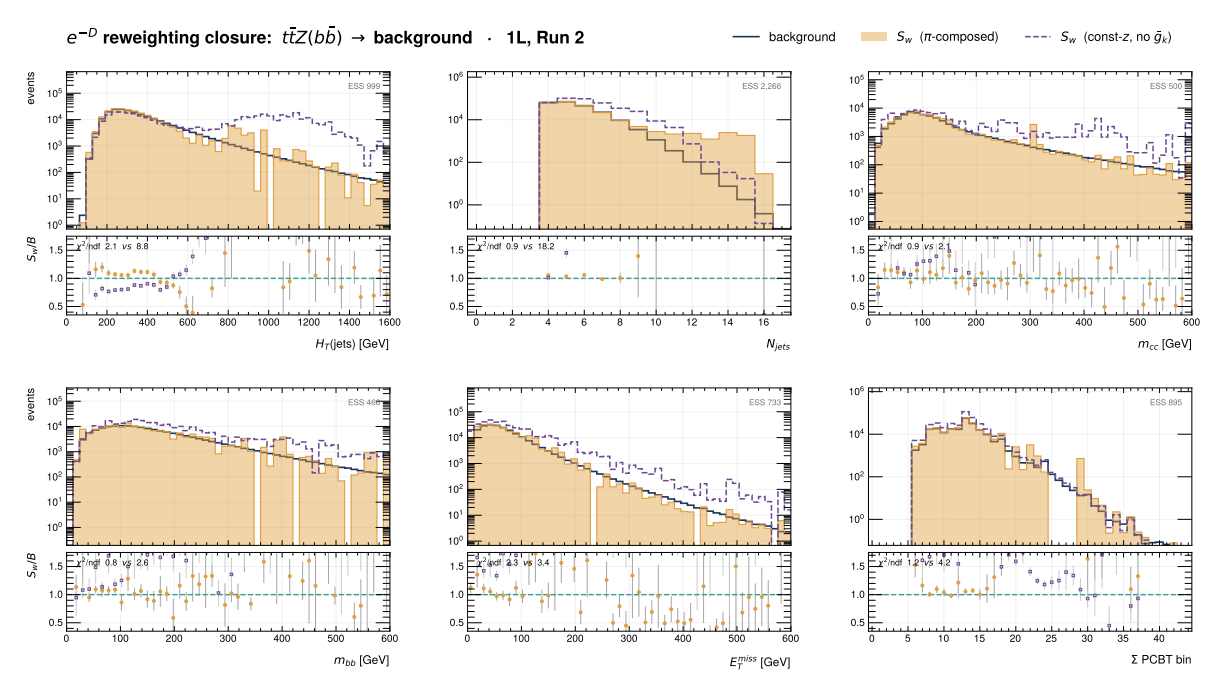


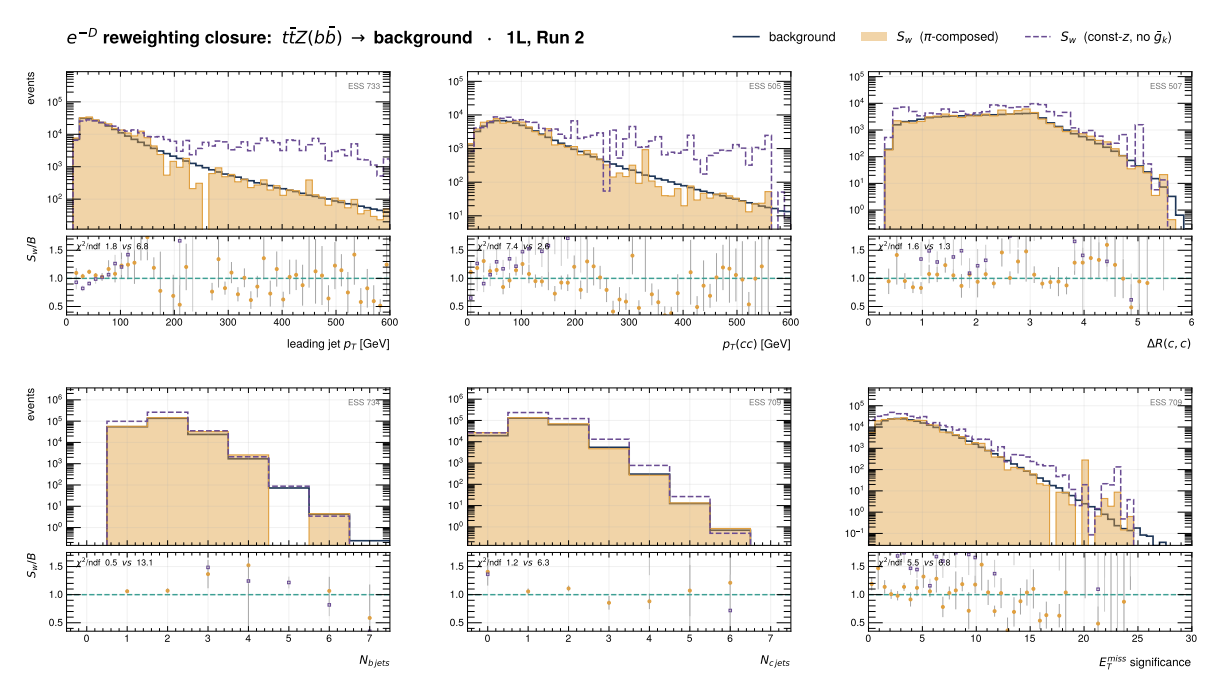

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 1,720
χ^2/ndf 2.5 vs 16.7
H_T(jets) [GeV]
ESS 3,427
χ^2/ndf 0.8 vs 43.6
N_{jets}
ESS 348
χ^2/ndf 1.4 vs 4.3
m_{cc} [GeV]
ESS 1,141
χ^2/ndf 1.5 vs 3.8
m_{bb} [GeV]
ESS 1,416
χ^2/ndf 4.2 vs 5.2
E_T^{miss} [GeV]
ESS 1,609
χ^2/ndf 1.3 vs 2.2
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 1,536
χ^2/ndf 3.2 vs 1.30
leading jet p_T [GeV]
ESS 314
χ^2/ndf 2.1 vs 6.6
$p_T(cc)$ [GeV]
ESS 336
χ^2/ndf 0.9 vs 1.5
$\Delta R(c,c)$
ESS 1,585
χ^2/ndf 2.4 vs 9.7
$N_{b\,jets}$
ESS 1,438
χ^2/ndf 0.3 vs 6.3
$N_{c\,jets}$
ESS 1,416
χ^2/ndf 1.4 vs 3.2
E_T^{miss} significance

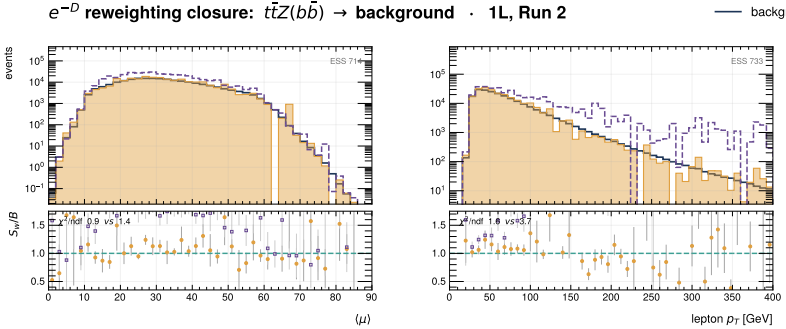
e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 2

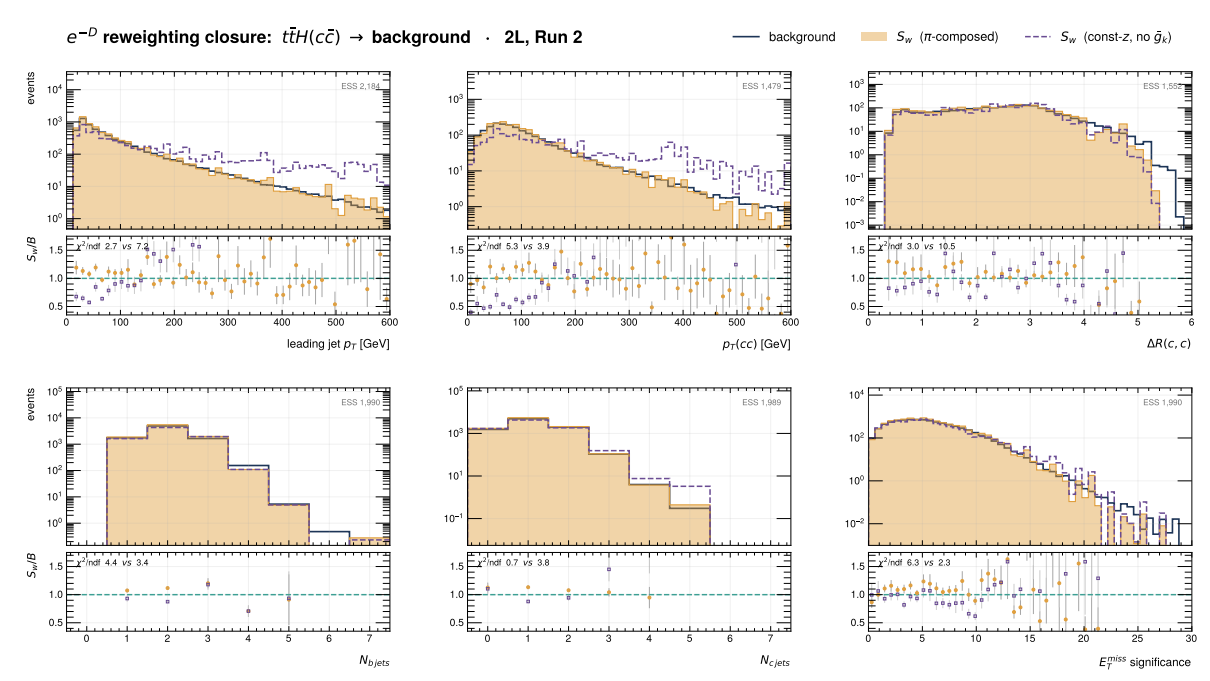



e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 999
ESS 2,266
ESS 500
S_w/B
χ^2/ndf 2.1 vs 8.8
χ^2/ndf 0.9 vs 18.2
χ^2/ndf 0.9 vs 2.1
H_T(jets) [GeV]
N_{jets}
m_{cc} [GeV]
ESS 469
ESS 733
ESS 895
χ^2/ndf 0.8 vs 2.6
χ^2/ndf 2.3 vs 3.4
χ^2/ndf 1.2 vs 4.2
m_{bb} [GeV]
E_T^{miss} [GeV]
Σ PCBT bin

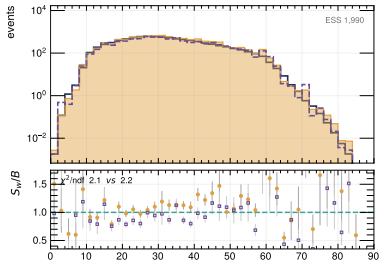

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 733
χ^2/ndf 1.8 vs 6.8
leading jet p_T [GeV]
ESS 505
χ^2/ndf 7.4 vs 2.6
$p_T(cc)$ [GeV]
ESS 507
χ^2/ndf 1.6 vs 1.3
$\Delta R(c,c)$
ESS 735
χ^2/ndf 0.5 vs 13.1
$N_{b\,jets}$
ESS 705
χ^2/ndf 1.2 vs 6.3
$N_{c\,jets}$
ESS 702
χ^2/ndf 5.5 vs 6.8
E_T^{miss} significance

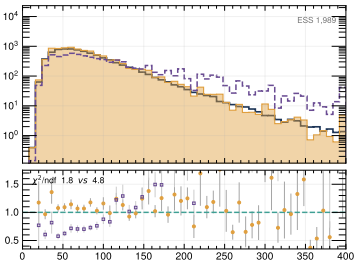
e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 2

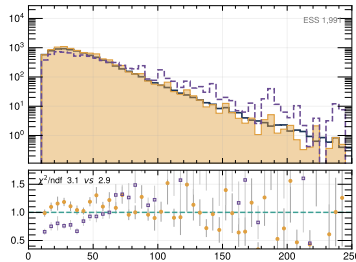


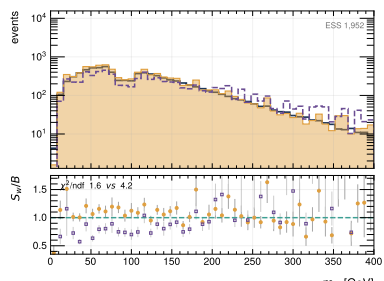

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 2,184
χ^2/ndf 2.7 vs 7.2
S_w/B
leading jet p_T [GeV]
ESS 1,479
χ^2/ndf 5.3 vs 3.9
$p_T(cc)$ [GeV]
ESS 1,558
χ^2/ndf 3.0 vs 10.5
$\Delta R(c,c)$
ESS 1,990
χ^2/ndf 4.4 vs 3.4
$N_{b\,jets}$
ESS 1,989
χ^2/ndf 0.7 vs 3.8
$N_{c\,jets}$
ESS 1,990
χ^2/ndf 6.3 vs 2.3
E_T^{miss} significance

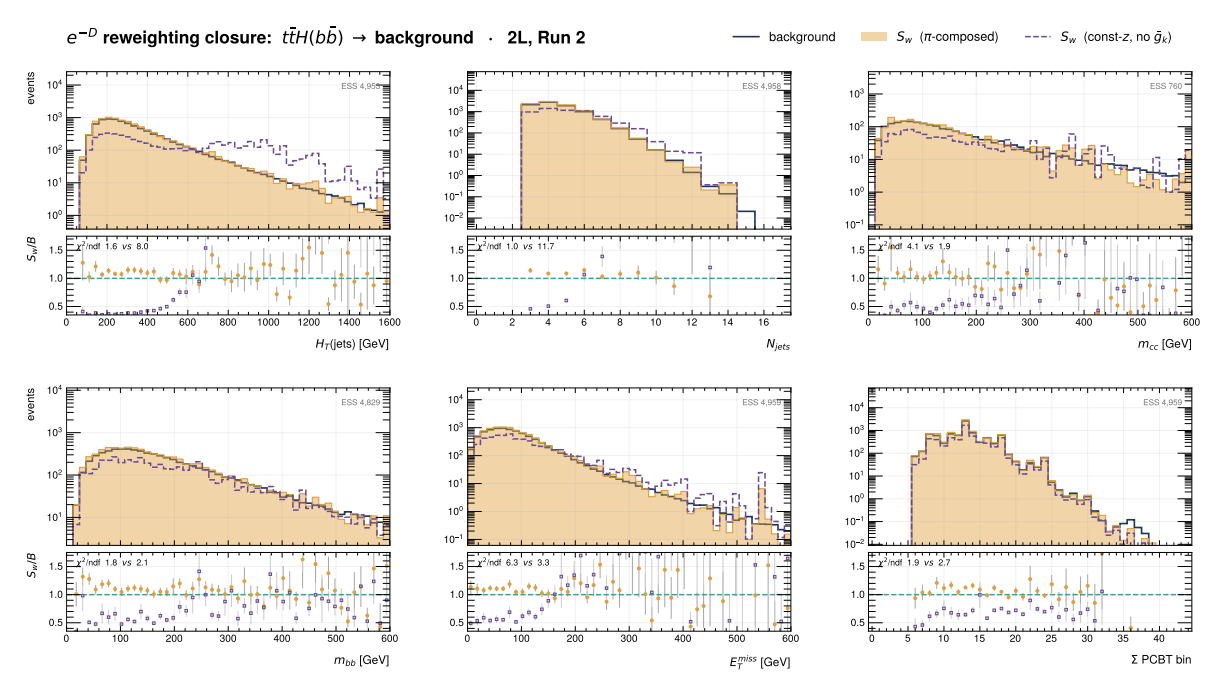
e^{-D} reweighting closure: $t\bar{t}H(c\bar{c}) \rightarrow$ background · 2L, Run 2

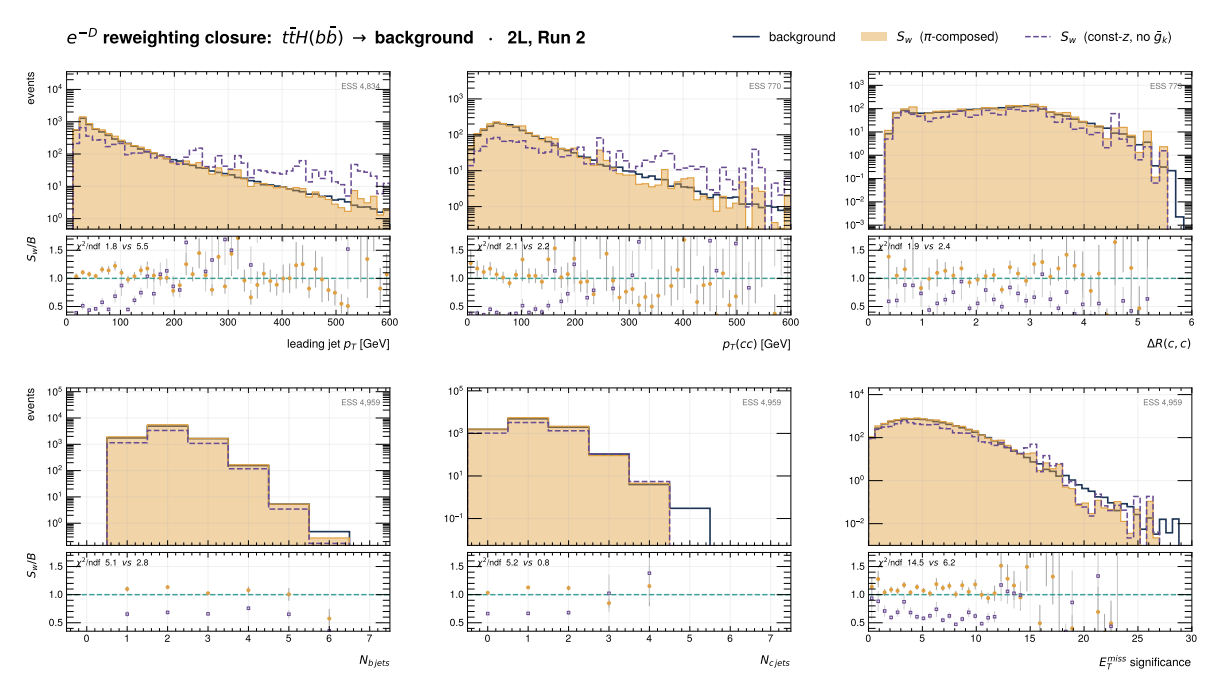


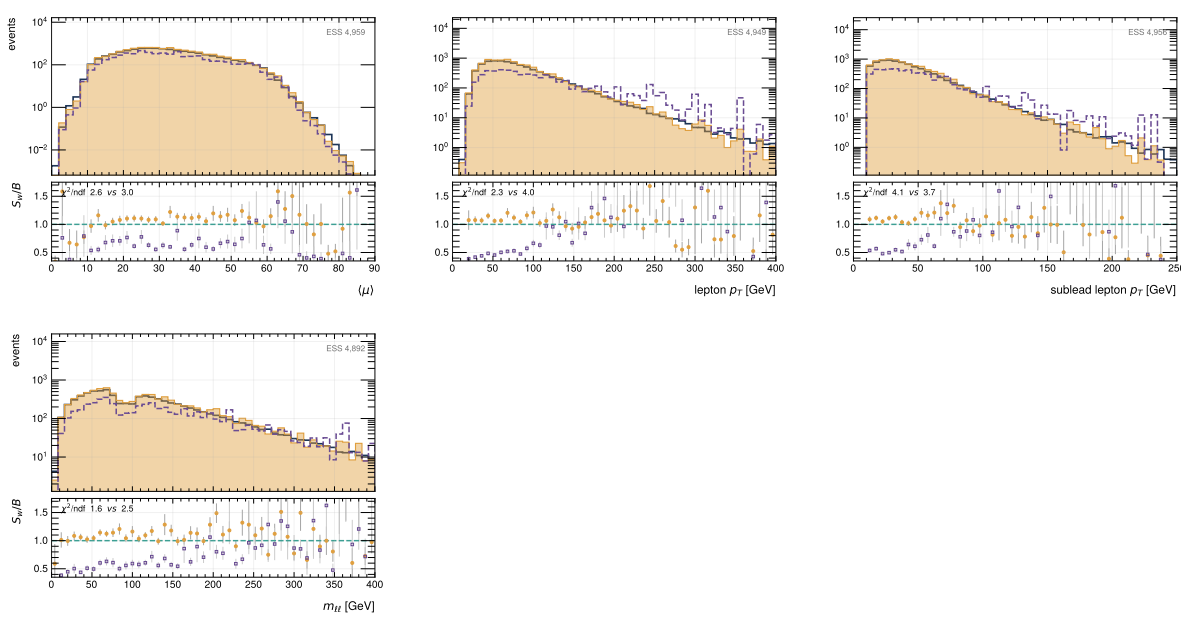


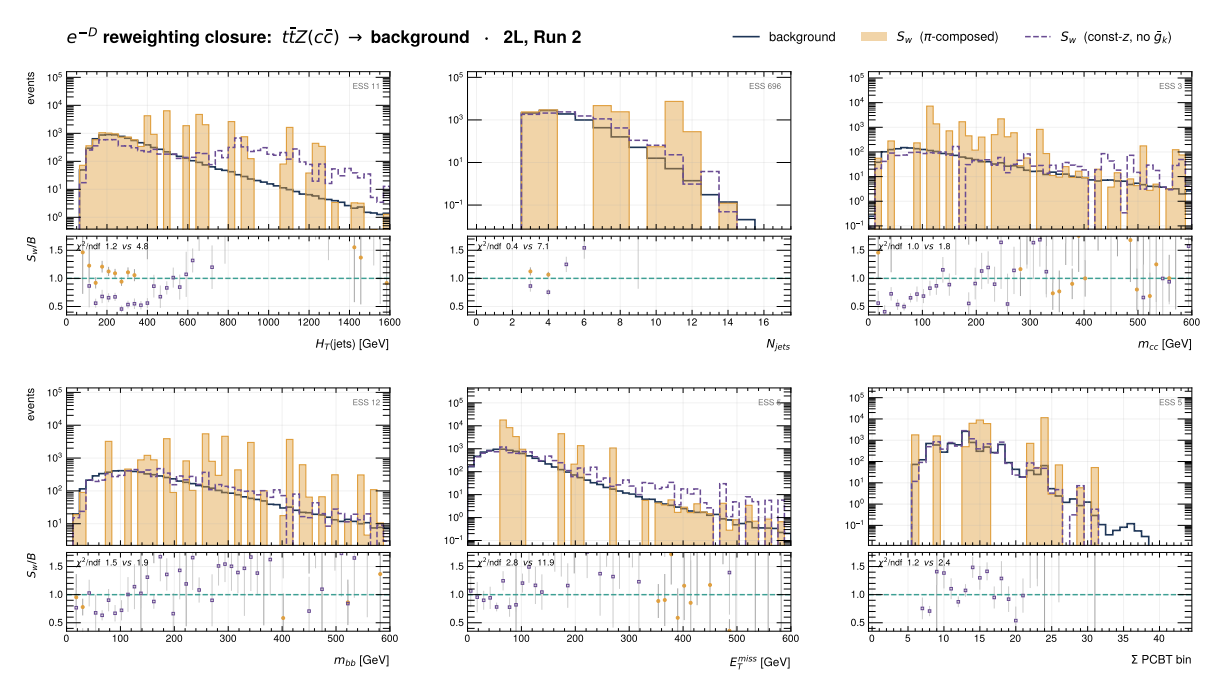


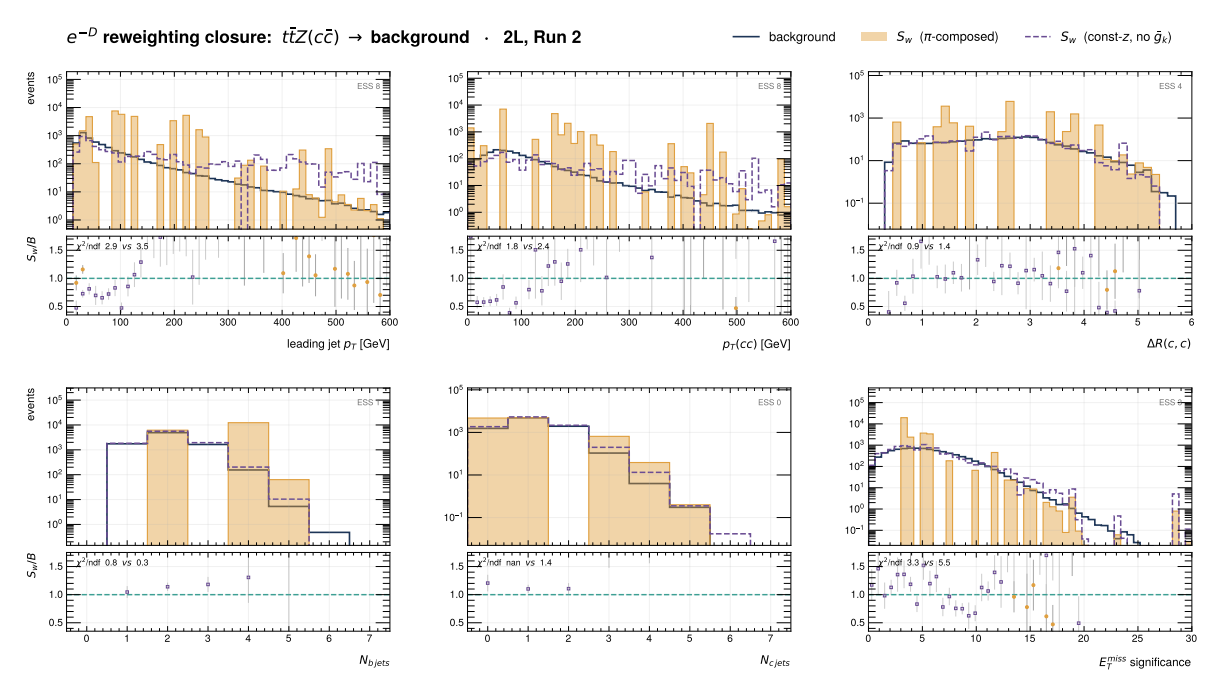



e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 4,958
χ^2/ndf 1.6 vs 8.0
H_T(jets) [GeV]
ESS 4,958
χ^2/ndf 1.0 vs 11.7
N_{jets}
ESS 760
χ^2/ndf 4.1 vs 1.9
m_{cc} [GeV]
ESS 4,829
χ^2/ndf 1.8 vs 2.1
m_{bb} [GeV]
ESS 4,957
χ^2/ndf 6.3 vs 3.3
E_T^{miss} [GeV]
ESS 4,959
χ^2/ndf 1.9 vs 2.7
Σ PCBT bin

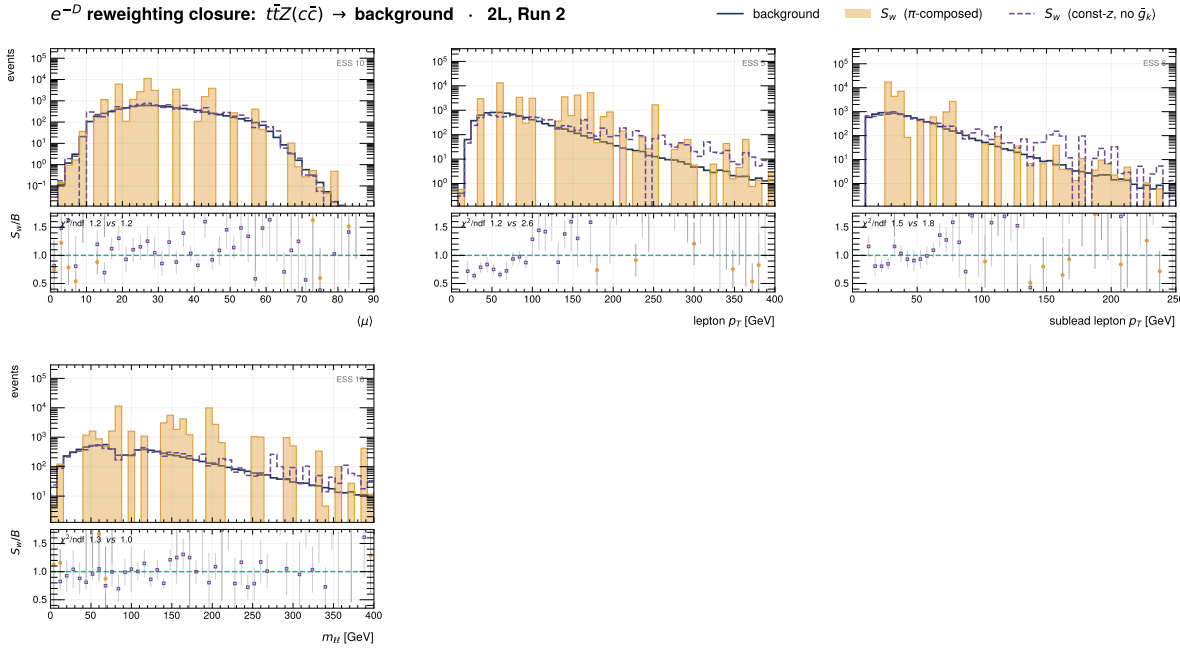

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b}) \rightarrow$ background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 4,83
ESS 770
ESS 779
ESS 4,959
ESS 4,959
ESS 4,959
χ^2/ndf 1.8 vs 5.5
χ^2/ndf 2.1 vs 2.2
χ^2/ndf 1.9 vs 2.4
χ^2/ndf 5.1 vs 2.8
χ^2/ndf 5.2 vs 0.8
χ^2/ndf 14.5 vs 6.2
leading jet p_T [GeV]
$p_T(cc)$ [GeV]
$\Delta R(c,c)$
$N_{b\,jets}$
$N_{c\,jets}$
E_T^{miss} significance

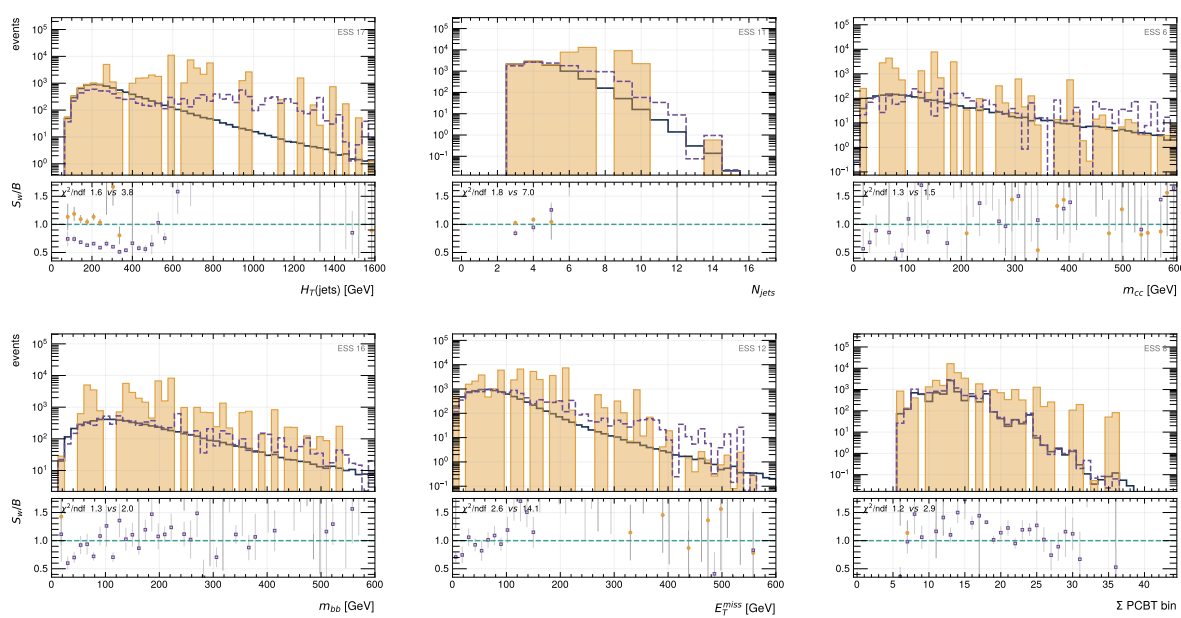

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 4,959
S_w/B
χ^2/ndf 2.6 vs 3.0
$\langle\mu\rangle$
ESS 4,949
χ^2/ndf 2.3 vs 4.0
lepton p_T [GeV]
ESS 4,959
χ^2/ndf 4.1 vs 3.7
sublead lepton p_T [GeV]
ESS 4,892
χ^2/ndf 1.6 vs 2.5
m_{tt} [GeV]


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 11
χ^2/ndf 1.2 vs 4.8
H_T(jets) [GeV]
ESS 696
χ^2/ndf 0.4 vs 7.1
N_{jets}
ESS 3
χ^2/ndf 1.0 vs 1.8
m_{cc} [GeV]
ESS 12
χ^2/ndf 1.5 vs 1.9
m_{bb} [GeV]
ESS 6
χ^2/ndf 2.8 vs 11.9
E_T^{miss} [GeV]
ESS 5
χ^2/ndf 1.2 vs 2.4
Σ PCBT bin

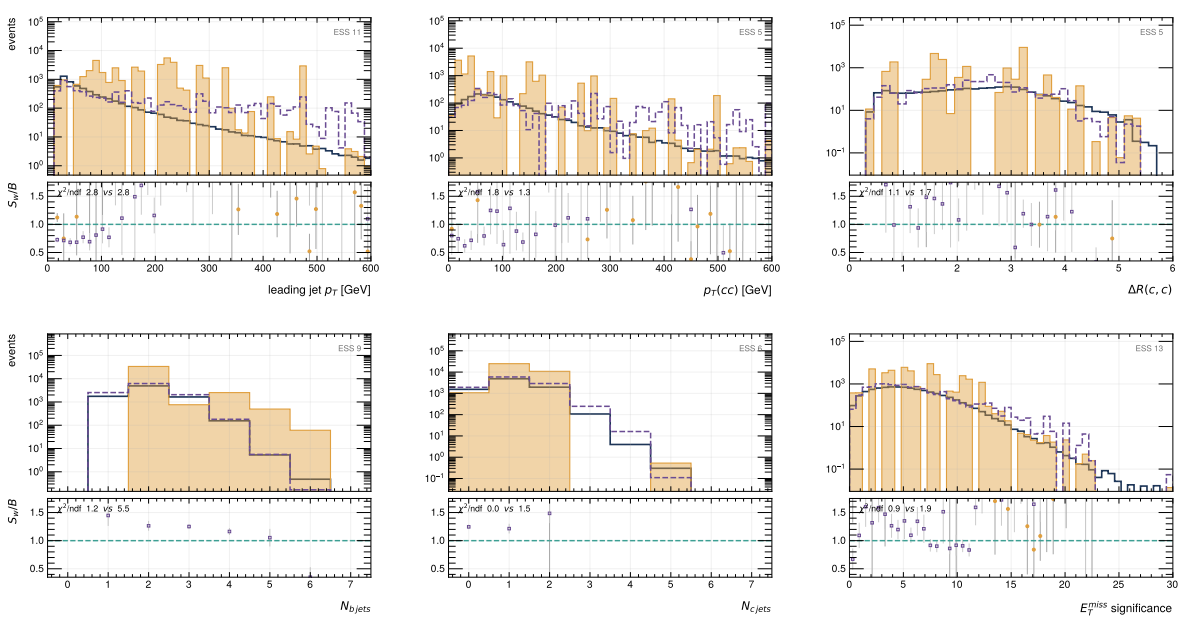

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 8
χ^2/ndf 2.9 vs 3.5
leading jet p_T [GeV]
ESS 8
χ^2/ndf 1.8 vs 2.4
$p_T(cc)$ [GeV]
ESS 4
χ^2/ndf 0.9 vs 1.4
$\Delta R(c,c)$
ESS 7
χ^2/ndf 0.8 vs 0.3
$N_{b\,jets}$
ESS 0
χ^2/ndf nan vs 1.4
$N_{c\,jets}$
ESS 5
χ^2/ndf 3.3 vs 5.5
E_T^{miss} significance

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 2

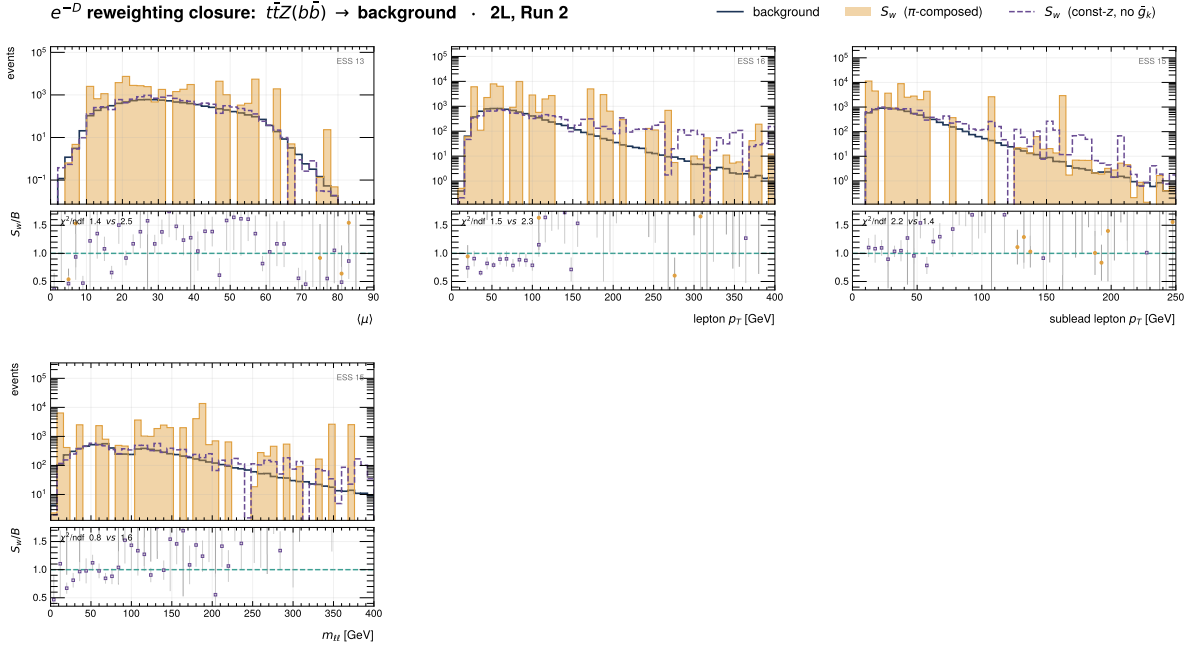


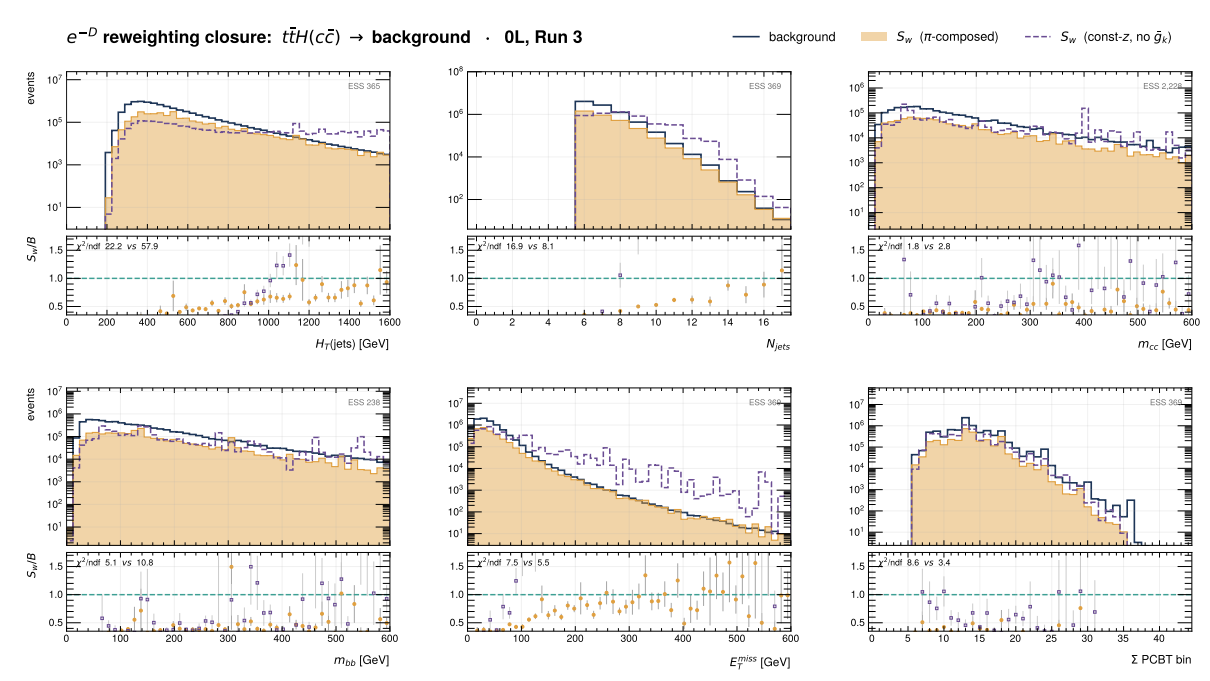

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 2
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 17
ESS 17
ESS 6
ESS 15
ESS 12
ESS 6
χ^2/ndf 1.6 vs 3.8
χ^2/ndf 1.8 vs 7.0
χ^2/ndf 1.3 vs 1.5
χ^2/ndf 1.3 vs 2.0
χ^2/ndf 2.6 vs 14.1
χ^2/ndf 1.2 vs 2.9
H_T(jets) [GeV]
N_{jets}
m_{cc} [GeV]
m_{bb} [GeV]
E_T^{miss} [GeV]
Σ PCBT bin

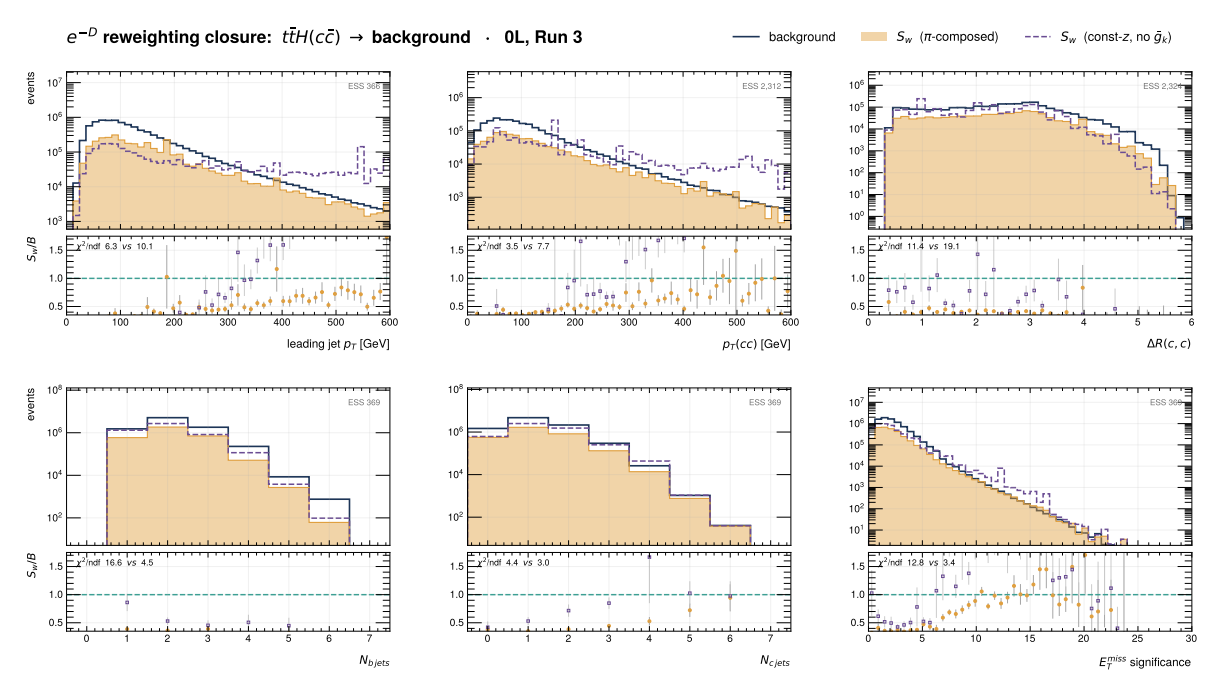
e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 2



e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 2

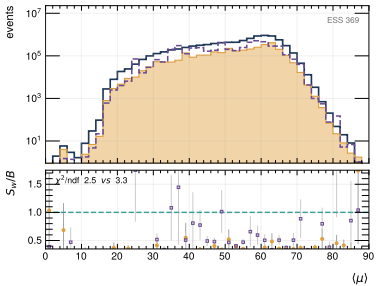


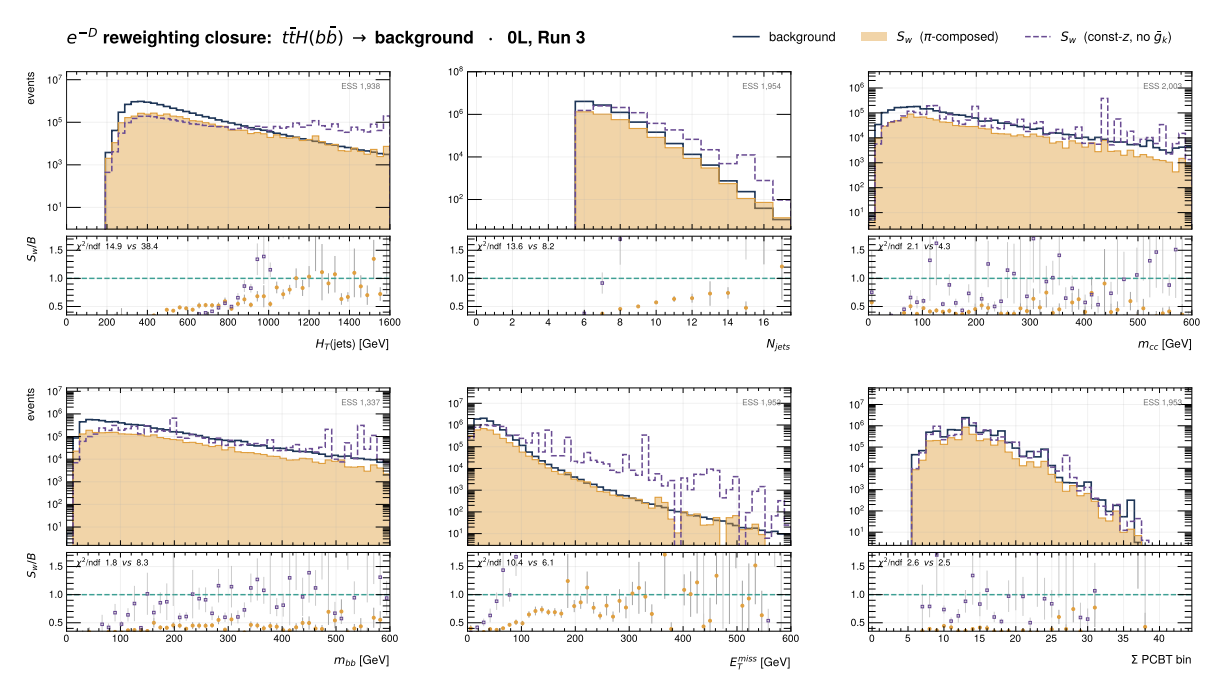

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 365
χ^2/ndf 22.2 vs 57.9
H_T(jets) [GeV]
ESS 369
χ^2/ndf 16.9 vs 8.1
N_{jets}
ESS 2,222
χ^2/ndf 1.8 vs 2.8
m_{cc} [GeV]
ESS 238
χ^2/ndf 5.1 vs 10.8
m_{bb} [GeV]
ESS 369
χ^2/ndf 7.5 vs 5.5
E_T^{miss} [GeV]
ESS 369
χ^2/ndf 8.6 vs 3.4
Σ PCBT bin

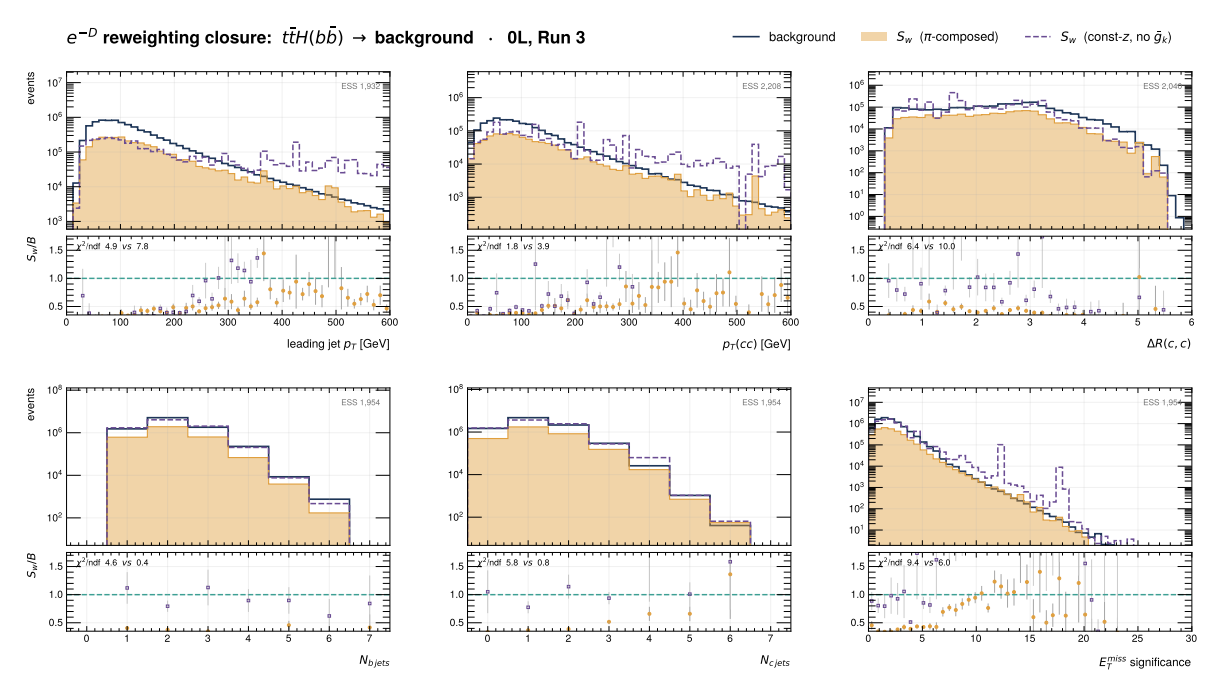

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 367
ESS 2,312
ESS 2,329
S_w/B
χ^2/ndf 6.3 vs 10.1
χ^2/ndf 3.5 vs 7.7
χ^2/ndf 11.4 vs 19.1
leading jet p_T [GeV]
$p_T(cc)$ [GeV]
$\Delta R(c,c)$
ESS 369
ESS 369
ESS 369
χ^2/ndf 16.6 vs 4.5
χ^2/ndf 4.4 vs 3.0
χ^2/ndf 12.8 vs 3.4
$N_{b\,jets}$
$N_{c\,jets}$
E_T^{miss} significance

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c}) \rightarrow$ background · 0L, Run 3

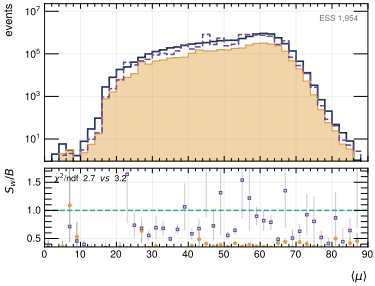
background S_w (π-composed) S_w (const-z, no $\tilde{g}_k$)

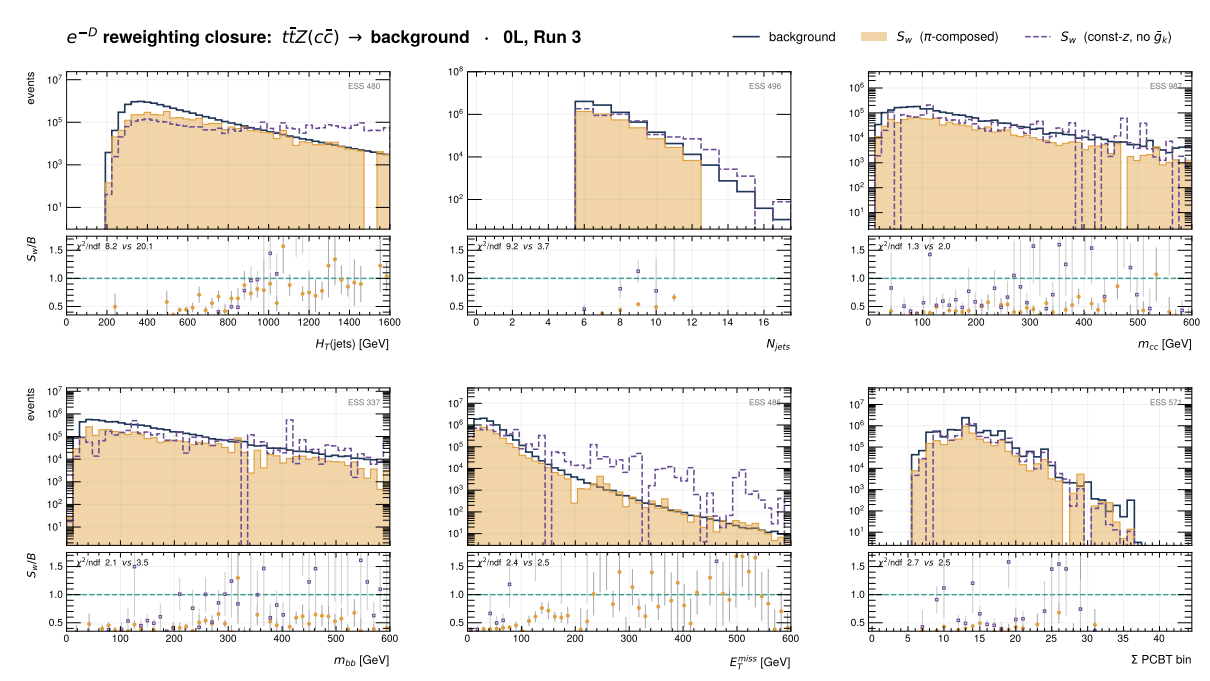


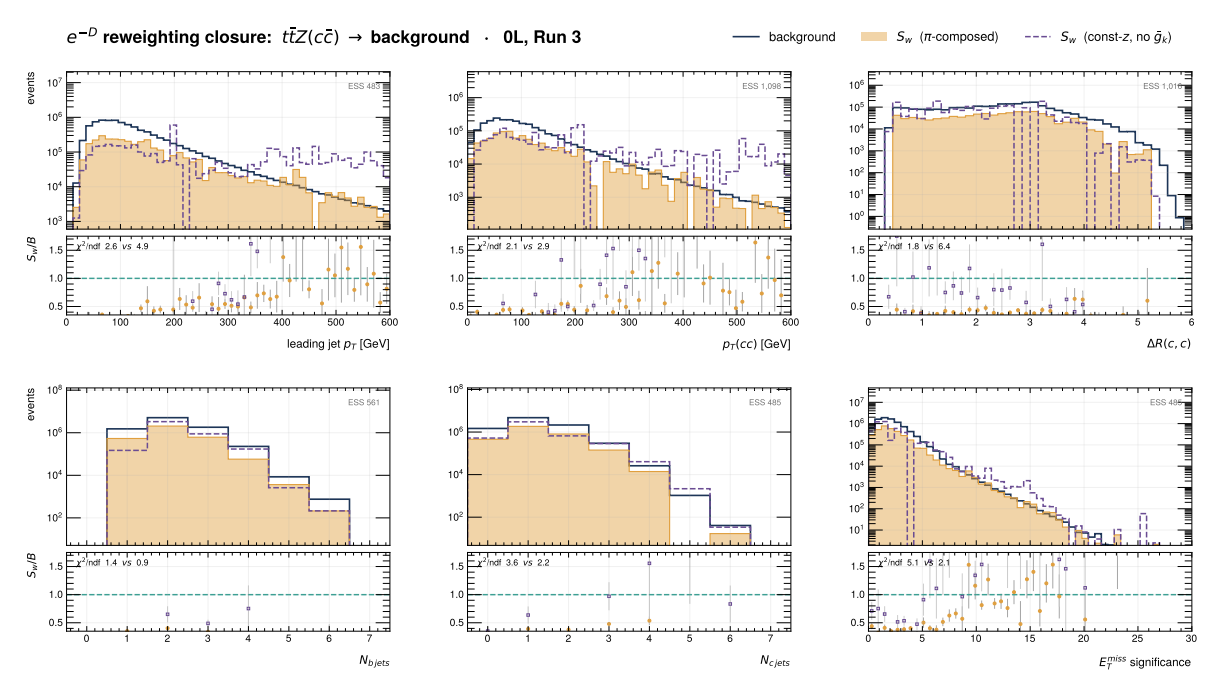

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
ESS 1,938
χ^2/ndf 14.9 vs 38.4
S_w/B
H_T(jets) [GeV]
ESS 1,954
χ^2/ndf 13.6 vs 8.2
N_{jets}
ESS 2,000
χ^2/ndf 2.1 vs 4.3
m_{cc} [GeV]
ESS 1,337
χ^2/ndf 1.8 vs 8.3
m_{bb} [GeV]
ESS 1,954
χ^2/ndf 10.4 vs 6.1
E_T^{miss} [GeV]
ESS 1,953
χ^2/ndf 2.6 vs 2.5
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 1,937
χ^2/ndf 4.9 vs 7.8
S_w/B
leading jet p_T [GeV]
ESS 2,208
χ^2/ndf 1.8 vs 3.9
$p_T(cc)$ [GeV]
ESS 2,040
χ^2/ndf 6.4 vs 10.0
$\Delta R(c,c)$
ESS 1,954
χ^2/ndf 4.6 vs 0.4
$N_{b\,jets}$
ESS 1,954
χ^2/ndf 5.8 vs 0.8
$N_{c\,jets}$
ESS 1,954
χ^2/ndf 9.4 vs 6.0
E_T^{miss} significance

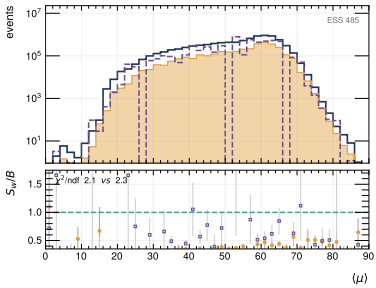
e^{-D} reweighting closure: $t\bar{t}H(b\bar{b}) \rightarrow$ background · 0L, Run 3

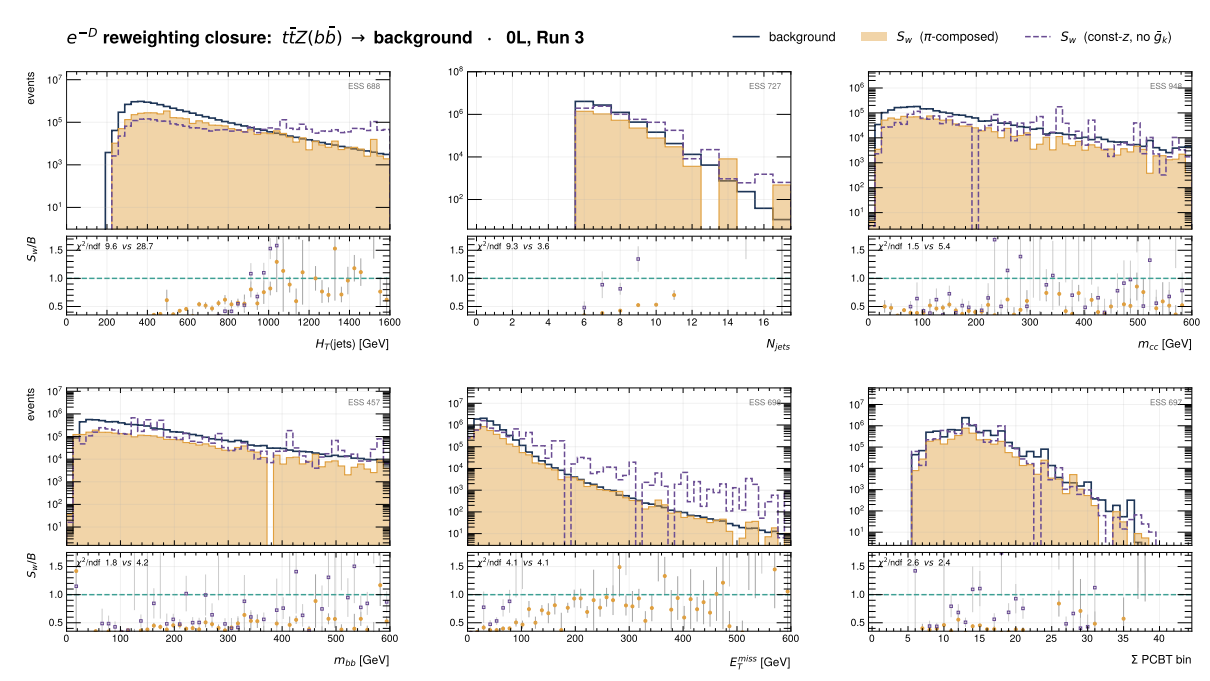


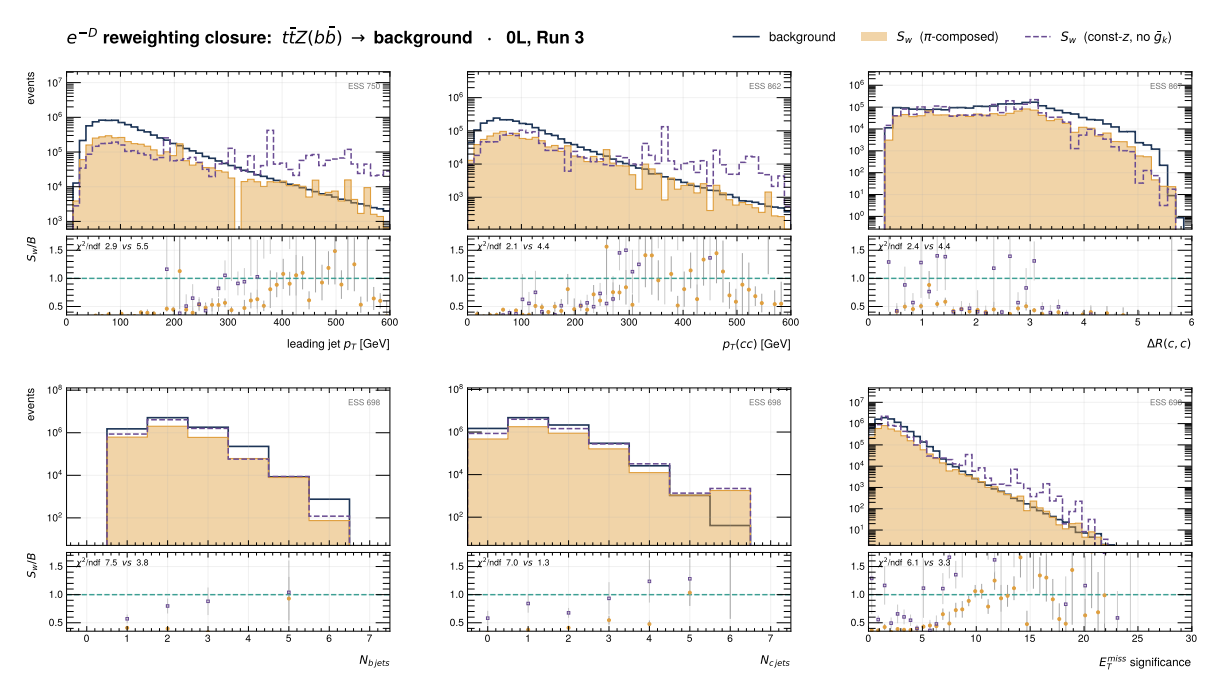

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
ESS 480
S_w/B
χ^2/ndf 8.2 vs 20.1
H_T(jets) [GeV]
ESS 496
χ^2/ndf 9.2 vs 3.7
N_{jets}
ESS 982
χ^2/ndf 1.3 vs 2.0
m_{cc} [GeV]
ESS 337
χ^2/ndf 2.1 vs 3.5
m_{bb} [GeV]
ESS 486
χ^2/ndf 2.4 vs 2.5
E_T^{miss} [GeV]
ESS 572
χ^2/ndf 2.7 vs 2.5
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 485
χ^2/ndf 2.6 vs 4.9
leading jet p_T [GeV]
ESS 1,098
χ^2/ndf 2.1 vs 2.9
$p_T(cc)$ [GeV]
ESS 1,019
χ^2/ndf 1.8 vs 6.4
$\Delta R(c,c)$
ESS 561
χ^2/ndf 1.4 vs 0.9
$N_{b\,jets}$
ESS 485
χ^2/ndf 3.6 vs 2.2
$N_{c\,jets}$
ESS 485
χ^2/ndf 5.1 vs 2.1
E_T^{miss} significance

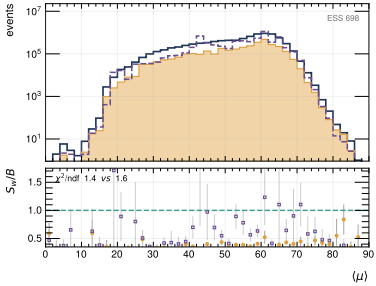
e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c}) \rightarrow$ background · 0L, Run 3

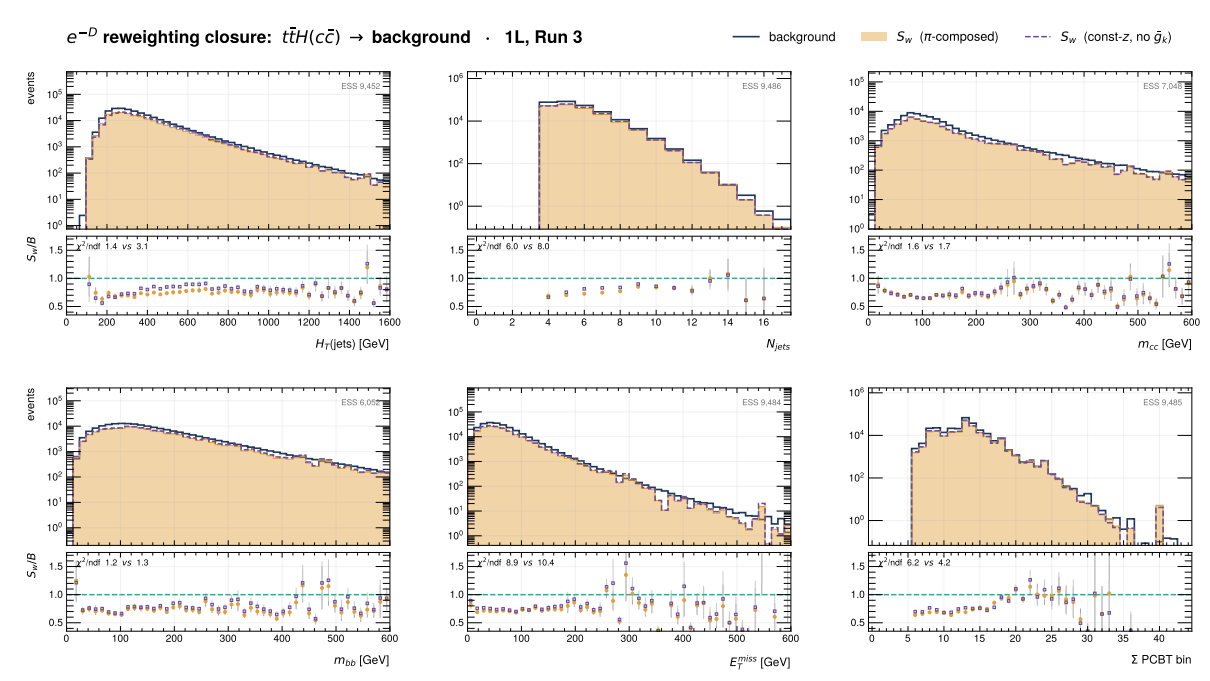


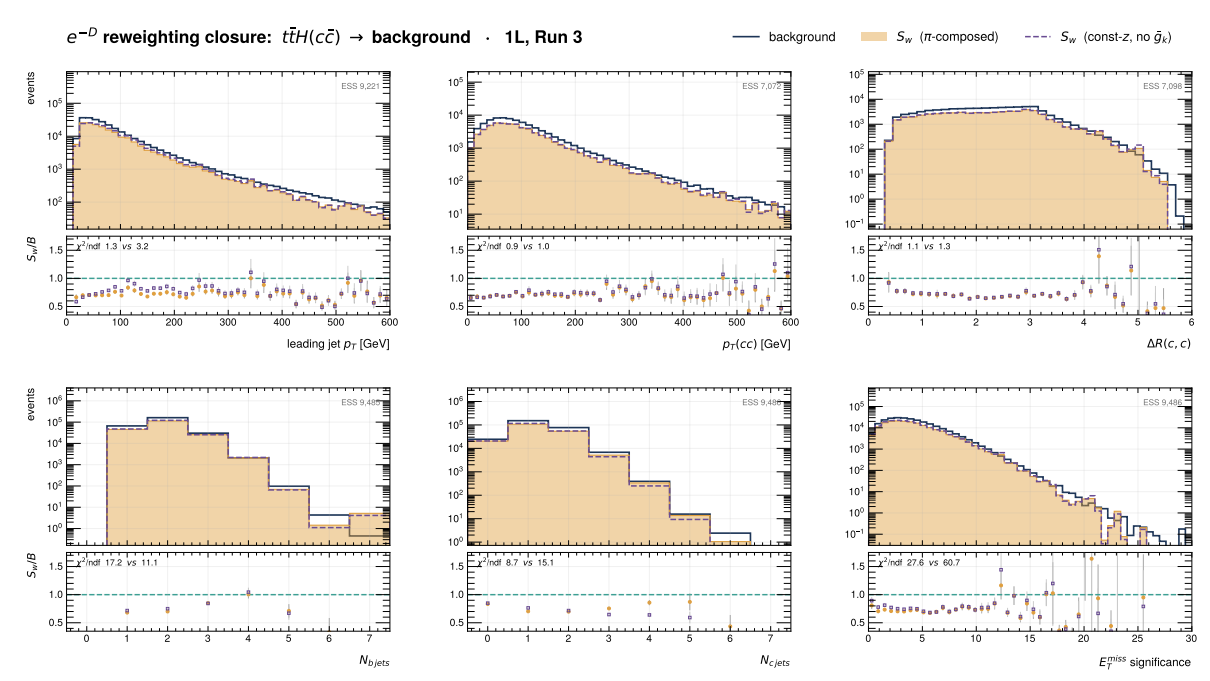

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 688
χ^2/ndf 9.6 vs 28.7
H_T(jets) [GeV]
ESS 727
χ^2/ndf 9.3 vs 3.6
N_{jets}
ESS 940
χ^2/ndf 1.5 vs 5.4
m_{cc} [GeV]
ESS 457
χ^2/ndf 1.8 vs 4.2
m_{bb} [GeV]
ESS 696
χ^2/ndf 4.1 vs 4.1
E_T^{miss} [GeV]
ESS 692
χ^2/ndf 2.6 vs 2.4
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 750
ESS 862
ESS 865
ESS 698
ESS 698
ESS 698
χ^2/ndf 2.9 vs 5.5
χ^2/ndf 2.1 vs 4.4
χ^2/ndf 2.4 vs 4.4
χ^2/ndf 7.5 vs 3.8
χ^2/ndf 7.0 vs 1.3
χ^2/ndf 6.1 vs 3.3
leading jet p_T [GeV]
$p_T(cc)$ [GeV]
$\Delta R(c,c)$
$N_{b\,jets}$
$N_{c\,jets}$
E_T^{miss} significance

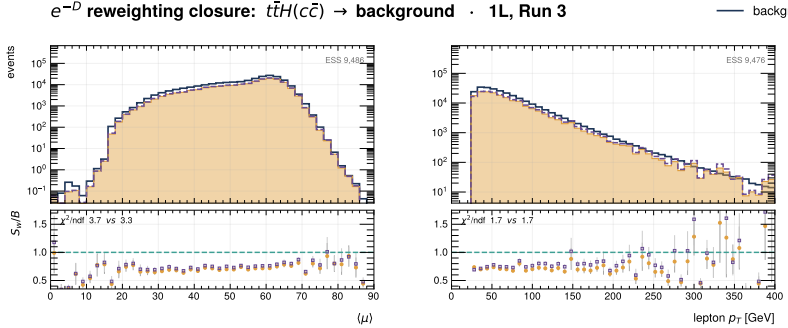
e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 0L, Run 3

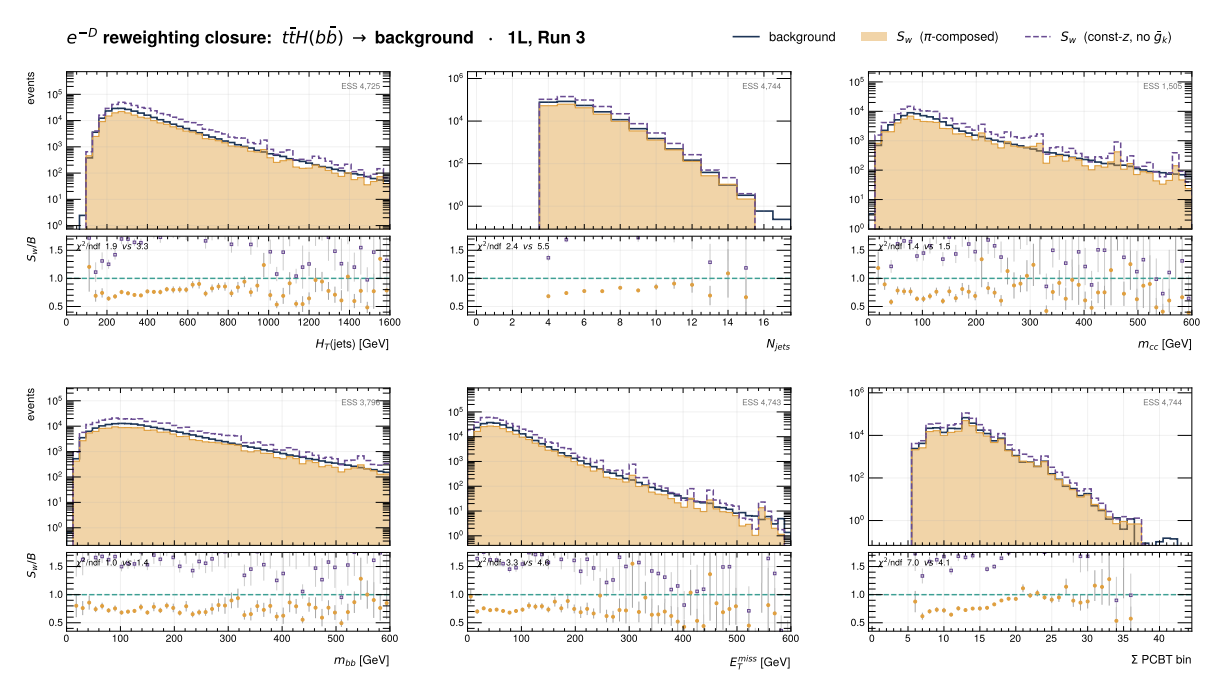


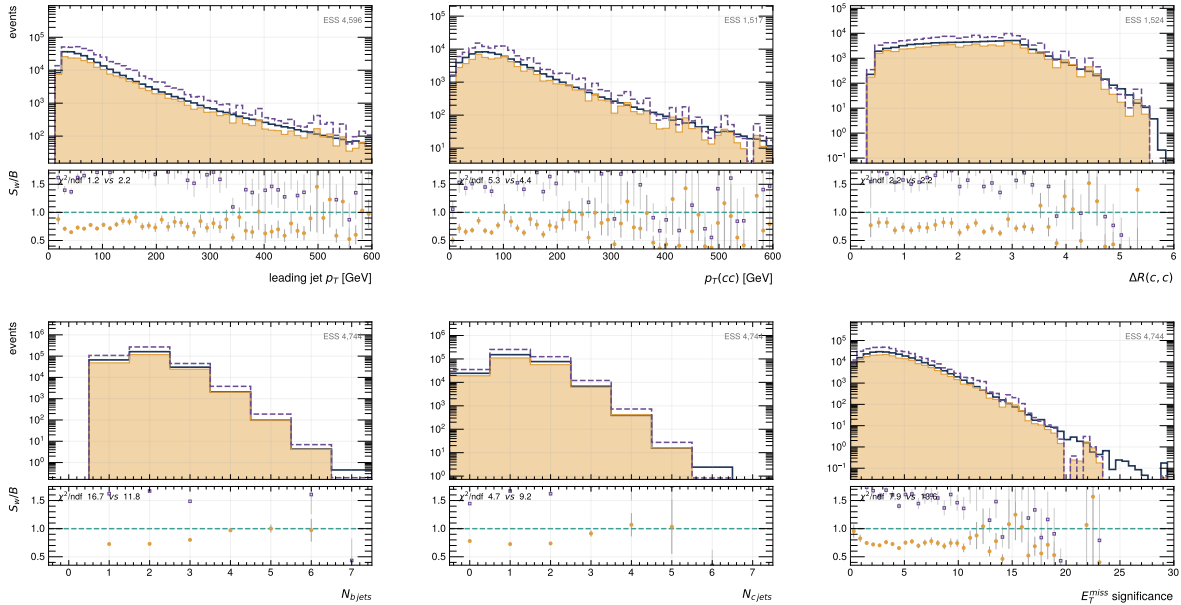

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 9,452
χ^2/ndf 1.4 vs 3.1
H_T(jets) [GeV]
ESS 9,486
χ^2/ndf 6.0 vs 8.0
N_{jets}
ESS 7,045
χ^2/ndf 1.6 vs 1.7
m_{cc} [GeV]
ESS 6,055
χ^2/ndf 1.2 vs 1.3
m_{bb} [GeV]
ESS 9,484
χ^2/ndf 8.9 vs 10.4
E_T^{miss} [GeV]
ESS 9,485
χ^2/ndf 6.2 vs 4.2
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\tilde{g}_k$)
events
ESS 9,221
χ^2/ndf 1.3 vs 3.2
leading jet p_T [GeV]
ESS 7,072
χ^2/ndf 0.9 vs 1.0
$p_T(cc)$ [GeV]
ESS 7,098
χ^2/ndf 1.1 vs 1.3
$\Delta R(c,c)$
S_w/B
ESS 9,485
χ^2/ndf 17.2 vs 11.1
$N_{b\,jets}$
ESS 9,486
χ^2/ndf 8.7 vs 15.1
$N_{c\,jets}$
ESS 9,486
χ^2/ndf 27.6 vs 60.7
E_T^{miss} significance

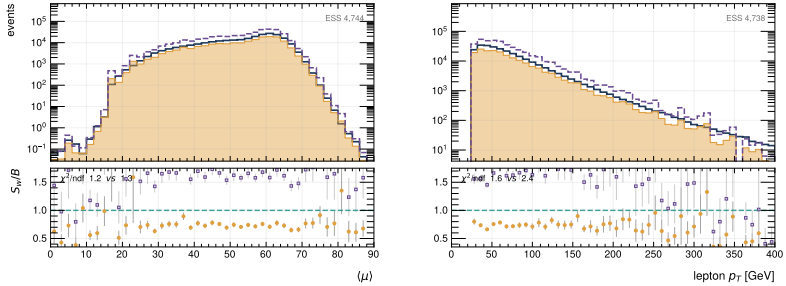
e^{-D} reweighting closure: $t\bar{t}H(c\bar{c}) \rightarrow$ background · 1L, Run 3

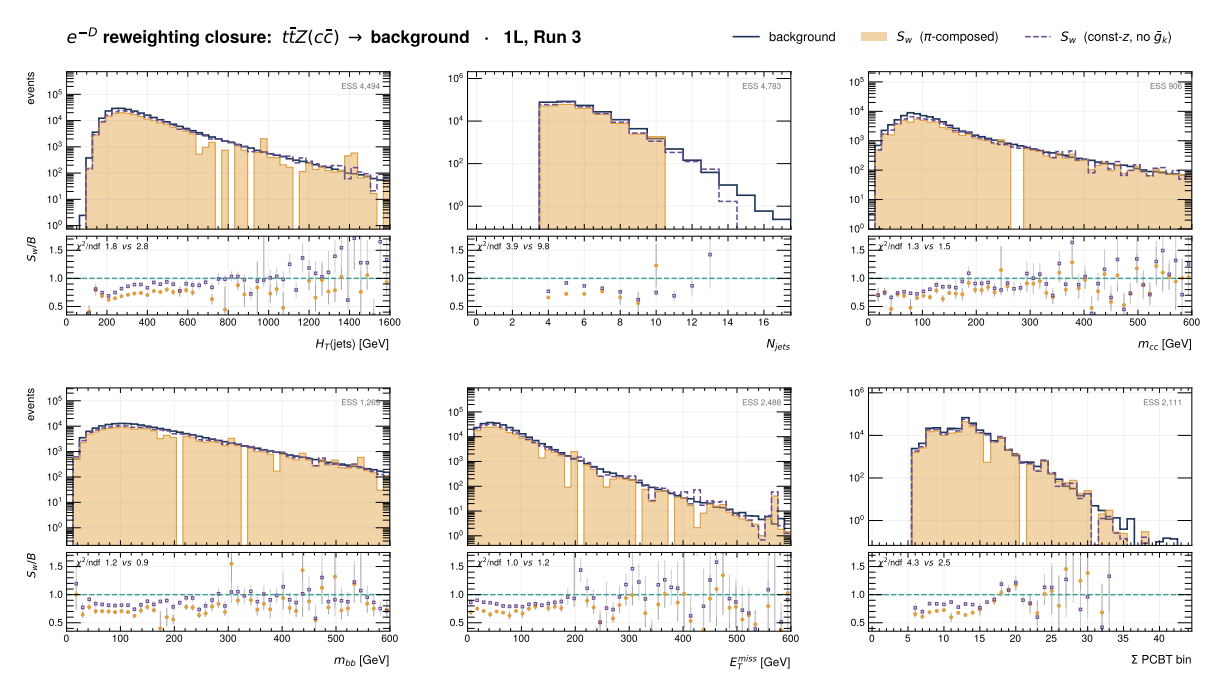


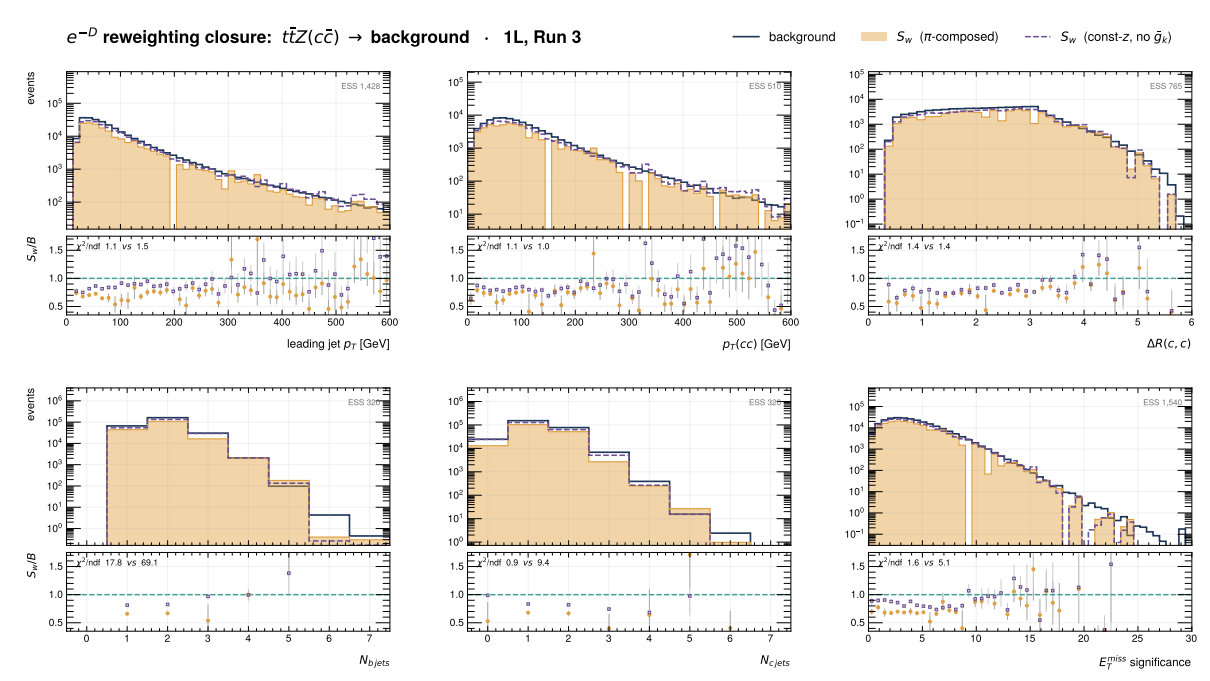

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 4,725
χ^2/ndf 1.9 vs 3.3
H_T(jets) [GeV]
ESS 4,744
χ^2/ndf 2.4 vs 5.5
N_{jets}
ESS 1,505
χ^2/ndf 1.4 vs 1.5
m_{cc} [GeV]
ESS 3,796
χ^2/ndf 1.0 vs 1.4
m_{bb} [GeV]
ESS 4,743
χ^2/ndf 3.3 vs 4.0
E_T^{miss} [GeV]
ESS 4,744
χ^2/ndf 7.0 vs 14.1
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
ESS 4,596
χ^2/ndf 1.2 vs 2.2
leading jet p_T [GeV]
ESS 1,517
χ^2/ndf 5.3 vs 4.4
$p_T(cc)$ [GeV]
ESS 1,524
χ^2/ndf 2.2 vs 2.2
$\Delta R(c,c)$
S_w/B
ESS 4,748
χ^2/ndf 16.7 vs 11.8
$N_{b\,jets}$
ESS 4,744
χ^2/ndf 4.7 vs 9.2
$N_{c\,jets}$
ESS 4,744
χ^2/ndf 7.9 vs 18.0
E_T^{miss} significance

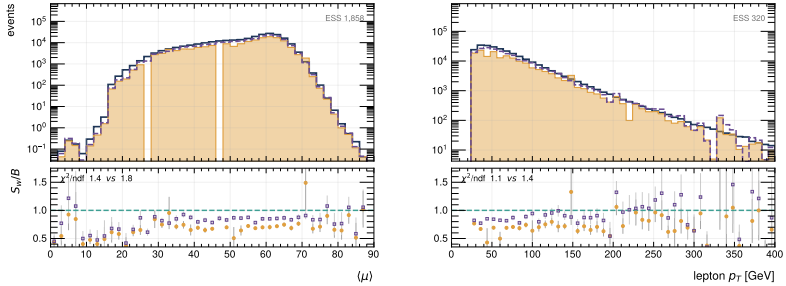
e^{-D} reweighting closure: $t\bar{t}H(b\bar{b}) \rightarrow$ background · 1L, Run 3

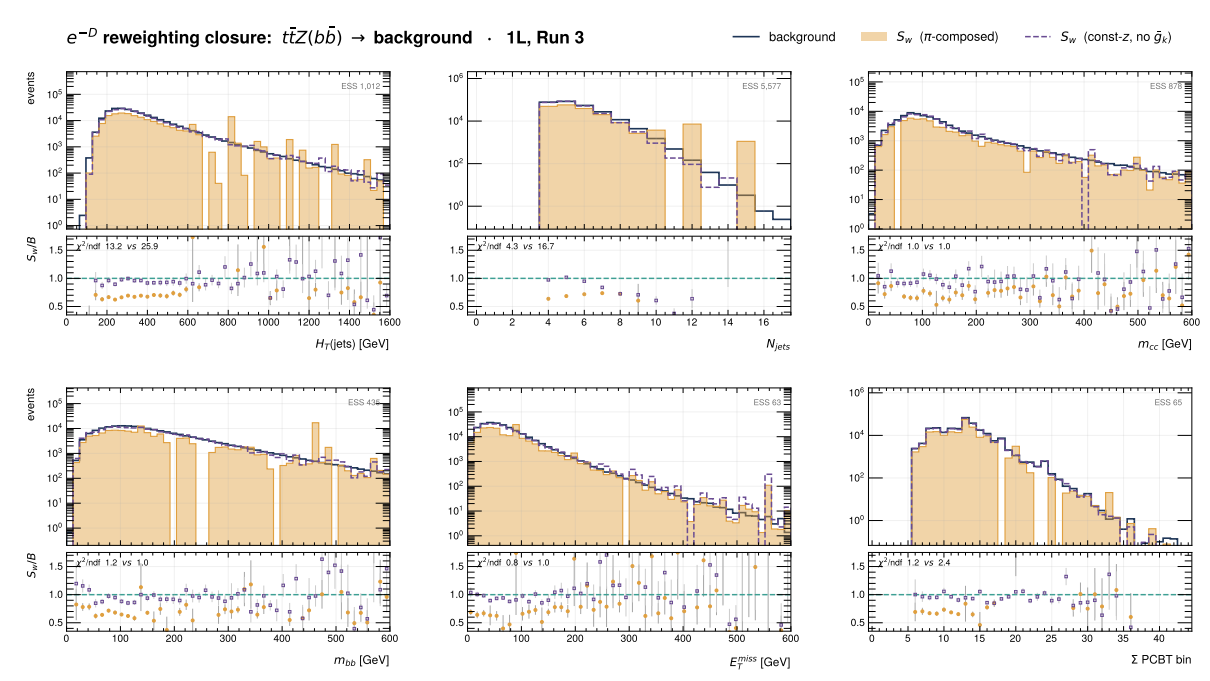


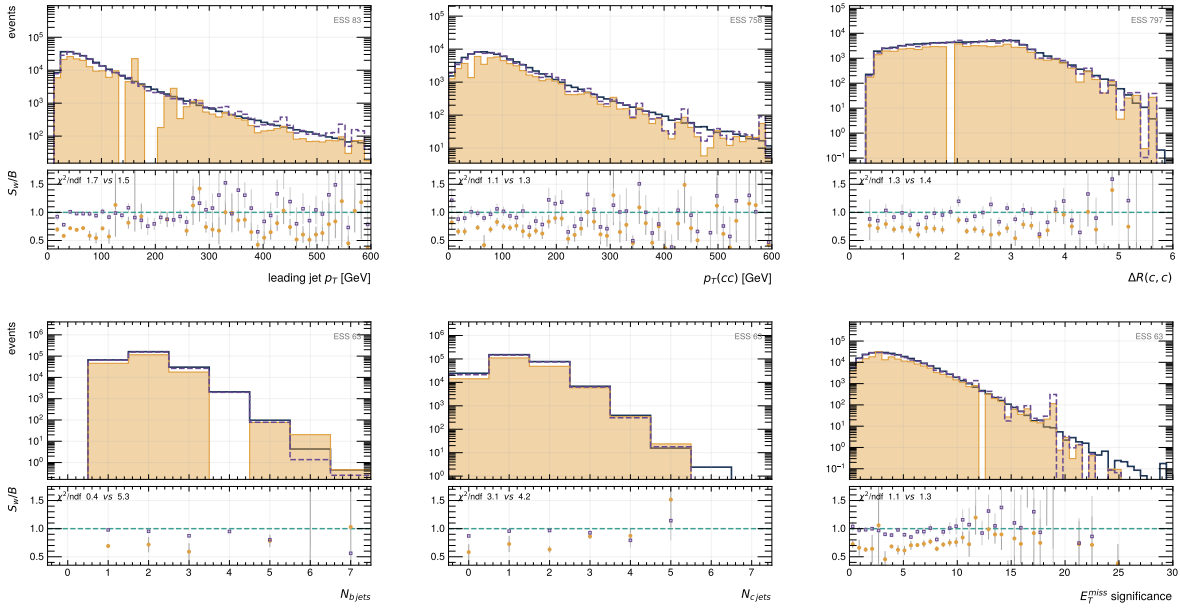

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 4,494
χ^2/ndf 1.8 vs 2.8
H_T(jets) [GeV]
ESS 4,783
χ^2/ndf 3.9 vs 9.8
N_{jets}
ESS 905
χ^2/ndf 1.3 vs 1.5
m_{cc} [GeV]
ESS 1,265
χ^2/ndf 1.2 vs 0.9
m_{bb} [GeV]
ESS 2,488
χ^2/ndf 1.0 vs 1.2
E_T^{miss} [GeV]
ESS 2,111
χ^2/ndf 4.3 vs 2.5
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 1,426
χ^2/ndf 1.1 vs 1.5
leading jet p_T [GeV]
ESS 510
χ^2/ndf 1.1 vs 1.0
$p_T(cc)$ [GeV]
ESS 765
χ^2/ndf 1.4 vs 1.4
$\Delta R(c,c)$
ESS 325
χ^2/ndf 17.8 vs 69.1
$N_{b\,jets}$
ESS 325
χ^2/ndf 0.9 vs 9.4
$N_{c\,jets}$
ESS 1,540
χ^2/ndf 1.6 vs 5.1
E_T^{miss} significance

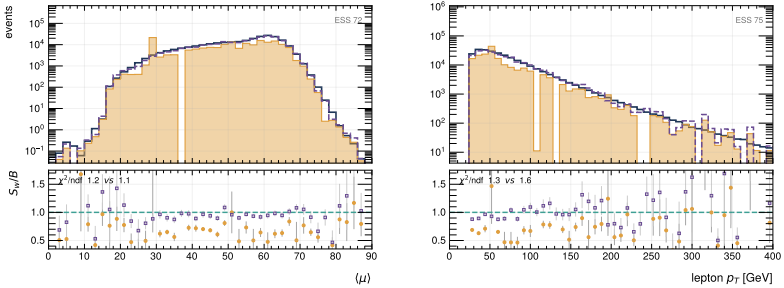
e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 1L, Run 3

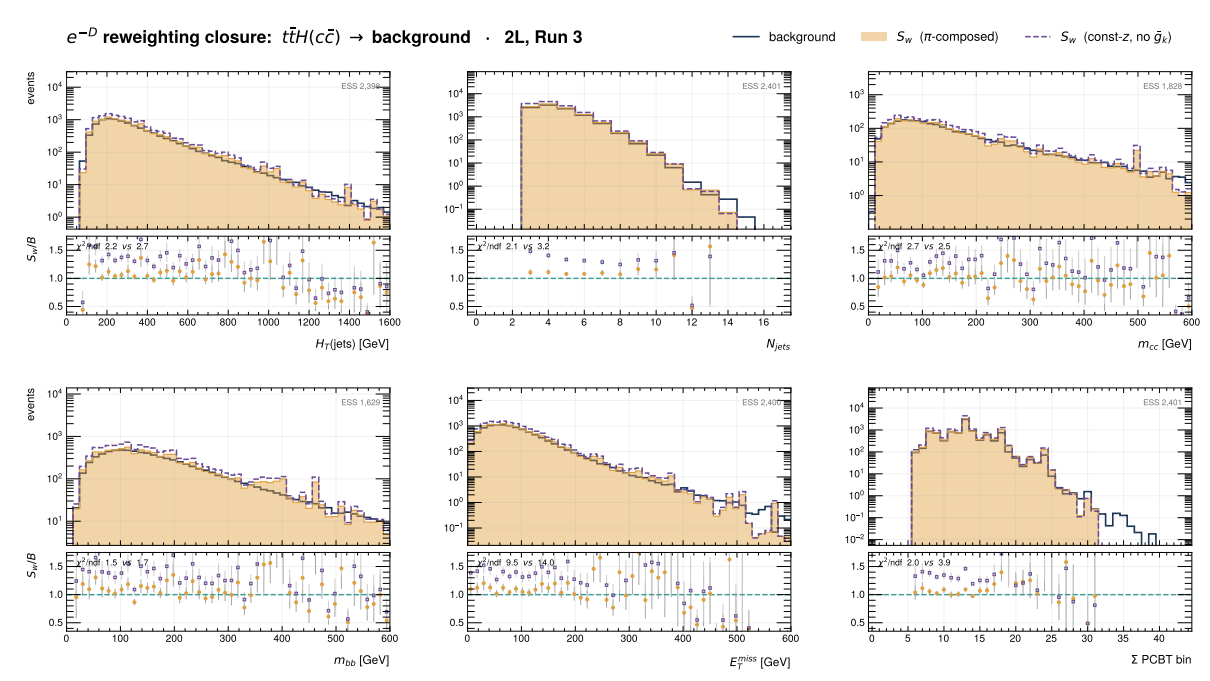


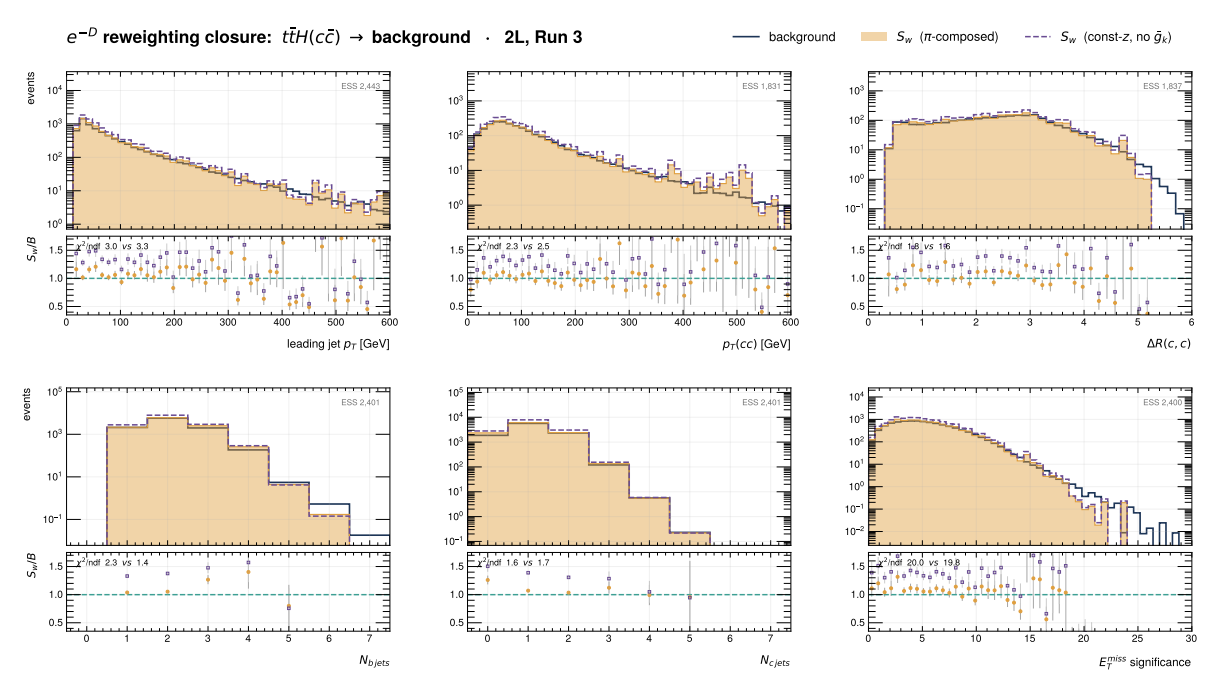

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 1,012
χ^2/ndf 13.2 vs 25.9
H_T(jets) [GeV]
ESS 5,577
χ^2/ndf 4.3 vs 16.7
N_{jets}
ESS 875
χ^2/ndf 1.0 vs 1.0
m_{cc} [GeV]
ESS 432
χ^2/ndf 1.2 vs 1.0
m_{bb} [GeV]
ESS 63
χ^2/ndf 0.8 vs 1.0
E_T^{miss} [GeV]
ESS 65
χ^2/ndf 1.2 vs 2.4
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 83
χ^2/ndf 1.7 vs 1.5
leading jet p_T [GeV]
ESS 758
χ^2/ndf 1.1 vs 1.3
$p_T(cc)$ [GeV]
ESS 797
χ^2/ndf 1.3 vs 1.4
$\Delta R(c,c)$
ESS 65
χ^2/ndf 0.4 vs 5.3
$N_{b\,jets}$
ESS 65
χ^2/ndf 3.1 vs 4.2
$N_{c\,jets}$
ESS 65
χ^2/ndf 1.1 vs 1.3
E_T^{miss} significance

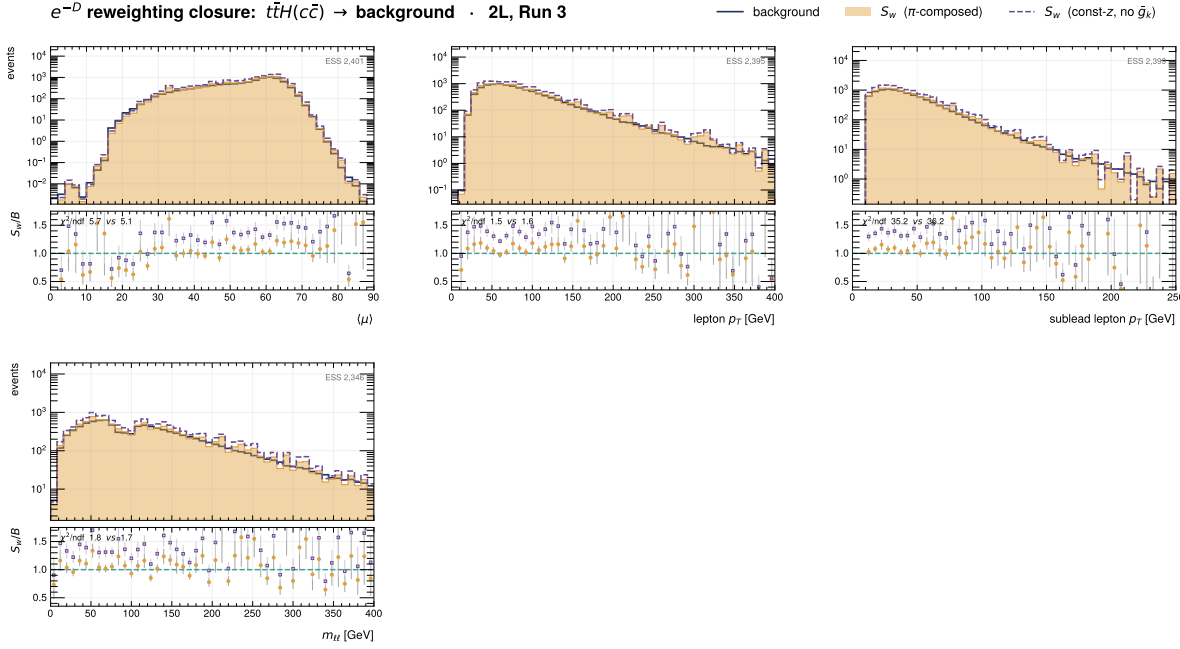
e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 1L, Run 3

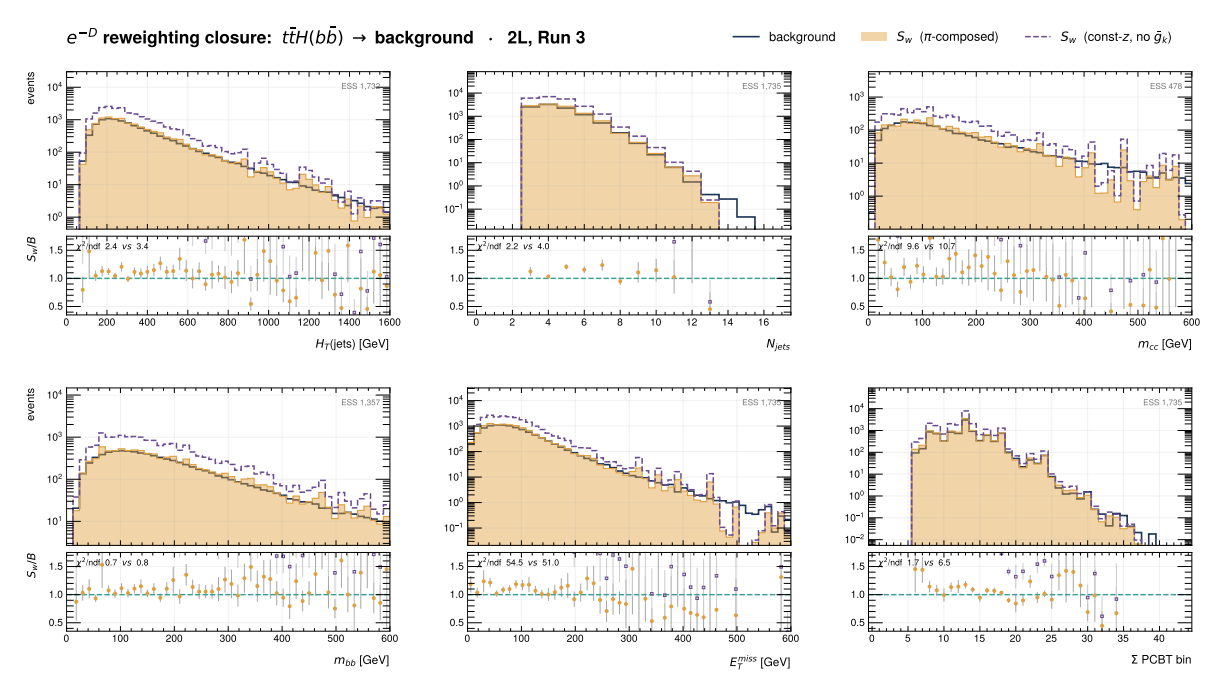


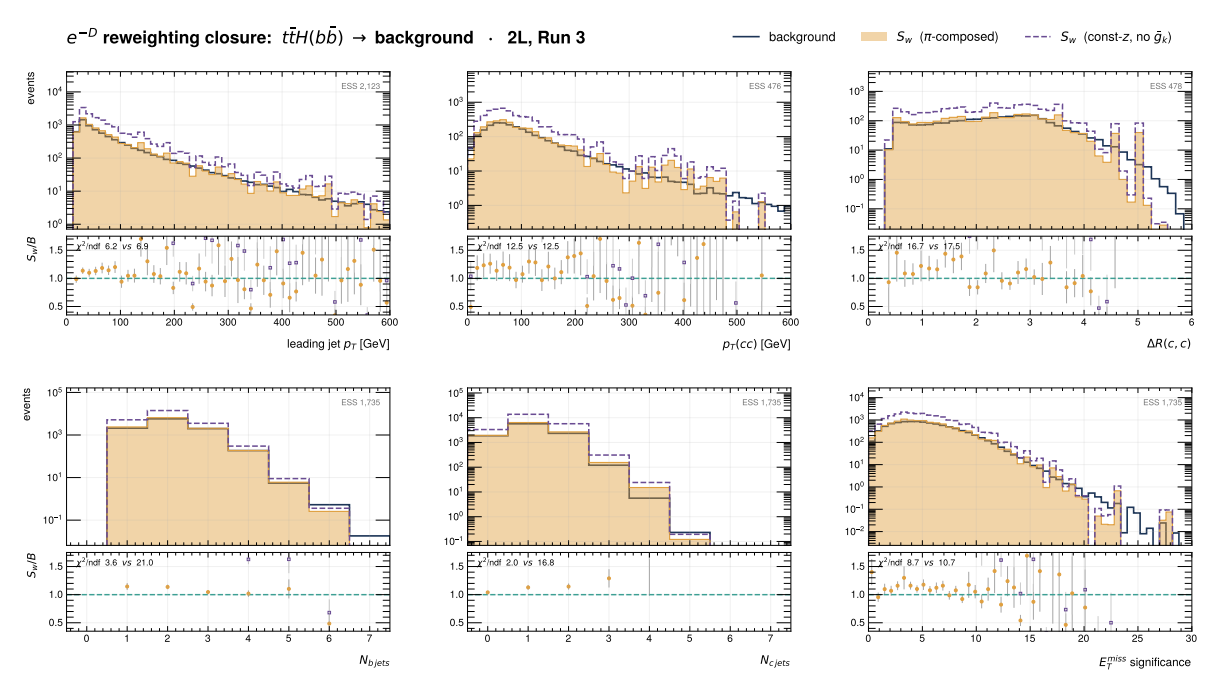

e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 2,394
χ^2/ndf 2.2 vs 2.7
H_T(jets) [GeV]
ESS 2,401
χ^2/ndf 2.1 vs 3.2
N_{jets}
ESS 1,828
χ^2/ndf 2.7 vs 2.5
m_{cc} [GeV]
ESS 1,629
χ^2/ndf 1.5 vs 1.7
m_{bb} [GeV]
ESS 2,400
χ^2/ndf 9.5 vs 14.0
E_T^{miss} [GeV]
ESS 2,401
χ^2/ndf 2.0 vs 3.9
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 2,442
χ^2/ndf 3.0 vs 3.3
leading jet p_T [GeV]
ESS 1,831
χ^2/ndf 2.3 vs 2.5
$p_T(cc)$ [GeV]
ESS 1,837
χ^2/ndf 1e8 vs 116
$\Delta R(c,c)$
ESS 2,401
χ^2/ndf 2.3 vs 1.4
$N_{b\,jets}$
ESS 2,401
χ^2/ndf 1.6 vs 1.7
$N_{c\,jets}$
ESS 2,400
χ^2/ndf 20.0 vs 19.8
E_T^{miss} significance

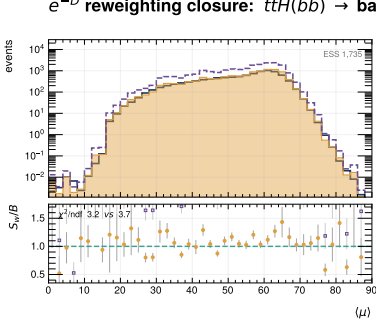
e^{-D} reweighting closure: $t\bar{t}H(c\bar{c})$ → background · 2L, Run 3

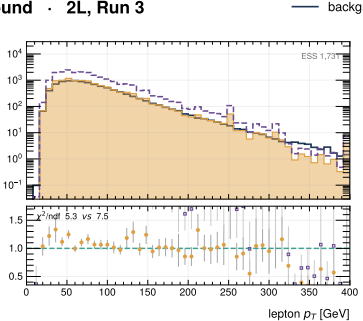


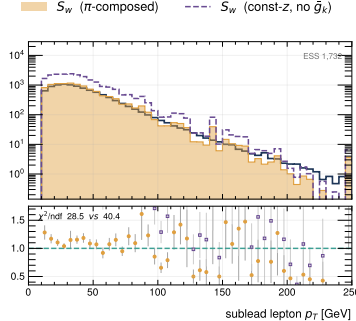

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 1,734
χ^2/ndf 2.4 vs 3.4
H_T(jets) [GeV]
ESS 1,735
χ^2/ndf 2.2 vs 4.0
N_{jets}
ESS 478
χ^2/ndf 9.6 vs 10.7
m_{cc} [GeV]
ESS 1,357
χ^2/ndf 0.7 vs 0.8
m_{bb} [GeV]
ESS 1,735
χ^2/ndf 54.5 vs 51.0
E_T^{miss} [GeV]
ESS 1,736
χ^2/ndf 1.7 vs 6.5
Σ PCBT bin

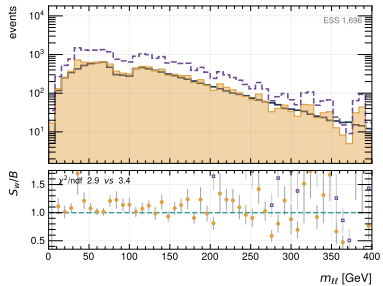

e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 2,129
χ^2/ndf 6.2 vs 6.9
leading jet p_T [GeV]
ESS 476
χ^2/ndf 12.5 vs 12.5
$p_T(cc)$ [GeV]
ESS 476
χ^2/ndf 16.7 vs 17.5
$\Delta R(c,c)$
ESS 1,735
χ^2/ndf 3.6 vs 21.0
$N_{b\,jets}$
ESS 1,735
χ^2/ndf 2.0 vs 16.8
$N_{c\,jets}$
ESS 1,735
χ^2/ndf 8.7 vs 10.7
E_T^{miss} significance

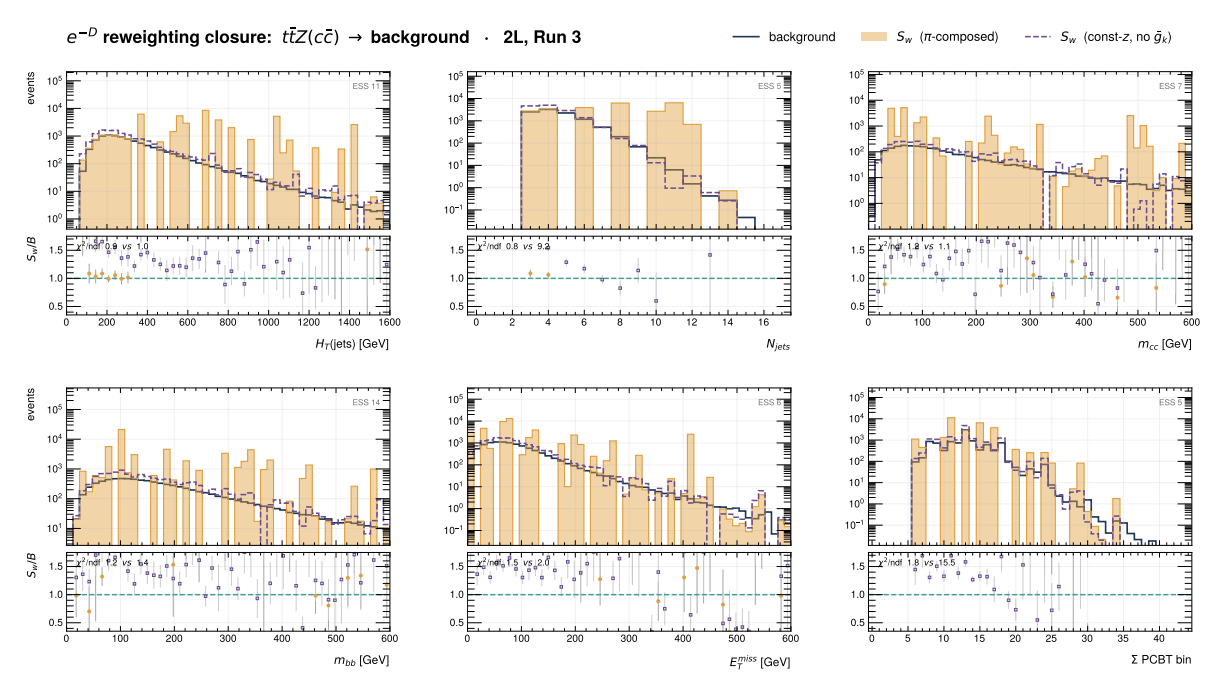
e^{-D} reweighting closure: $t\bar{t}H(b\bar{b})$ → background · 2L, Run 3

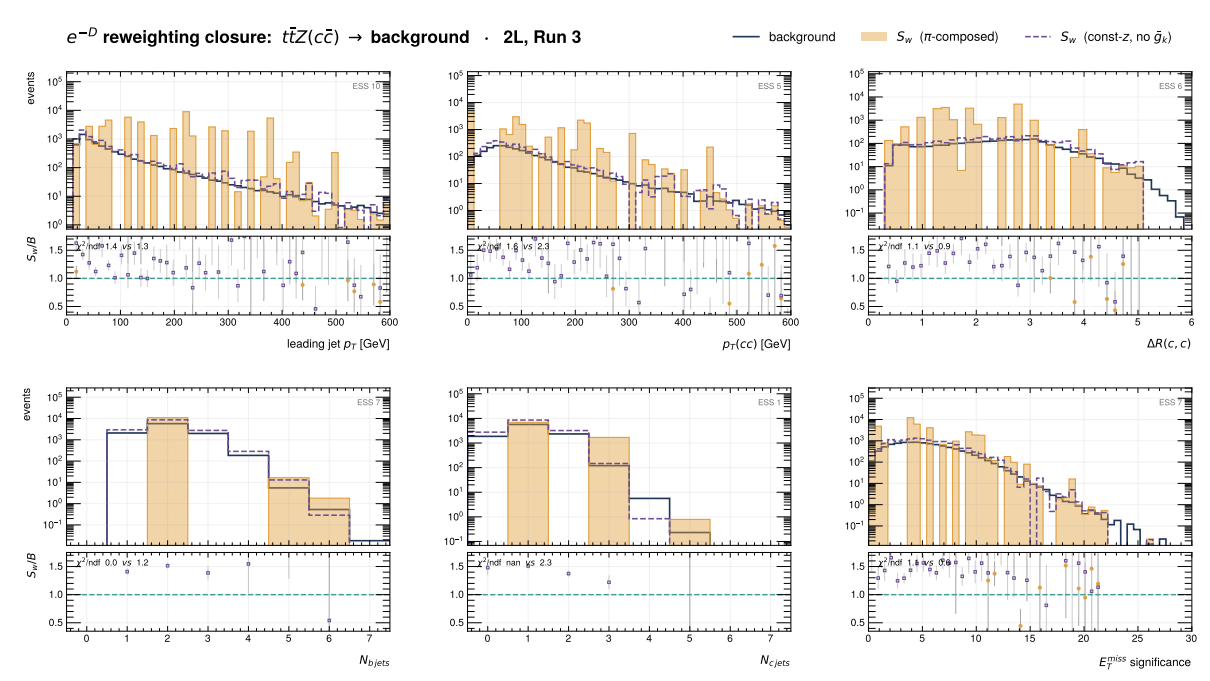


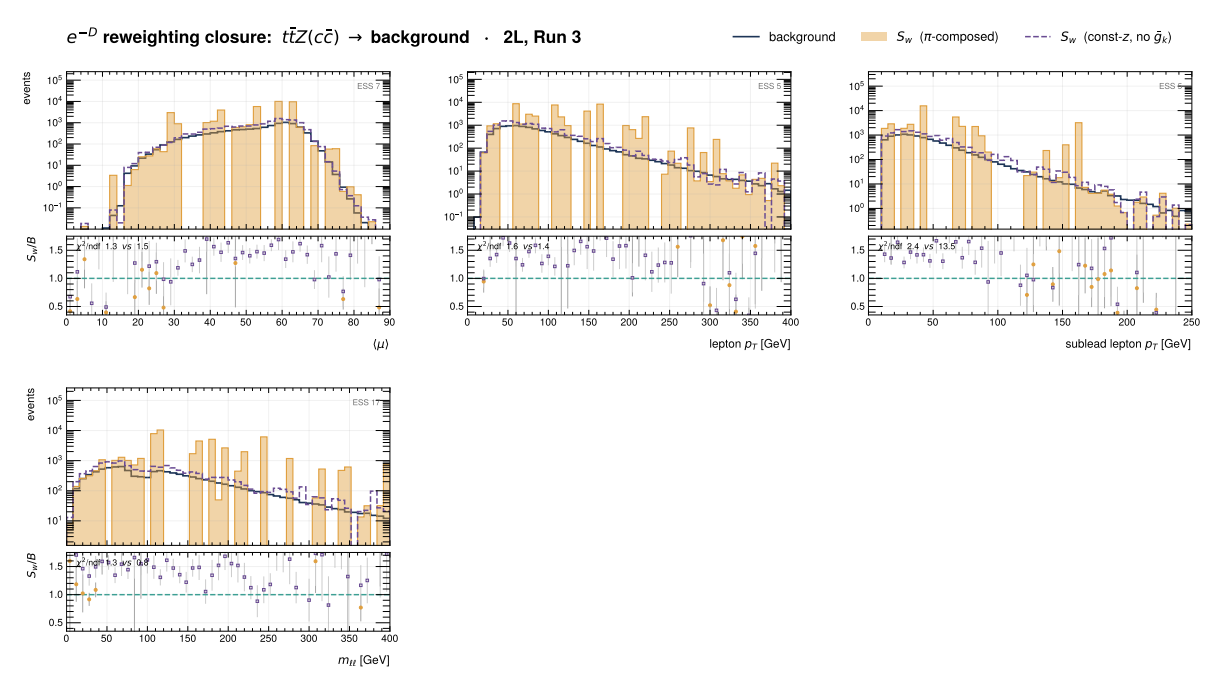


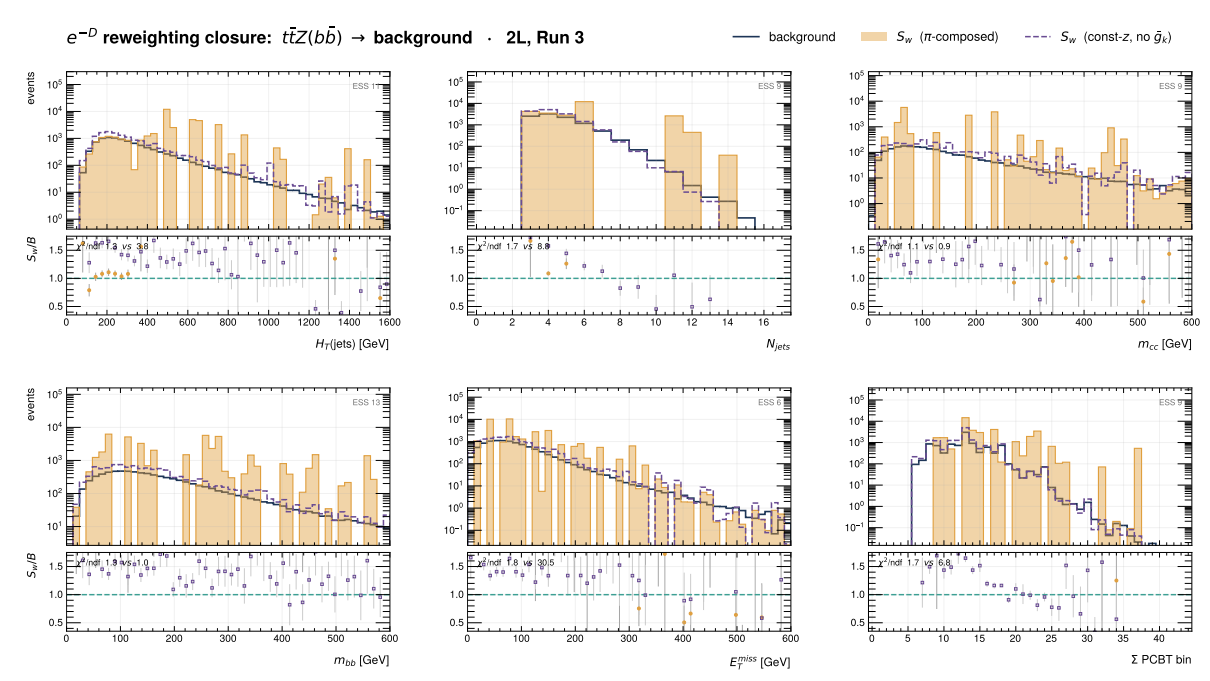


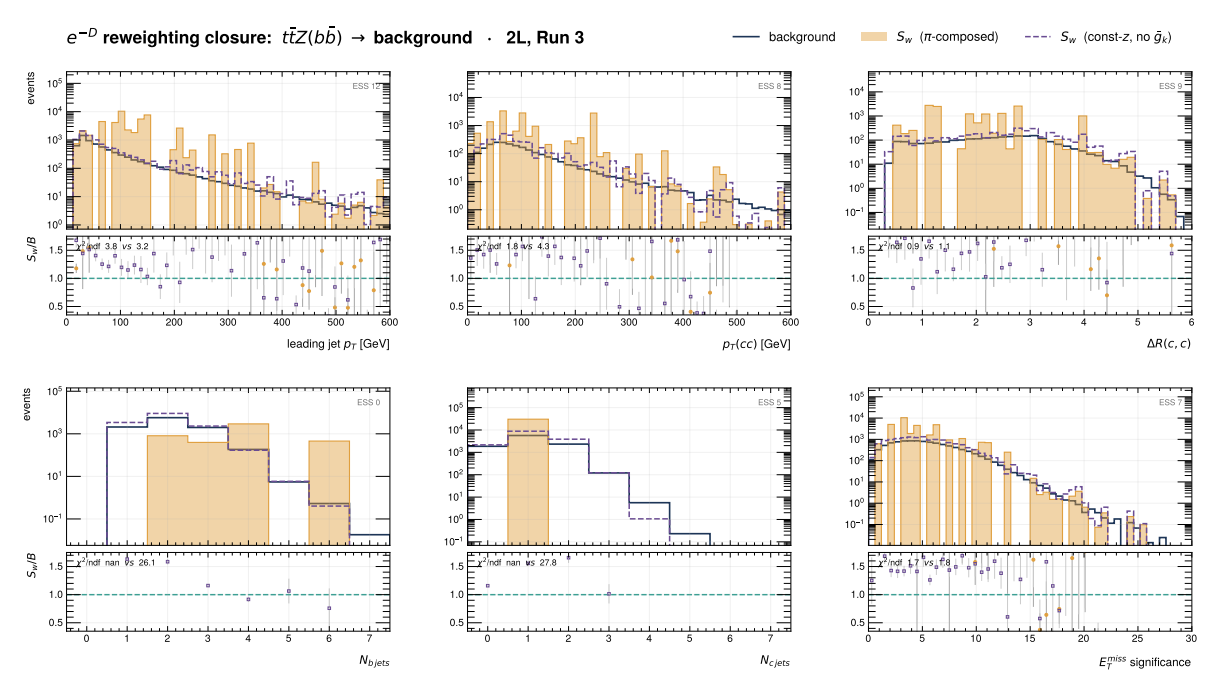


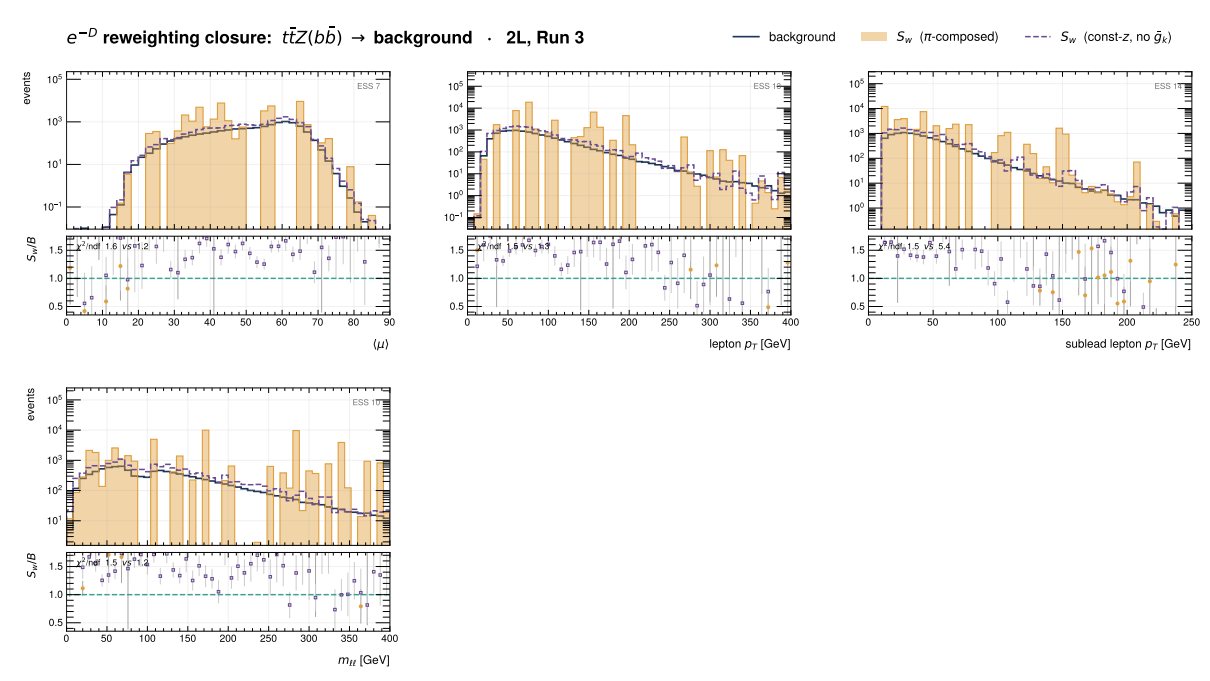

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 11
χ^2/ndf 0.9 vs 1.0
H_T(jets) [GeV]
ESS 5
χ^2/ndf 0.8 vs 9.2
N_{jets}
ESS 7
χ^2/ndf 1.8 vs 1.1
m_{cc} [GeV]
ESS 14
χ^2/ndf 1.2 vs 5.4
m_{bb} [GeV]
ESS 5
χ^2/ndf 1.5 vs 2.0
E_T^{miss} [GeV]
ESS 5
χ^2/ndf 1.8 vs 15.5
Σ PCBT bin


e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 10
χ^2/ndf 1.4 vs 1.3
leading jet p_T [GeV]
ESS 5
χ^2/ndf 1.6 vs 2.3
$p_T(cc)$ [GeV]
ESS 6
χ^2/ndf 1.1 vs 0.9
$\Delta R(c,c)$
ESS 7
χ^2/ndf 0.0 vs 1.2
$N_{b\,jets}$
ESS 1
χ^2/ndf nan vs 2.3
$N_{c\,jets}$
ESS 7
χ^2/ndf 1.1 vs 0.6
E_T^{miss} significance

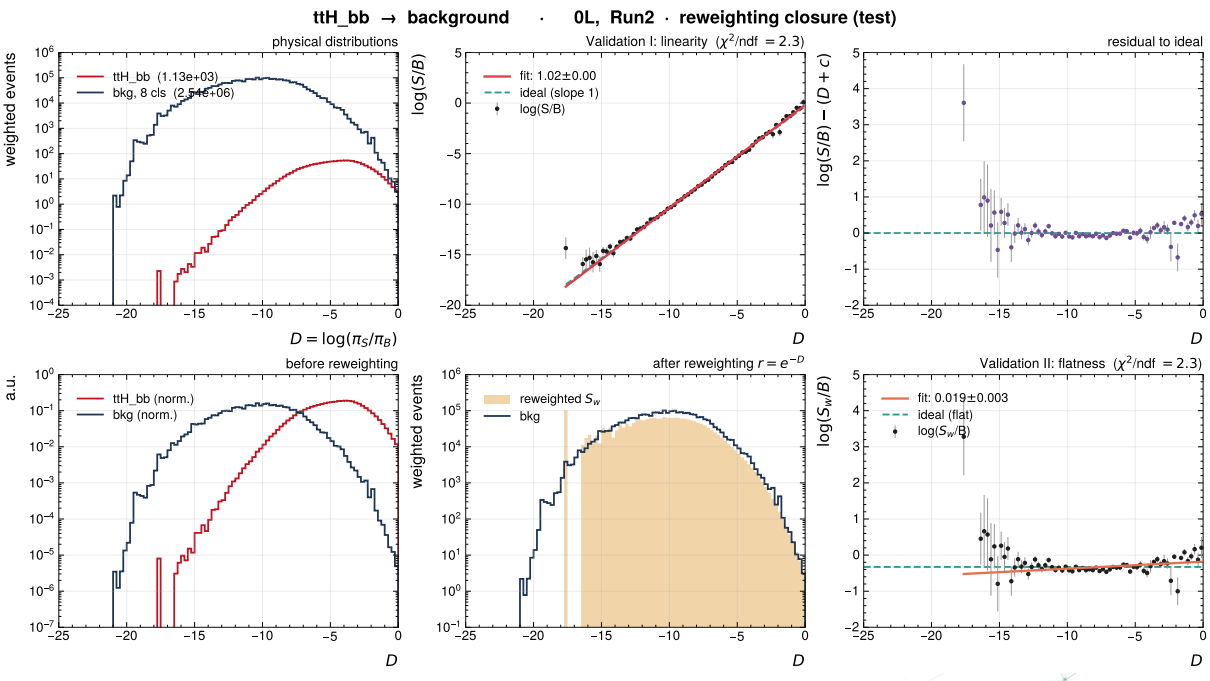

e^{-D} reweighting closure: $t\bar{t}Z(c\bar{c})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 7
χ^2/ndf 1.3 vs 1.5
$\langle\mu\rangle$
ESS 5
χ^2/ndf 1.6 vs 1.4
lepton p_T [GeV]
ESS 5
χ^2/ndf 2.4 vs 13.5
sublead lepton p_T [GeV]
ESS 17
χ^2/ndf 1.3 vs 0.8
m_{tt} [GeV]

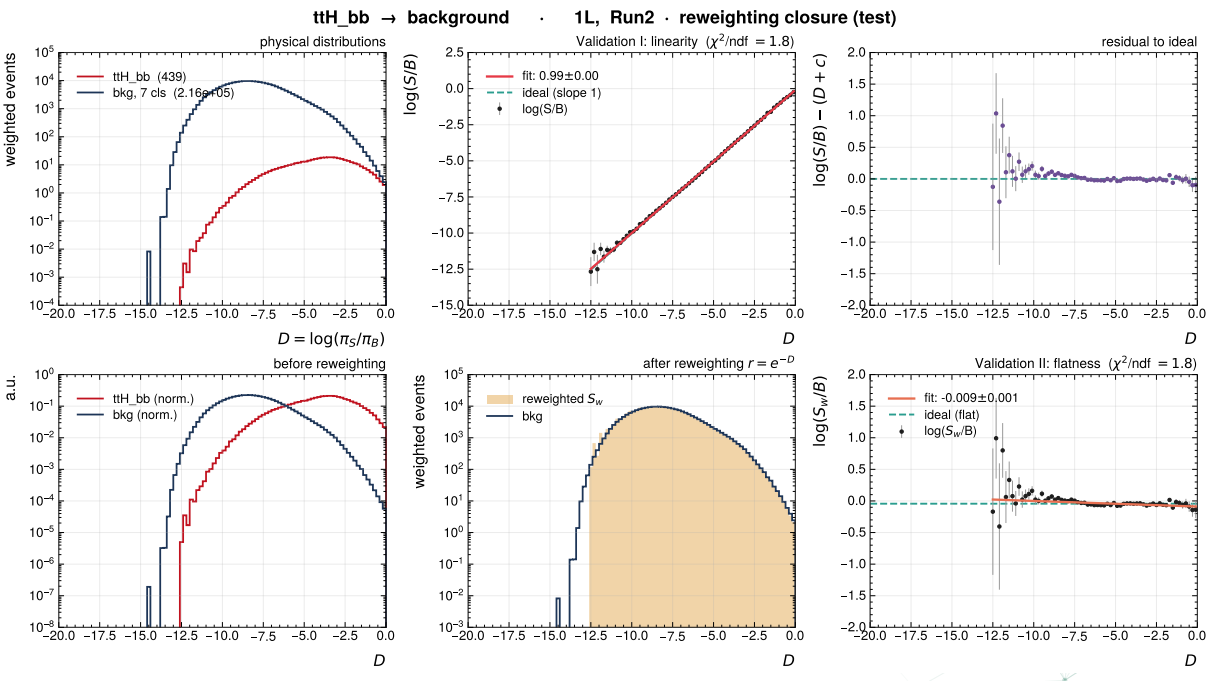

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\tilde{g}_k$)
events
S_w/B
ESS 11
χ^2/ndf 1.3 vs 3.8
H_T(jets) [GeV]
ESS 9
χ^2/ndf 1.7 vs 8.9
N_{jets}
ESS 9
χ^2/ndf 1.1 vs 0.9
m_{cc} [GeV]
ESS 13
χ^2/ndf 1.8 vs 1.0
m_{bb} [GeV]
ESS 6
χ^2/ndf 1.8 vs 30.5
E_T^{miss} [GeV]
ESS 7
χ^2/ndf 1.7 vs 6.8
Σ PCBT bin

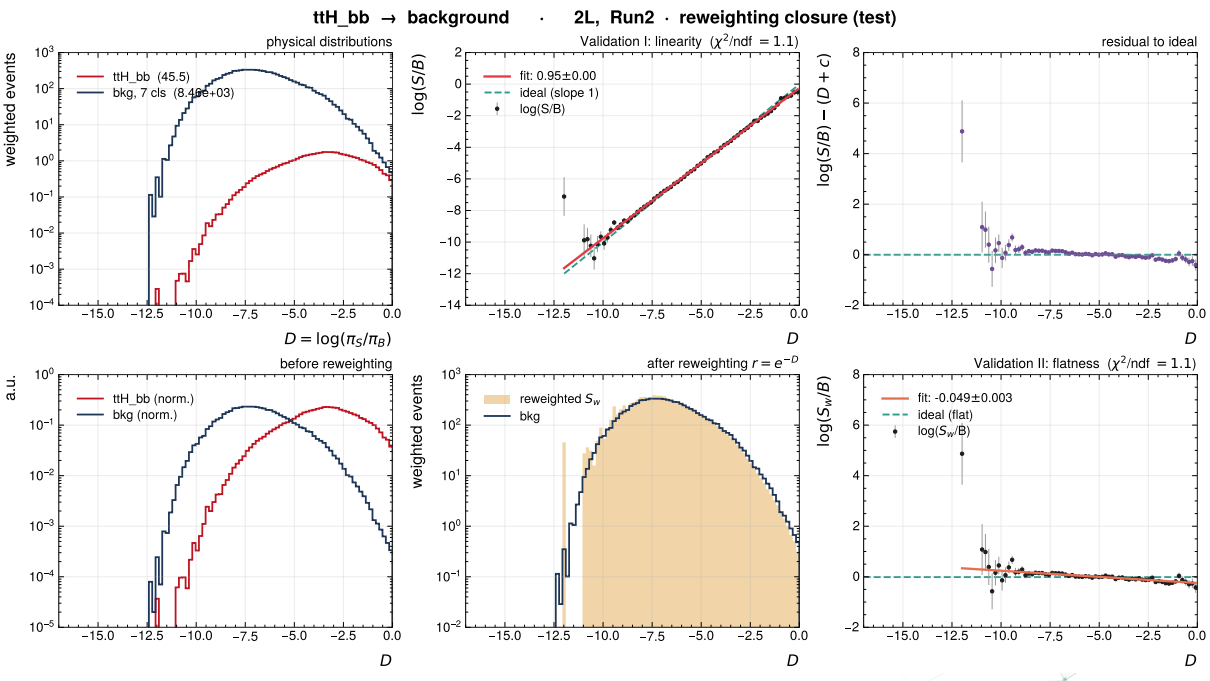

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\vec{g}_k$)
events
S_w/B
ESS 12
χ^2/ndf 3.8 vs 3.2
leading jet p_T [GeV]
ESS 8
χ^2/ndf 1.8 vs 4.3
$p_T(cc)$ [GeV]
ESS 0
χ^2/ndf 0.9 vs 1.1
$\Delta R(c,c)$
ESS 0
χ^2/ndf nan vs 26.1
$N_{b\,jets}$
ESS 5
χ^2/ndf nan vs 27.8
$N_{c\,jets}$
ESS 7
χ^2/ndf 1.7 vs 1.8
E_T^{miss} significance

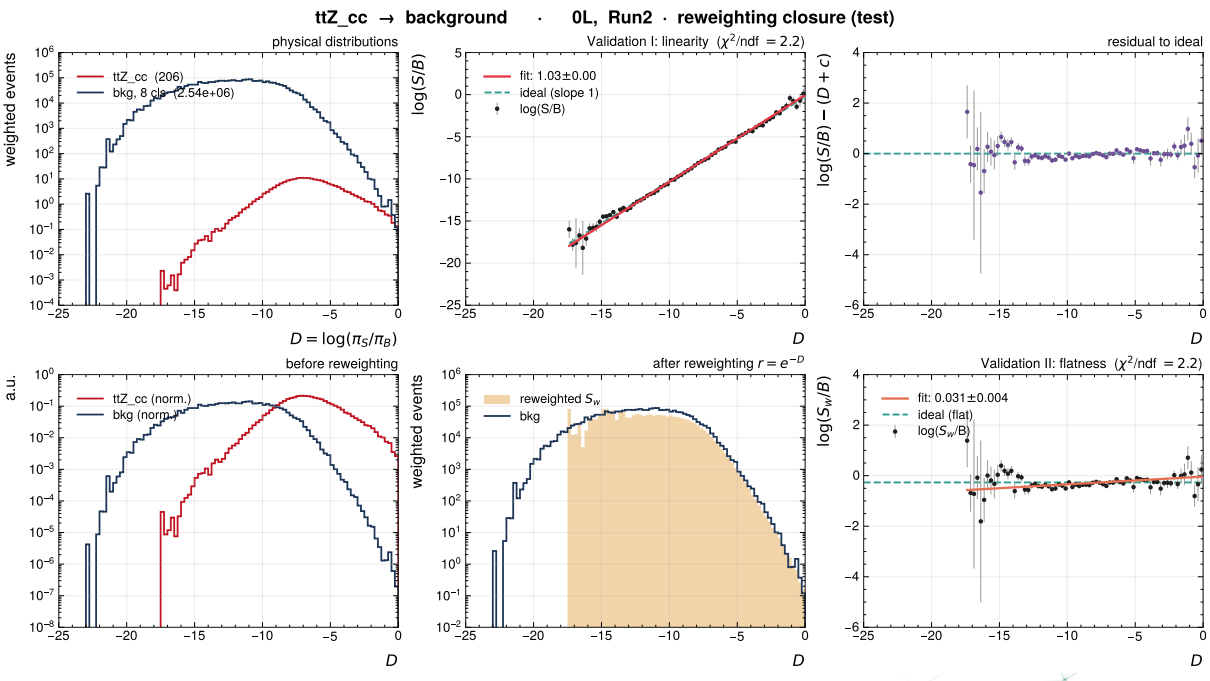

e^{-D} reweighting closure: $t\bar{t}Z(b\bar{b})$ → background · 2L, Run 3
background
S_w (π-composed)
S_w (const-z, no $\bar{g}_k$)
events
S_w/B
ESS 7
χ^2/ndf 1.6 vs 1.2
$\langle\mu\rangle$
ESS 18
χ^2/ndf 1.5 vs 1.3
lepton p_T [GeV]
ESS 17
χ^2/ndf 1.5 vs 5.4
sublead lepton p_T [GeV]
ESS 17
χ^2/ndf 1.5 vs 1.4
$m_{\ell\ell}$ [GeV]

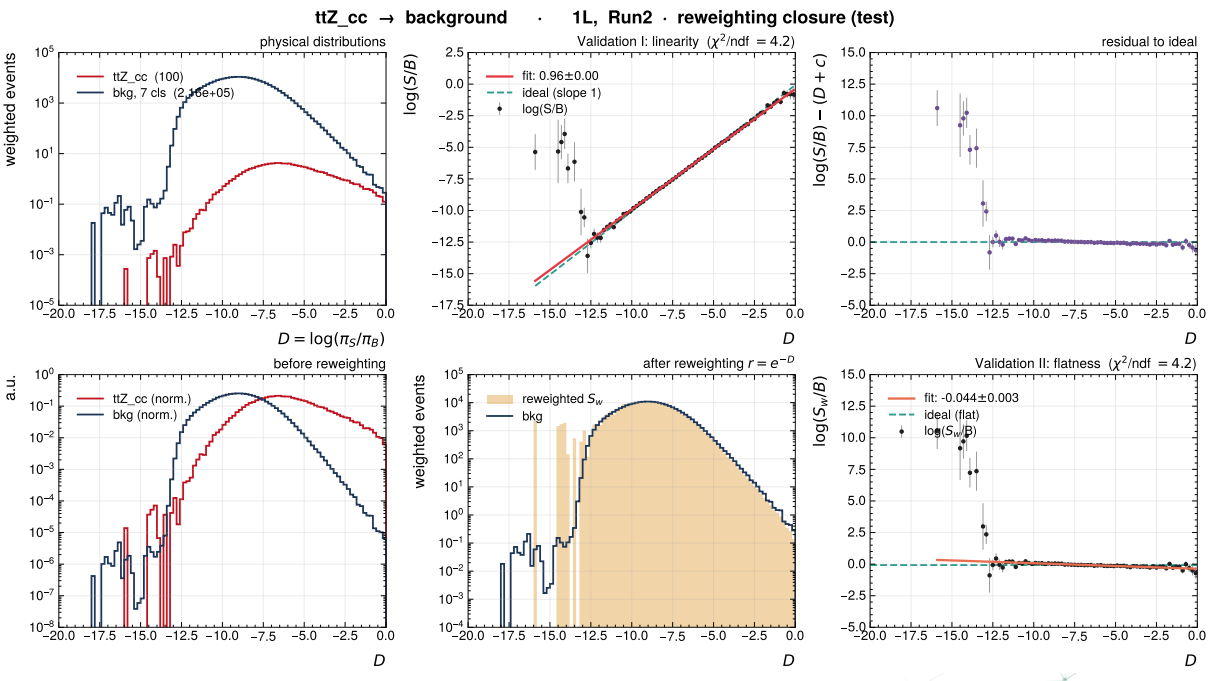
Backup 2: Linearity and reweighting closure

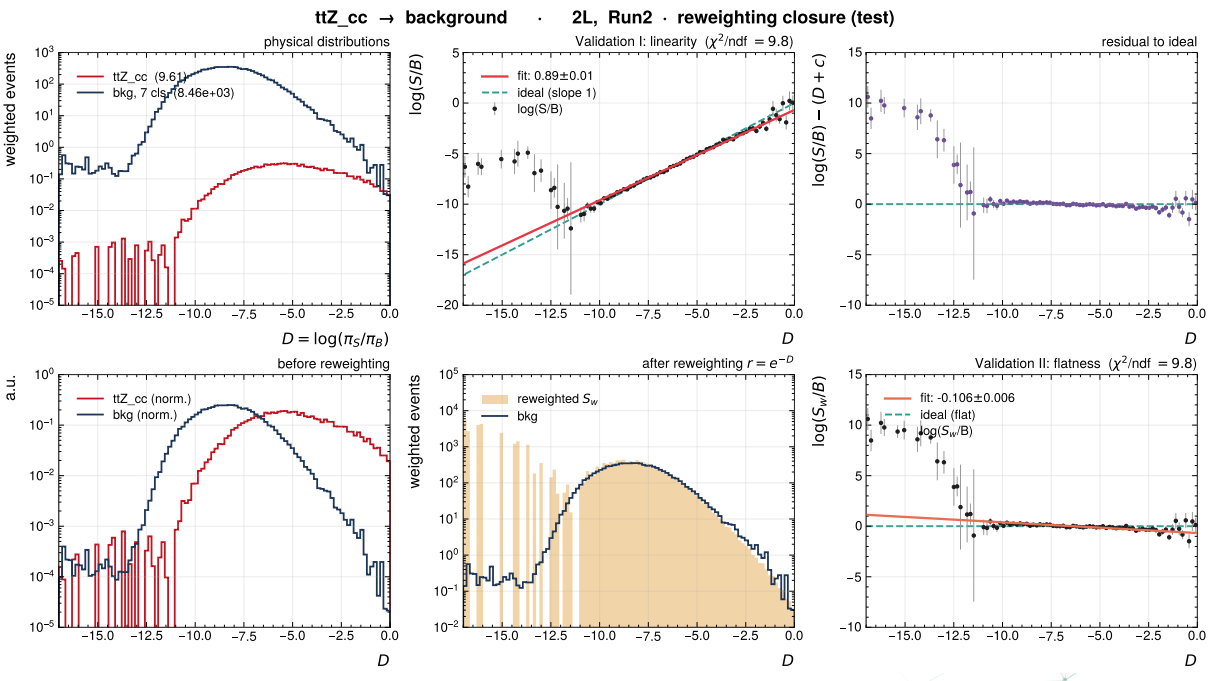

ttH_bb → background · 0L, Run2 · reweighting closure (test)
physical distributions
ttH_bb (1.13e+03)
bkg, 8 cls (2.54E+06)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 2.3)
fit: 1.02±0.00
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttH_bb (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 2.3)
fit: 0.019±0.003
ideal (flat)
log(S_w/B)
log(S_w/B)
D

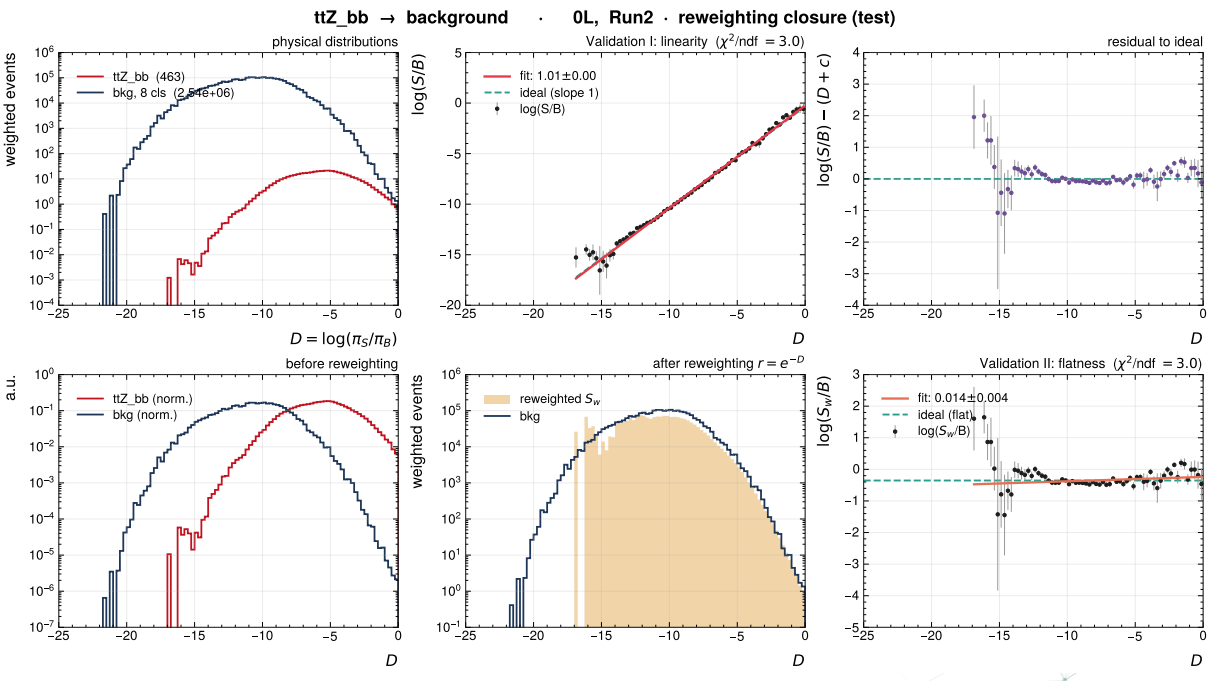

ttH_bb → background · 1L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttH_bb (439)
bkg, 7 cls (2.16e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 1.8)
$\log(S/B)$
fit: 0.99±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttH_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 1.8)
$\log(S_w/B)$
fit: -0.009±0.001
ideal (flat)
log(S_w/B)
D

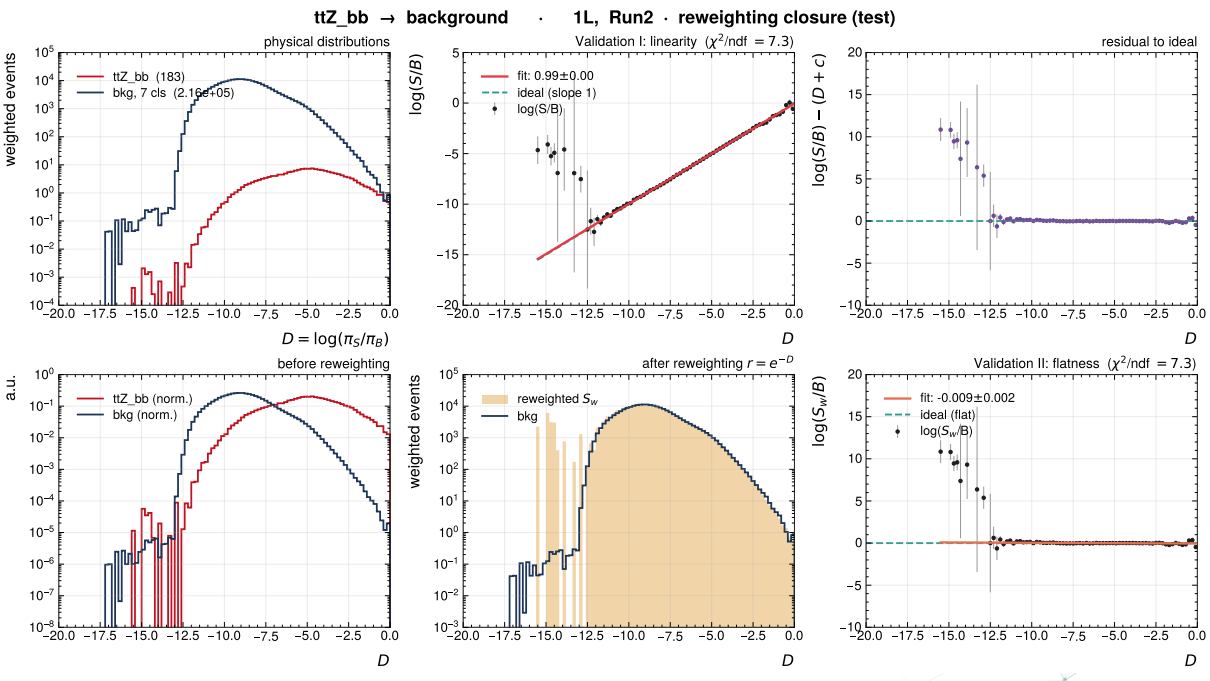

ttH_bb → background · 2L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttH_bb (45.5)
bkg, 7 cls (8.486+03)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 1.1)
$\log(S/B)$
fit: 0.95±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttH_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 1.1)
$\log(S_w/B)$
fit: -0.049±0.003
ideal (flat)
log(S_w/B)
D

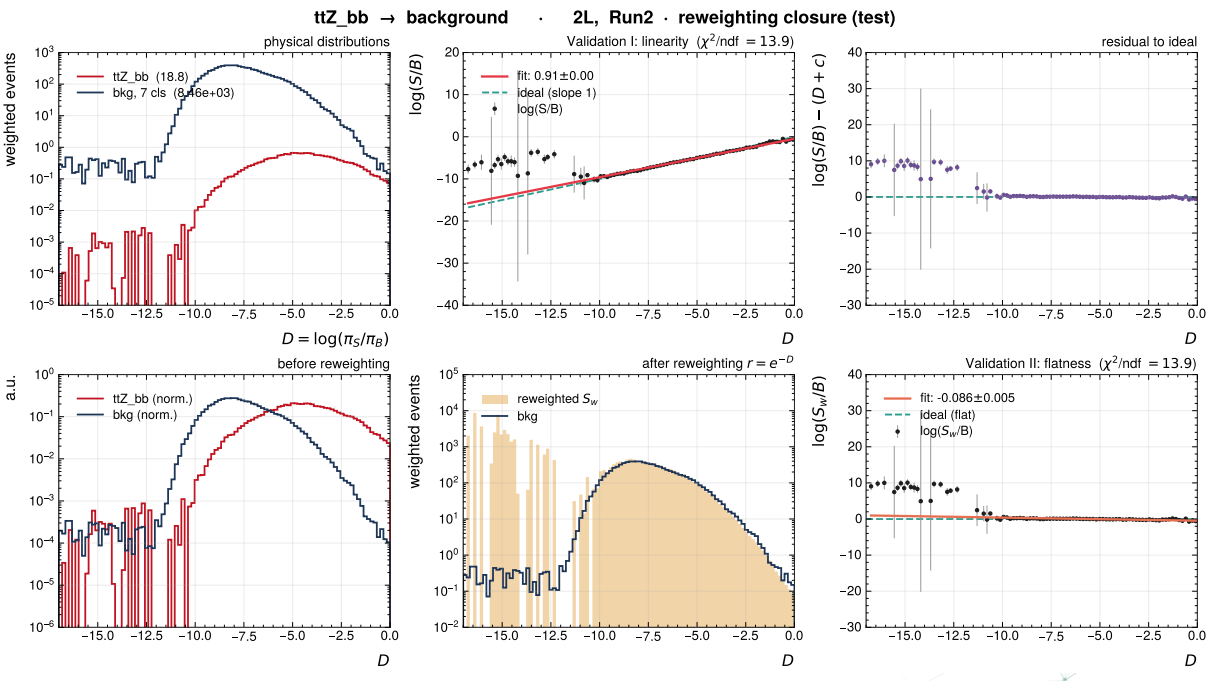

ttZ_cc → background · 0L, Run2 · reweighting closure (test)
physical distributions
ttZ_cc (206)
bkg, 8 cls (2.54e+06)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 2.2)
fit: 1.03±0.00
ideal (slope 1)
log(S/B)
$\log(S/B)$
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
ttZ_cc (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 2.2)
fit: 0.031±0.004
ideal (flat)
log(S_w/B)
$\log(S_w/B)$
D

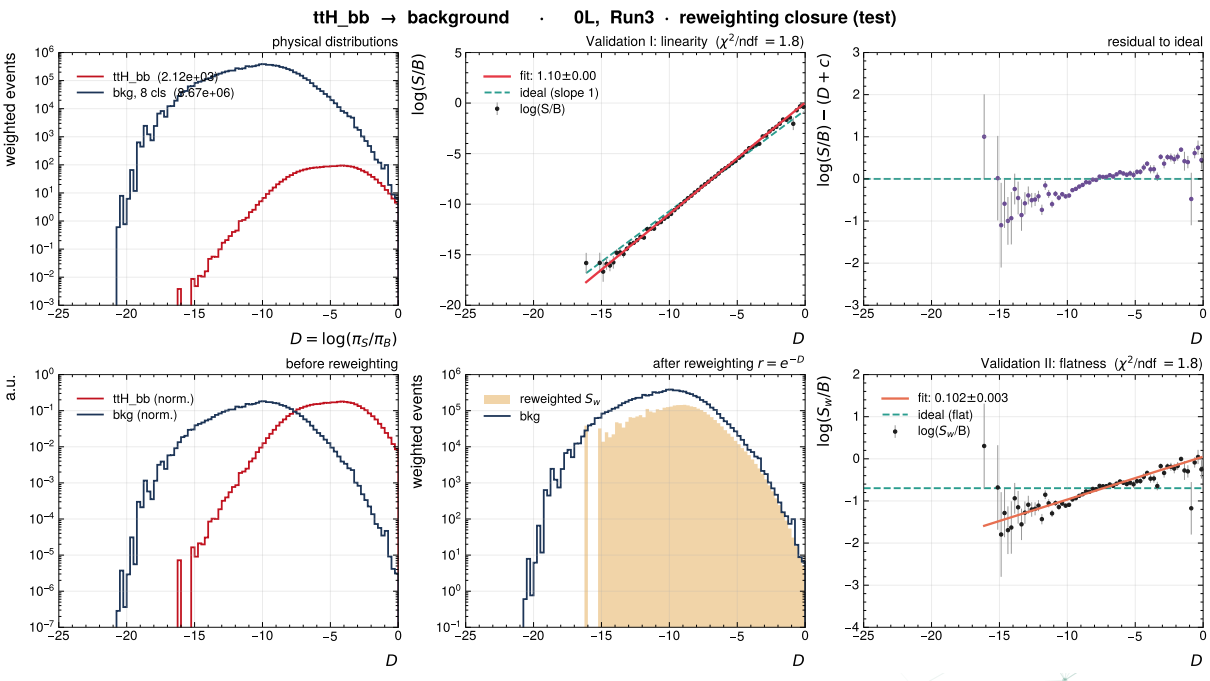

ttZ_cc → background · 1L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttZ_cc (100)
bkg, 7 cls (2.16e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 4.2)
$\log(S/B)$
fit: 0.96±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_cc (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 4.2)
$\log(S_w/B)$
fit: -0.044±0.003
ideal (flat)
log(S_w/B)
D

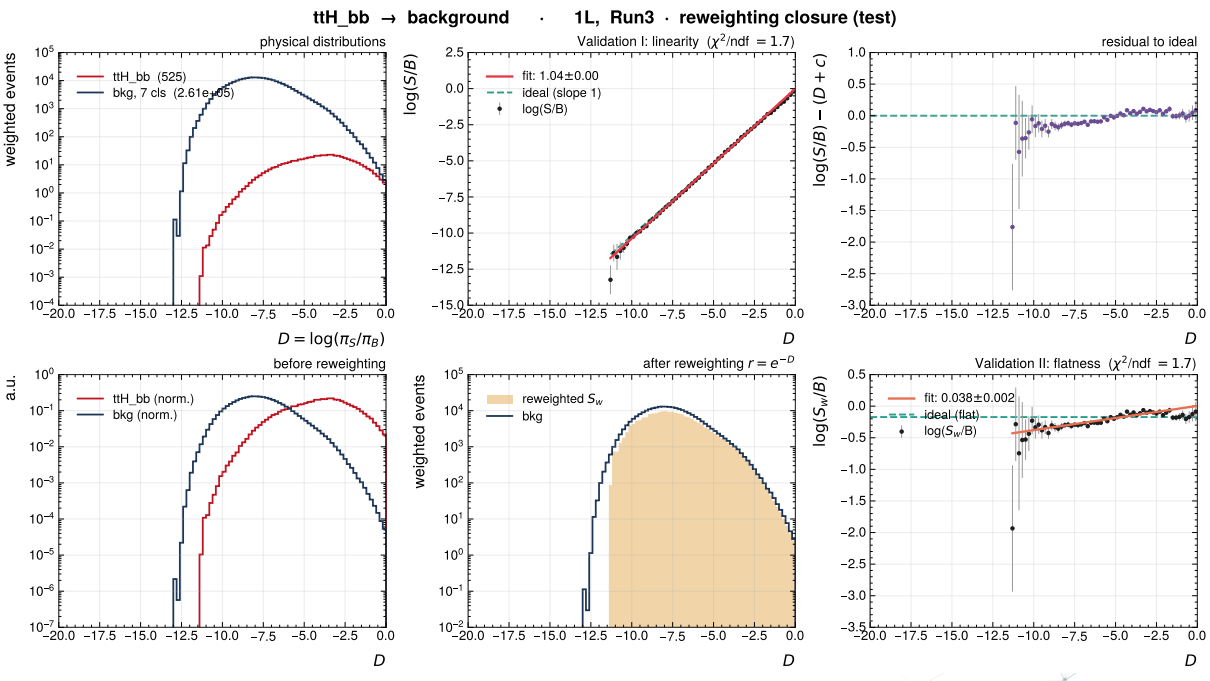

ttZ_cc → background · 2L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttZ_cc (9.61)
bkg, 7 cls (8.46e+03)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 9.8)
$\log(S/B)$
fit: 0.89±0.01
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_cc (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 9.8)
$\log(S_w/B)$
fit: -0.106±0.006
ideal (flat)
log(S_w/B)
D

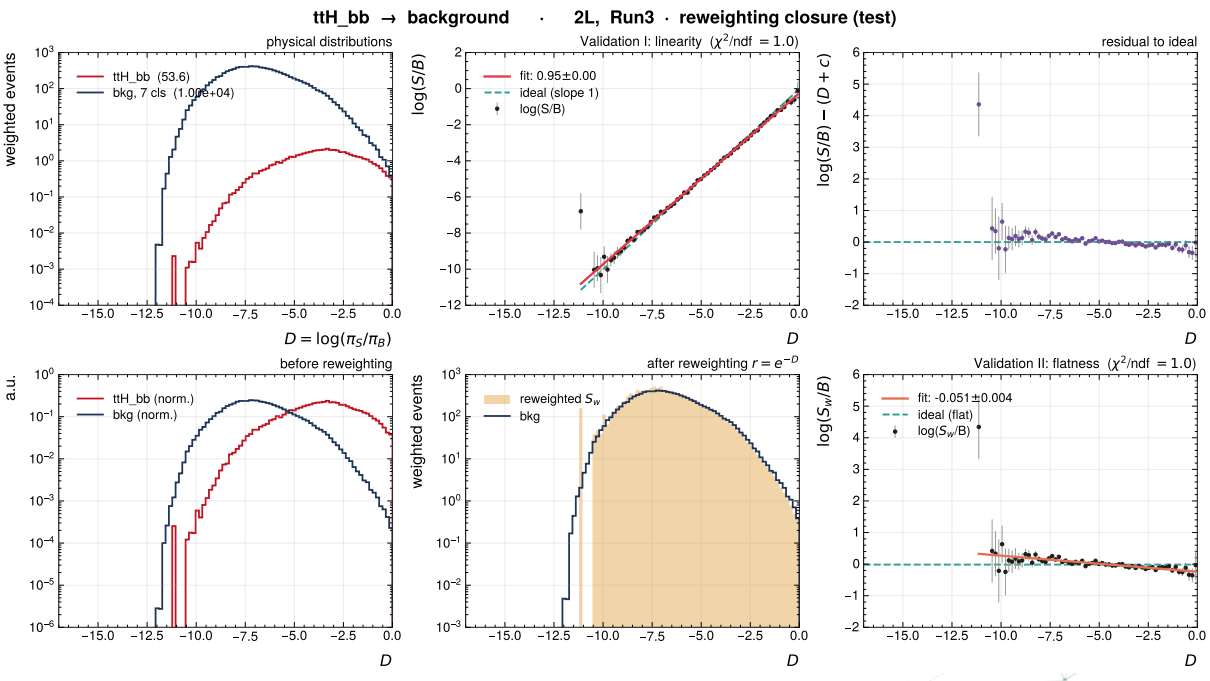

ttZ_bb → background · 0L, Run2 · reweighting closure (test)
physical distributions
ttZ_bb (463)
bkg, 8 cls (2.54e+06)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 3.0)
fit: 1.01±0.00
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttZ_bb (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 3.0)
fit: 0.014±0.004
ideal (flat)
log(S_w/B)
log(S_w/B)
D

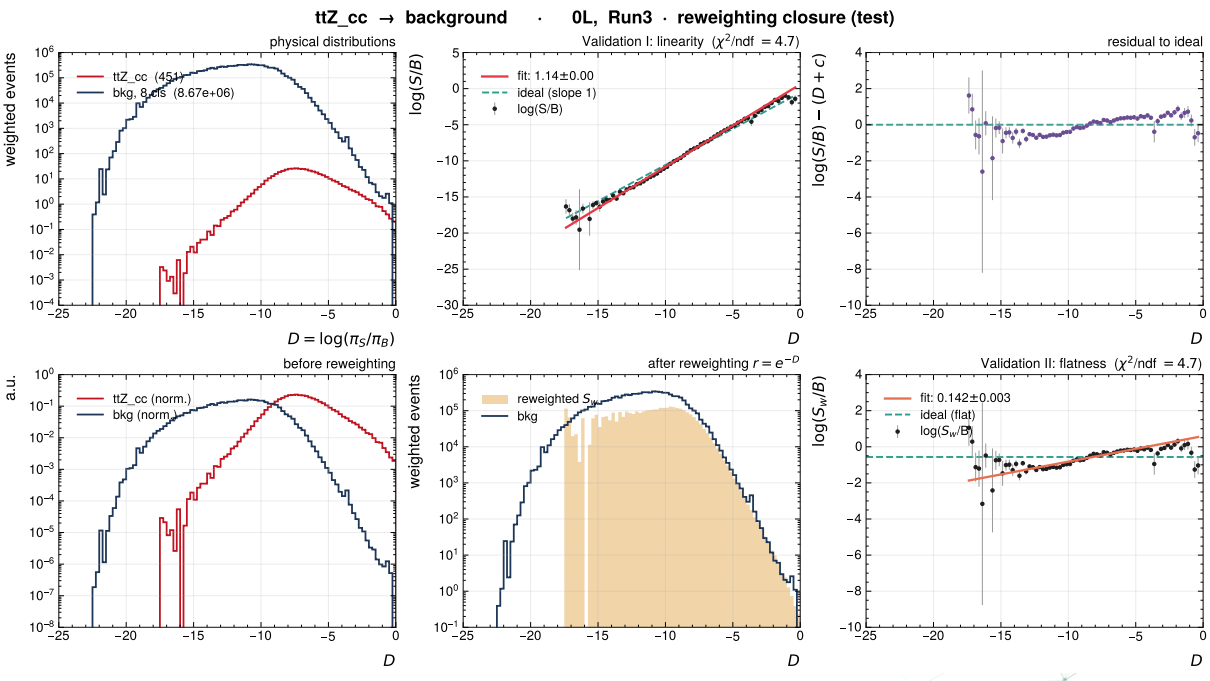

ttZ_bb → background · 1L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttZ_bb (183)
bkg, 7 cls (2.16e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 7.3)
$\log(S/B)$
fit: 0.99±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 7.3)
$\log(S_w/B)$
fit: -0.009±0.002
ideal (flat)
log(S_w/B)
D

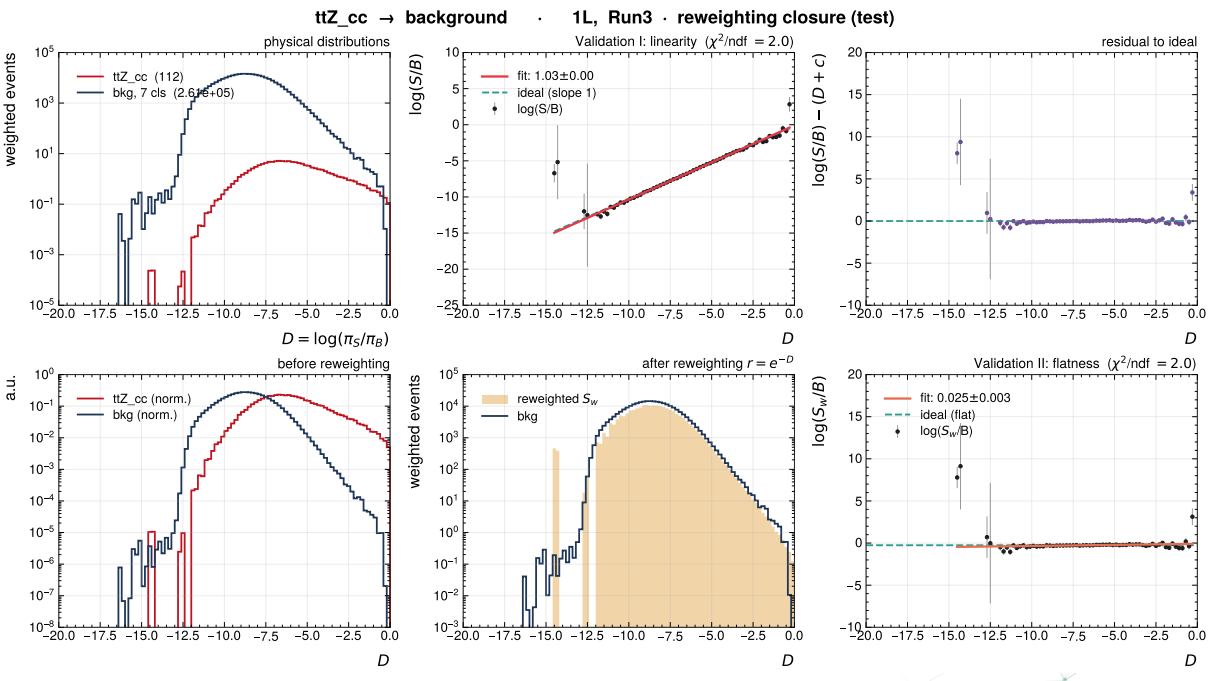

ttZ_bb → background · 2L, Run2 · reweighting closure (test)
physical distributions
weighted events
ttZ_bb (18.8)
bkg, 7 cls (8.16e+03)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 13.9)
$\log(S/B)$
fit: 0.91±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 13.9)
$\log(S_w/B)$
fit: -0.086±0.005
ideal (flat)
log(S_w/B)
D

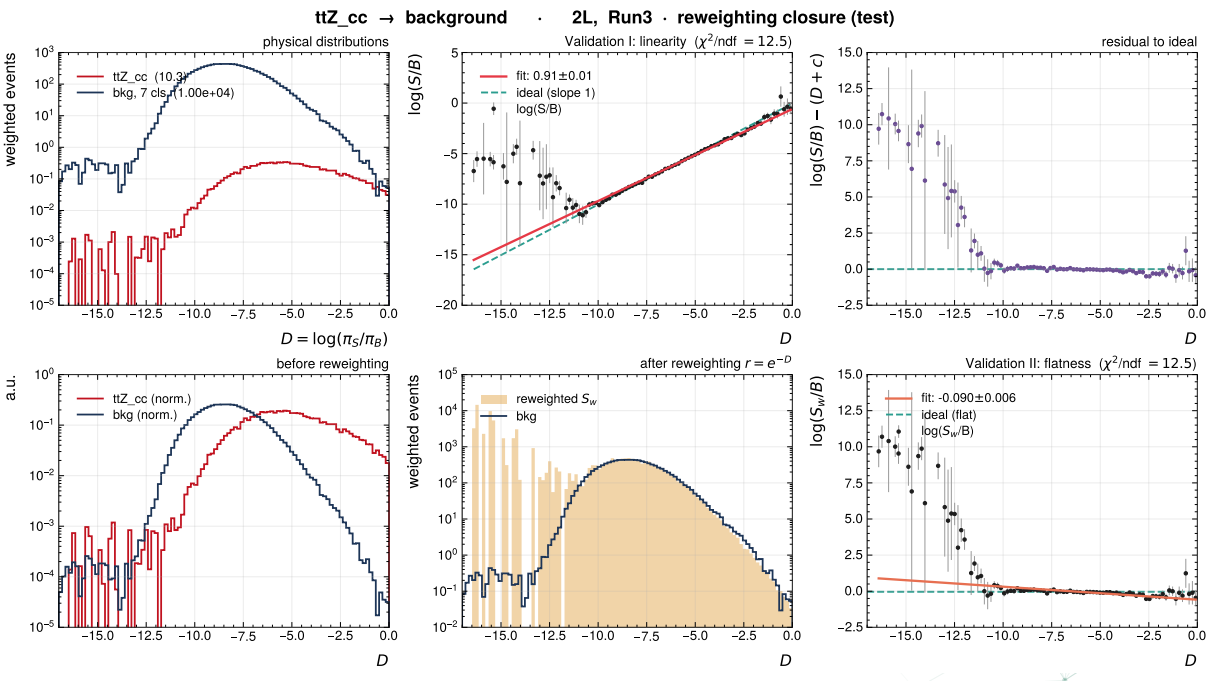

ttH_bb → background · 0L, Run3 · reweighting closure (test)
physical distributions
ttH_bb (2.12e+03)
bkg, 8 cls (3.67e+06)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 1.8)
fit: 1.10±0.00
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttH_bb (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 1.8)
fit: 0.102±0.003
ideal (flat)
log(S_w/B)
log(S_w/B)
D

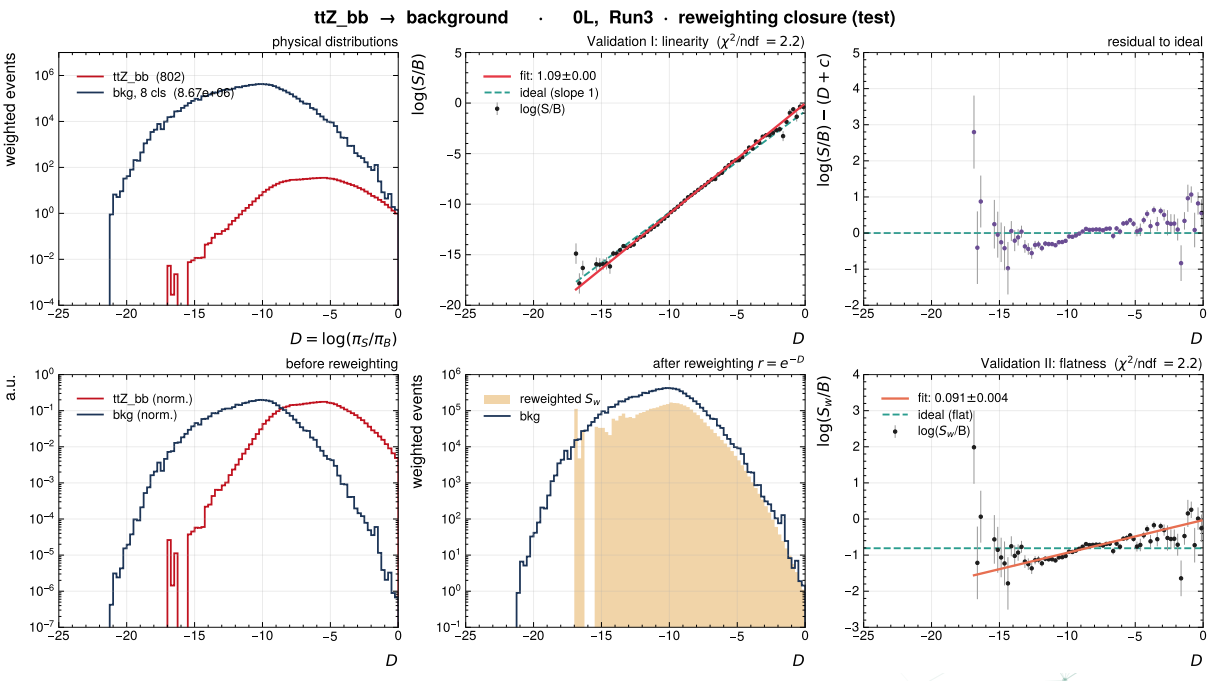

ttH_bb → background · 1L, Run3 · reweighting closure (test)
physical distributions
weighted events
ttH_bb (525)
bkg, 7 cls (2.61e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 1.7)
$\log(S/B)$
fit: 1.04±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttH_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 1.7)
$\log(S_w/B)$
fit: 0.038±0.002
ideal (flat)
log(S_w/B)
D

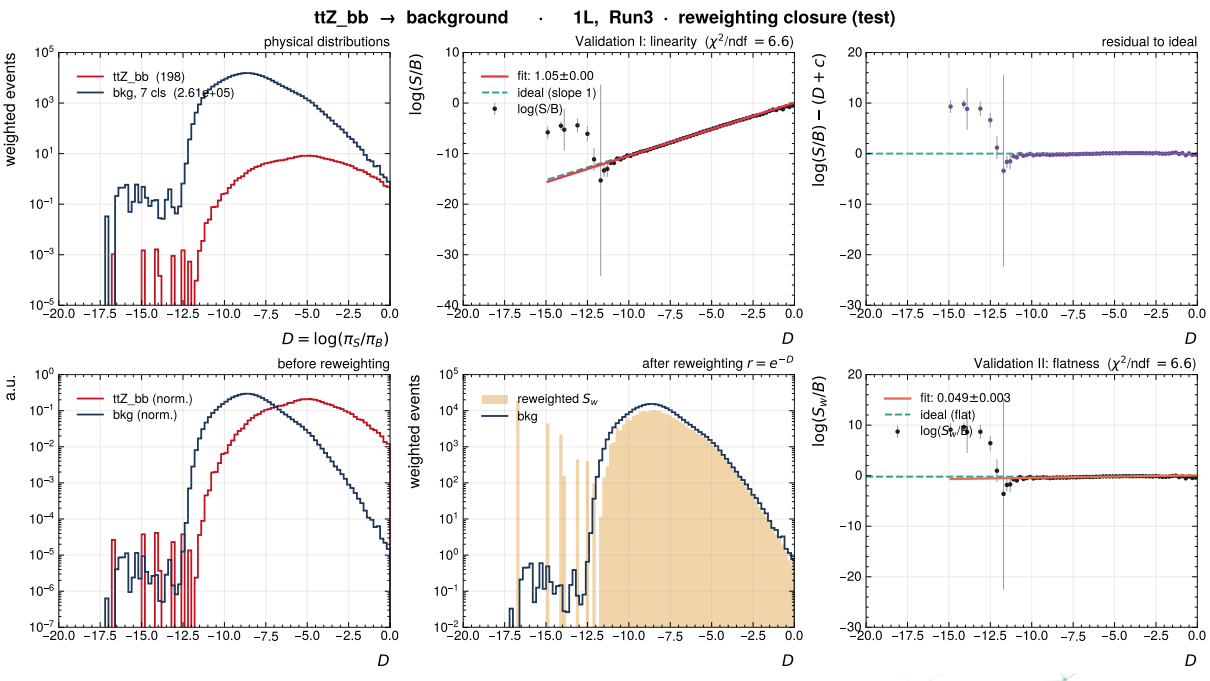

ttH_bb → background · 2L, Run3 · reweighting closure (test)
physical distributions
ttH_bb (53.6)
bkg, 7 cls (1.00e+04)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 1.0)
fit: 0.95±0.00
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttH_bb (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 1.0)
fit: -0.051±0.004
ideal (flat)
log(S_w/B)
log(S_w/B)
D

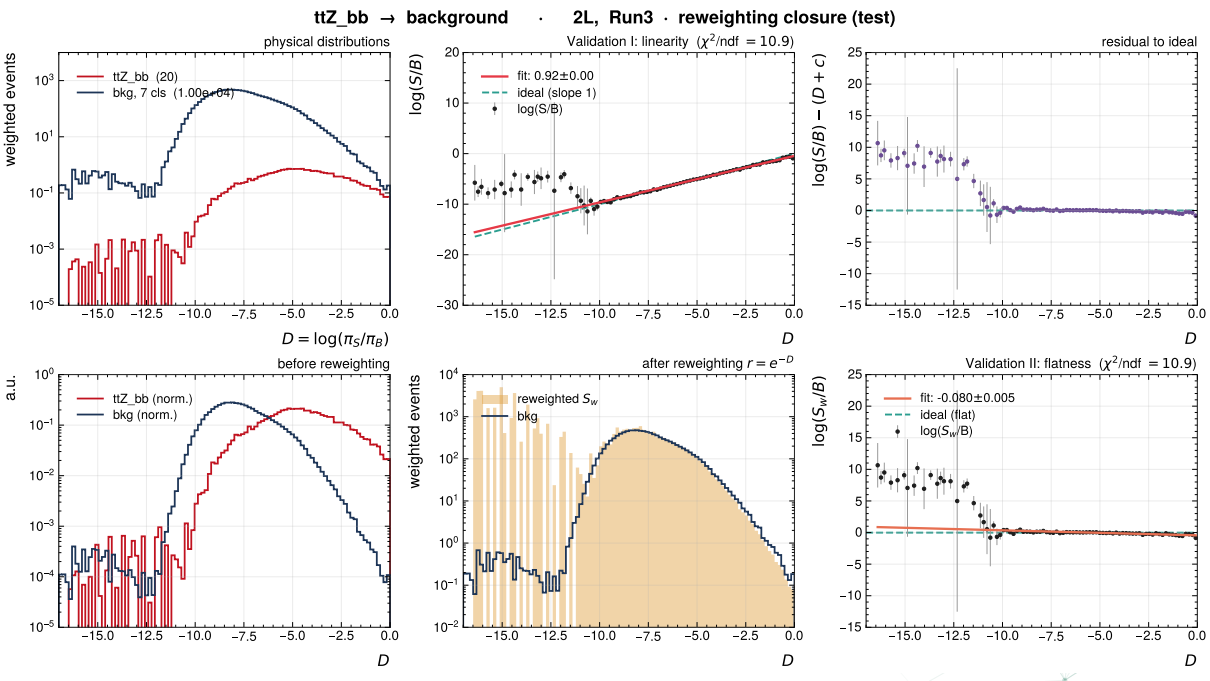

ttZ_cc → background · 0L, Run3 · reweighting closure (test)
physical distributions
ttZ_cc (451)
bkg, 8 cls (8.67e+06)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 4.7)
fit: 1.14±0.00
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttZ_cc (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 4.7)
fit: 0.142±0.003
ideal (flat)
log(S_w/B)
log(S_w/B)
D

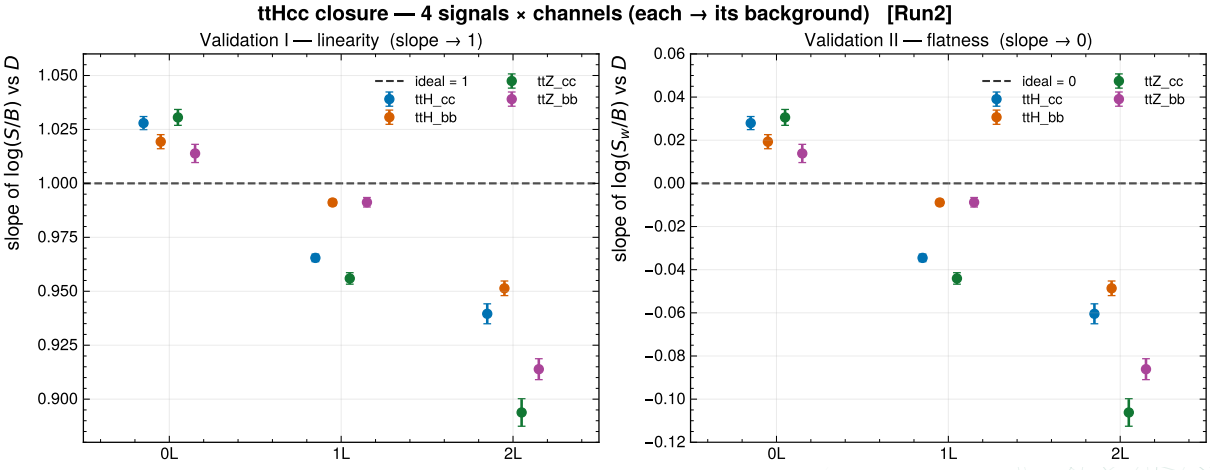

ttZ_cc → background · 1L, Run3 · reweighting closure (test)
physical distributions
weighted events
ttZ_cc (112)
bkg, 7 cls (2.61e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 2.0)
$\log(S/B)$
fit: 1.03±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_cc (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 2.0)
$\log(S_w/B)$
fit: 0.025±0.003
ideal (flat)
log(S_w/B)
D

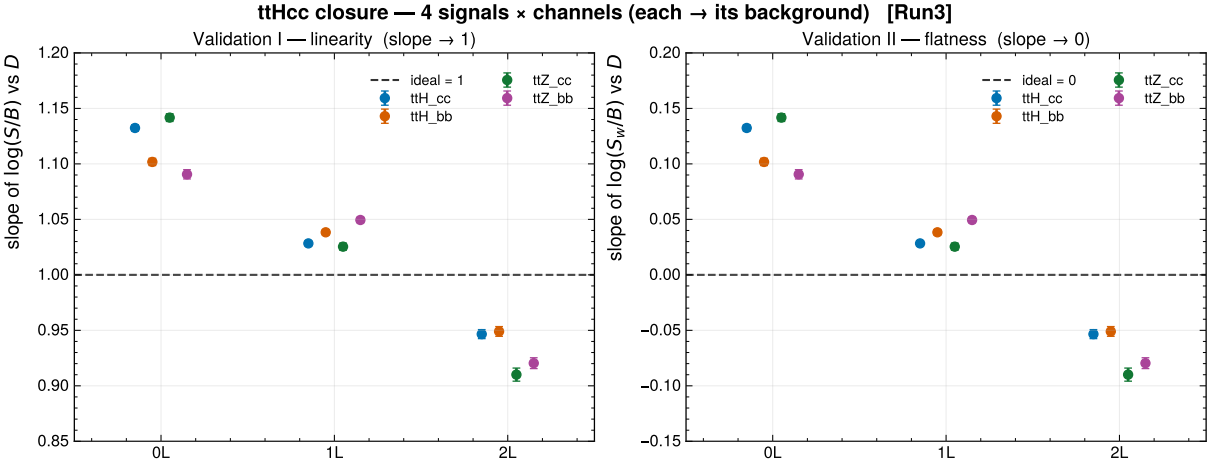

ttZ_cc → background · 2L, Run3 · reweighting closure (test)
physical distributions
ttZ_cc (10.3)
bkg, 7 cls (1.00e+04)
weighted events
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 12.5)
fit: 0.91±0.01
ideal (slope 1)
log(S/B)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
ttZ_cc (norm.)
bkg (norm.)
a.u.
D
after reweighting $r = e^{-D}$
reweighted S_w
bkg
weighted events
D
Validation II: flatness (χ^2/ndf = 12.5)
fit: -0.090±0.006
ideal (flat)
log(S_w/B)
log(S_w/B)
D


ttZ_bb → background · 0L, Run3 · reweighting closure (test)
physical distributions
weighted events
ttZ_bb (802)
bkg, 8 cls (8.67e+06)
D = log(πS/πB)
Validation I: linearity (χ²/ndf = 2.2)
log(S/B)
fit: 1.09±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
log(S/B) − (D + c)
D
before reweighting
a.u.
ttZ_bb (norm.)
bkg (norm.)
D
after reweighting r = e^−D
weighted events
reweighted Sw
bkg
D
Validation II: flatness (χ²/ndf = 2.2)
log(Sw/B)
fit: 0.091±0.004
ideal (flat)
log(Sw/B)
D


ttZ_bb → background · 1L, Run3 · reweighting closure (test)
physical distributions
weighted events
ttZ_bb (198)
bkg, 7 cls (2.61e+05)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 6.6)
$\log(S/B)$
fit: 1.05±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 6.6)
$\log(S_w/B)$
fit: 0.049±0.003
ideal (flat)
$\log(S_w/B)$
D


ttZ_bb → background · 2L, Run3 · reweighting closure (test)
physical distributions
weighted events
ttZ_bb (20)
bkg, 7 cls (1.00e+04)
$D = \log(\pi_S/\pi_B)$
Validation I: linearity (χ^2/ndf = 10.9)
$\log(S/B)$
fit: 0.92±0.00
ideal (slope 1)
log(S/B)
D
residual to ideal
$\log(S/B) - (D + c)$
D
before reweighting
a.u.
ttZ_bb (norm.)
bkg (norm.)
D
after reweighting $r = e^{-D}$
weighted events
reweighted S_w
bkg
D
Validation II: flatness (χ^2/ndf = 10.9)
$\log(S_w/B)$
fit: -0.080±0.005
ideal (flat)
log(S_w/B)
D


ttHcc closure — 4 signals × channels (each → its background) [Run2]
Validation I — linearity (slope → 1)
slope of log(S/B) vs D
ideal = 1
ttH_cc
ttH_bb
ttZ_cc
ttZ_bb
1.050
1.025
1.000
0.975
0.950
0.925
0.900
0L
1L
2L
Validation II — flatness (slope → 0)
slope of log(S_w/B) vs D
ideal = 0
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
−0.12


ttHcc closure — 4 signals × channels (each → its background) [Run3]
Validation I — linearity (slope → 1)
slope of log(S/B) vs D
ideal = 1
ttH_cc
ttH_bb
ttZ_cc
ttZ_bb
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
0L
1L
2L
Validation II — flatness (slope → 0)
slope of log(S_w/B) vs D
ideal = 0
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15

Backup 3: Probability and derivations

Backup: generation and physical laws

- *Definition (generation & physical laws)*: the generation law Q is the law of the weighted events (X, Y, W) the simulator emits ($\mathbb{E}_Q[\cdot]$ counts, ignoring weights). The physical expectation is the weighted average

$$\mathbb{E}_P[f(X,Y)] := \frac{\mathbb{E}_Q[\,W\,f(X,Y)\,]}{\mathbb{E}_Q[W]} = \frac{\mathbb{E}_Q[Wf]}{Z}, \qquad Z := \mathbb{E}_Q[W],$$

  which *defines* the physical law P
- Two readings of "the distribution": Q is what the network sees when it *counts*; P is what an analyst means by "the distribution." They differ only through the weights
- Standing assumptions: **(1)** physicality $\bar{w}_y(x) = \mathbb{E}_Q[W \mid X{=}x,\, Y{=}y] \geq 0$ and $Z_y = \mathbb{E}_Q[W \mid Y{=}y] > 0$; **(2)** finite variance $\mathbb{E}_Q[W^2 \mid X, Y] < \infty$; **(3)** normalisation $Z = \mathbb{E}_Q[W] > 0$

Backup: KL divergence

- Events drawn from the generation law Q (density q); how far is a model q_θ from it?
- *Kullback–Leibler divergence*:

$$\mathrm{KL}(Q \,\|\, q_\theta) \;=\; \mathbb{E}_Q\Big[\log \tfrac{q(X)}{q_\theta(X)}\Big] \;=\; \int q(x)\,\log \frac{q(x)}{q_\theta(x)}\,\mathrm{d}x,$$

  the mean log-likelihood-ratio under the truth — information lost when q_θ stands in for q [Cover & Thomas]
- ≥ 0, zero iff $q_\theta = q$ a.e. (Gibbs / Jensen on the convex $-\log$) — so "fit the truth" = "drive KL $\to 0$"

Backup: from KL to the cross-entropy loss

– Only *samples* from Q, not its density — KL itself is unknowable, but its θ-dependence is the cross-entropy. Split the log:

$$\mathrm{KL}(Q \,\|\, q_\theta) \;=\; \underbrace{\int q \,\log q \,\mathrm{d}x}_{=-H(Q)} \;-\; \underbrace{\int q \,\log q_\theta \,\mathrm{d}x}_{=-H(Q,\,q_\theta)} \;=\; H(Q, q_\theta) - H(Q),$$

with cross-entropy $H(Q, q_\theta) = -\,\mathbb{E}_Q[\log q_\theta]$

– Only $H(Q, q_\theta)$ depends on θ — minimise its sample estimate, the cross-entropy loss

$$\mathcal{L}_{\mathrm{CE}}(\theta) \;=\; -\frac{1}{n}\sum_{i=1}^{n} \log q_\theta(y_i \,|\, x_i) \;\xrightarrow[n\to\infty]{}\; H(Q, q_\theta).$$

– A normalised negative log-likelihood (maximum likelihood); *strictly proper* [Gneiting & Raftery 2007] — optimum is the generation posterior $q(\cdot \,|\, x)$

Backup: change of measure — deriving the master identity

- To reach P *without* weighting a loss, tilt Q by the mean weight. By Def. 2.1 and the tower rule, for any (X, Y)-measurable f,

$$\mathbb{E}_P[f] \;=\; \frac{\mathbb{E}_Q[\,Wf\,]}{Z} \;=\; \frac{\mathbb{E}_Q[\,\bar{w}_Y(X)\,f\,]}{Z} \;\Longrightarrow\; \frac{\mathrm{d}P}{\mathrm{d}Q}(x,y) = \frac{\bar{w}_y(x)}{Z}.$$

- Its posterior — Z and the marginal $q(x)$ cancel:

$$\pi(y \mid x) \;=\; \frac{q(y \mid x)\,\bar{w}_y(x)}{\sum_{y'} q(y' \mid x)\,\bar{w}_{y'}(x)}. \qquad (\star)$$

- KL reading: unweighted CE drives $\mathrm{KL}(Q \,\|\, q_\theta) \to 0 \Rightarrow$ the generation posterior q; the tilt $\bar{w}/Z$ turns it into the physical π via $(\star)$. The weight is a *predicted target*, never a loss coefficient

Backup: standardisation and the label-shift limit

- $\bar{w}_y(x)$ can span many orders of magnitude across classes — poorly conditioned. Using $\bar{w}_y(x) = Z_y\big(1 + \delta_y(x)\big)$, train on the *standardised* target W/Z_y, estimating

$$r_y(x) = \frac{\bar{w}_y(x)}{Z_y} = 1 + \delta_y(x),$$

  an $O(1)$ quantity; de-standardise at inference, $\widehat{\bar{w}}_y(x) = r_y(x)\, Z_y$
- *Exact reduction to label-shift*: if $\bar{w}_y(x) = Z_y$ for all x ($\delta_y \equiv 0$), then $\pi(y \mid x) = \mathrm{softmax}_y\big(\log q(y \mid x) + \log Z_y\big)$
- Where the genuinely new modelling lives: everything beyond Z_y is the feature-dependent $\delta_y(x) = \bar{w}_y(x)/Z_y - 1$. CMOK strictly generalises label-shift to a per-event correction

Backup 4: Weight-head methods and next steps

Backup: heteroscedastic β-NLL for the weight head

- **We use β-NLL.** Plain MSE lets heavy-tailed noise $v(x, y) = \mathrm{Var}(W \mid x, y)$ swamp the small learnable $\delta_y(x)$. Fix: emit a per-class log-variance $s_y = \log \sigma_y^2$ beside μ_y and train with β-NLL (Seitzer 2022),

$$\ell_i = (\sigma_{y_i}^2)^\beta \Big[\tfrac{1}{2} \frac{(t_i - \mu_{y_i})^2}{\sigma_{y_i}^2} + \tfrac{1}{2} \log \sigma_{y_i}^2 \Big], \qquad (\sigma^2)^\beta \text{ stop-gradient.}$$

  - **No mean bias (the point):** at fixed (x, y) the stop-gradient prefactor $(\sigma^2)^\beta$ is constant, so σ^2 reweights the *gradient*, not the *location* — $\partial_\mu \mathbb{E}[\ell \mid x, y] \propto (\mu - \bar{w}_y(x))$, hence the μ-minimiser stays $\mathbb{E}[W \mid x, y] = \bar{w}_y(x)$ for every $\beta \in [0, 1]$. A robust location loss (Huber / Student-t) *would* bias the mean for skewed weights
  - Composition $(\star)$ still uses only μ_y; σ_y^2 is a free per-event uncertainty
  - β interpolates: $0 \to$ Gaussian NLL ($1/\sigma^2$ tail down-weighting, fragile), $1 \to$ plain MSE up to a scale, 0.5 the default (bounded below, but not by 0: $\min \approx -1/e$)

Backup: no class weights, and further plans

- **We do not use class weights.** Two distinct "weights": the signed event weight W — forbidden in the loss ($-\frac{1}{n}\sum_i w_i \log q$ is unbounded below, no minimiser) — and a positive class weight c_y for imbalance. The latter is bounded, *but* class-weighted CE has population minimiser $s^\star_y(x) \propto c_y\, q(y \mid x)$: a constant-per-class tilt living in the same slot as Z_y. Composing through $(\star)$ gives $\pi \propto c_y\, q(y \mid x)\, \bar{w}_y(x)$, i.e. $Z_y \to c_y Z_y$ — it silently corrupts the per-class scale the weight head owns, while $(\star)$ *already* rebalances classes through Z_y. ($Z_y = \mathbb{E}[W \mid Y{=}y]$ is a within-class average, immune to c_y; recoverable by subtracting $\log c_y$ from the logits, but for zero benefit.) $\Rightarrow$ plain unweighted CE, and let $(\star)$ rebalance
- **Further plans** (details in backup):
  - *Weight-head stability across a* 10^{10} *dynamic range* — bin $|W|$ and train a binned CE (bounded, scale-free), then compose the reconstructed mean $\widehat{\bar{w}}(x) = \sum_b p_b(x)\, c_b$ in linear weight-space (binning is make-or-break)
  - *Rescue the dead class* — give the weight head its own encoder (Nachman–Thaler), since the classifier collapses the within-class $\delta_y(x)$ it needs

Backup: a binned-CE weight head for a 10^{10} dynamic range

- Why plain MSE struggles: rare huge-$|W|$ events dominate every gradient $\Rightarrow$ weighted MSE is almost all the irreducible floor $\mathbb{E}[v]$ with δ_y a sliver on top; minima wide and flat (the "can't learn 10^{10}" / dead-class symptom)
- Binned CE (Jan's instinct) — binning is make-or-break: CE over $|W|$-bins is bounded and scale-free, cares only *which* bin, so 10^5 outliers stop steering. But $(\star)$ needs $\mathbb{E}[W \mid x]$ in linear space, so compose the reconstructed mean

$$\widehat{\bar{w}}(x) = \sum_b p_b(x)\, c_b, \quad c_b = \langle W \mid \text{bin } b \rangle \quad \text{(never bin index / quantile / CDF)}.$$

- Bin in $\log|W|$ with a sign split — *not* quantile/uniform: $\mathbb{E}[W \mid x]$ is partly *tail* (a $|W| \sim 10^5$ bin at prob 10^{-5} adds ~ 1). Quantile bins lump the tail into one wide top bin whose centroid is ill-determined — moving the dynamic range into the centroid. Certify with high-$|W|$ region-resolved closure ($\sum_b p_b c_b$ vs. $\langle W \rangle$)

Backup: a separate encoder for the weight head (the dead class)

- Frozen-trunk + imbalance, not a fundamental wall: classification is trained to *collapse* within-class variation (that is what makes a representation discriminative). But $\delta_y(x) = \bar{w}_y(x)/Z_y - 1$ is within-class variation, so the CE objective destroys exactly the structure the weight head needs
- Why enlarging the regression head cannot help: once classification plateaus, the shared trunk freezes at a representation that already threw that structure away — the information is no longer in the trunk to enlarge
- Fix — give the weight regressor its own encoder: a small kinematics→scalar net, own LR — dissolving the plateau-freeze, multitask imbalance and LR mismatch at once. The two-separately-optimised-network construction of Nachman–Thaler; stop-gradient on the trunk is the lighter version

Backup: weighted CE = unweighted CE, weights at inference

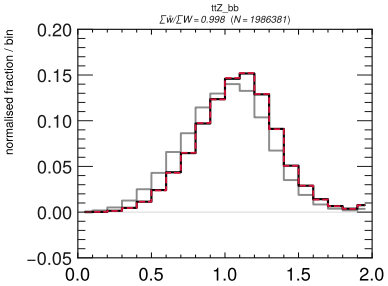
- Laws related by $p(x,y) = q(x,y)\,\bar{w}_y(x)/Z$, with $\bar{w}_y(x) = \mathbb{E}[W \mid x, y]$ and $Z = \mathbb{E}[W]$
- Physical CE from KL: $\mathrm{KL}(P\|P_\theta) = H(P,P_\theta) - H(P)$, $H(P)$ θ-independent $\Rightarrow$ minimise $L_{\text{phys}} = -\int p \log p_\theta$. The sufficiency lemma gives $\mathbb{E}_P[f] = \frac{1}{Z}\,\mathbb{E}_Q[Wf]$, so with $f = \log p_\theta$,

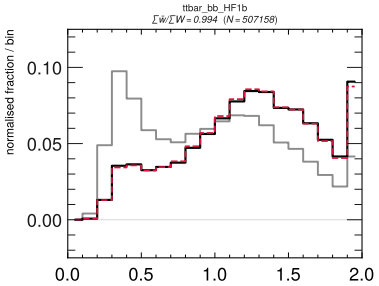
$$L_{\text{phys}} = -\tfrac{1}{Z}\,\mathbb{E}_Q[W \log p_\theta] \;\propto\; \text{weighted CE}.$$

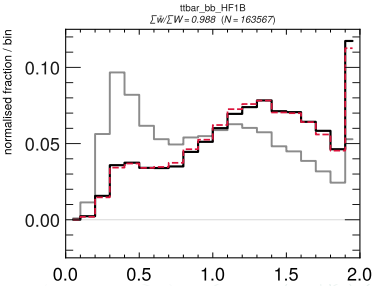
- The point that does not cancel: weighting the loss and weighting the data minimise the *same* objective ($p_\theta \to p$), but the fit needs $p = q\,\bar{w}/Z$ — the classifier gives q; you multiply by $\bar{w}$ at inference to recover p
- (Population picture; the *empirical* weighted loss still collapses — counts go negative, $-W \log p_\theta \to -\infty$. Hence predict $\bar{w}$ rather than multiply by W.)

Backup 5: Detailed class diagnostics

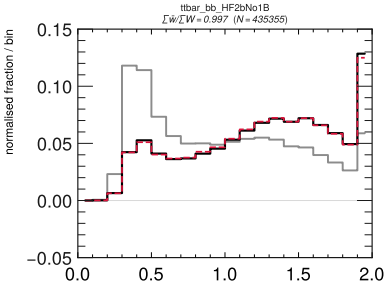
Weight-shape closure, 0L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

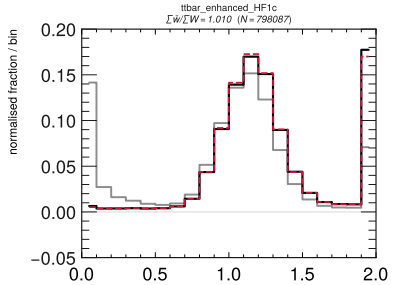


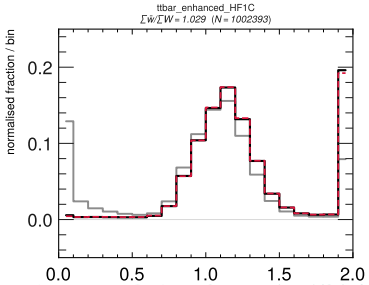




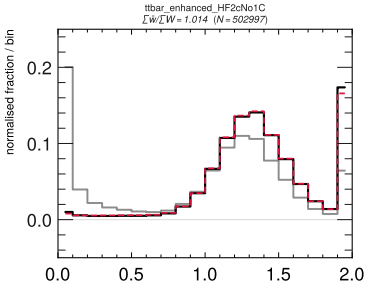
Weight-shape closure, 0L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C

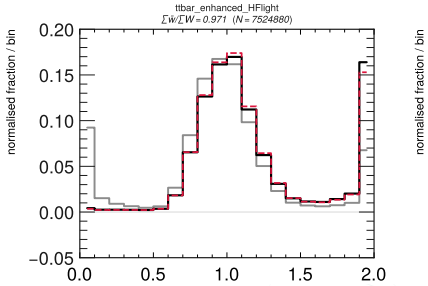




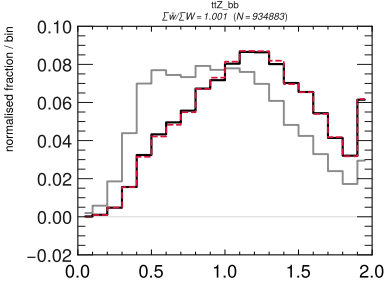


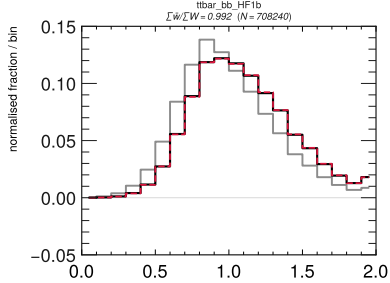
Weight-shape closure, 0L — $t\bar{t}$ HF2c, $t\bar{t}$ light

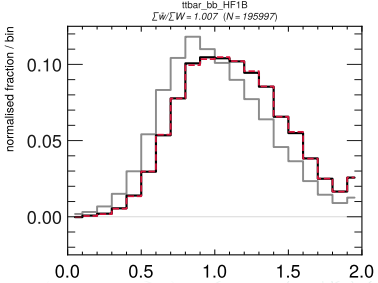




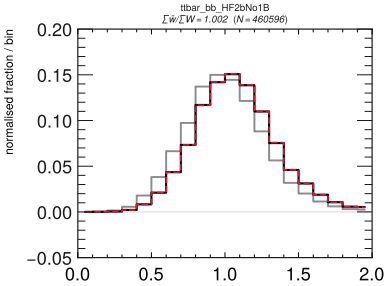
Weight-shape closure, 1L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

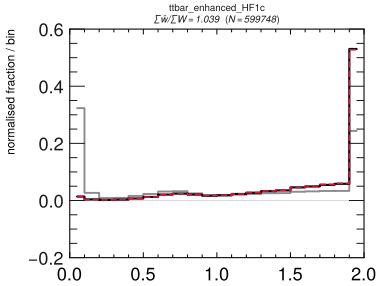


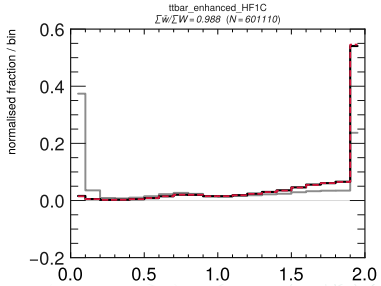




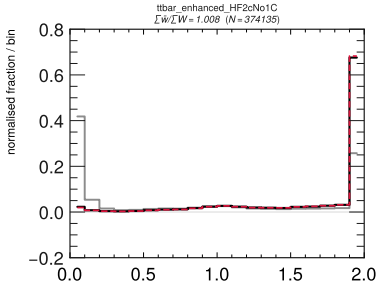
Weight-shape closure, 1L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C

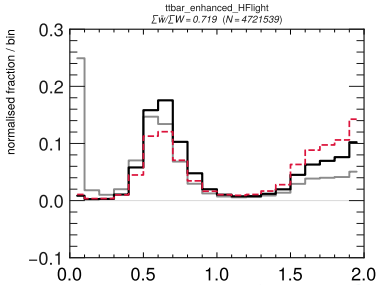




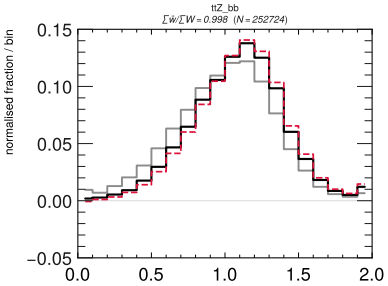


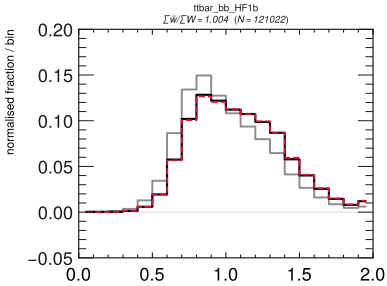
Weight-shape closure, 1L — $t\bar{t}$ HF2c, $t\bar{t}$ light

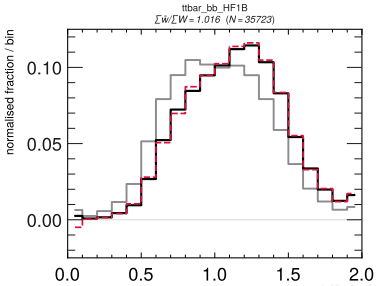




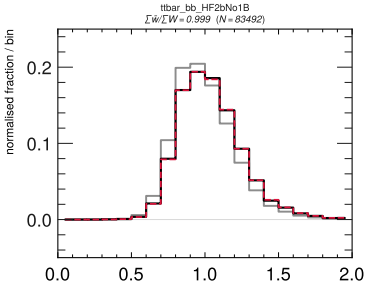
Weight-shape closure, 2L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

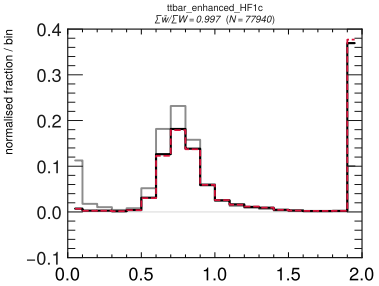


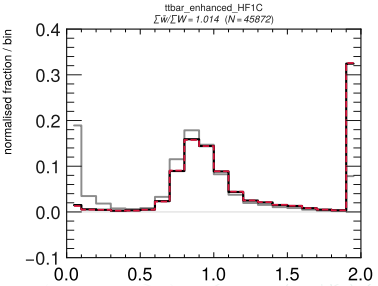




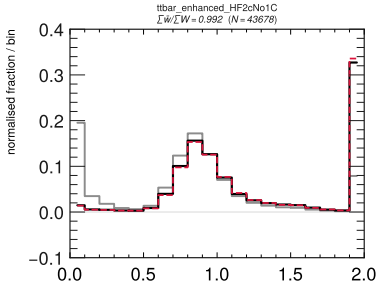
Weight-shape closure, 2L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C

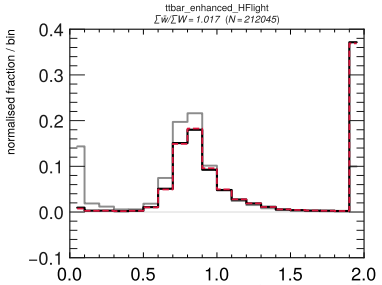




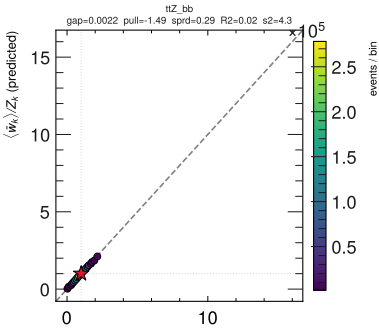


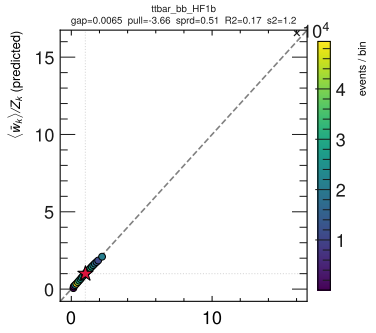
Weight-shape closure, 2L — $t\bar{t}$ HF2c, $t\bar{t}$ light

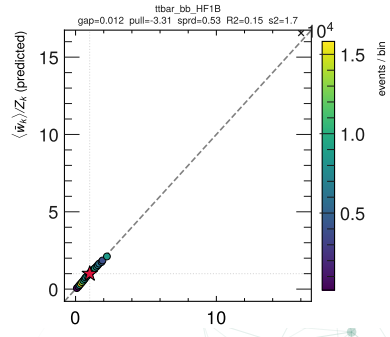




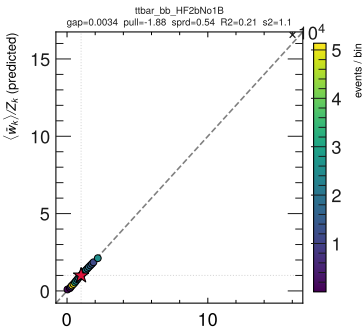
Weight-head closure, 0L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

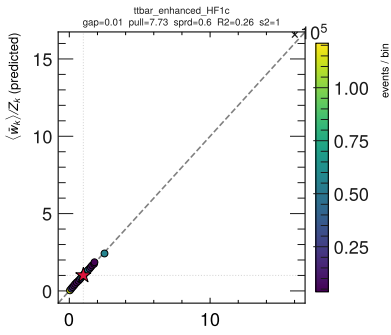


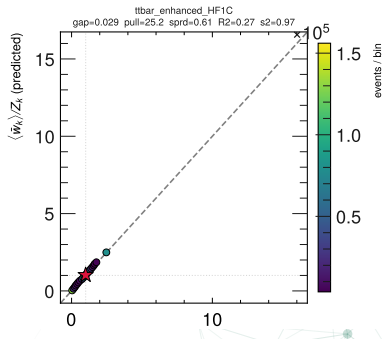




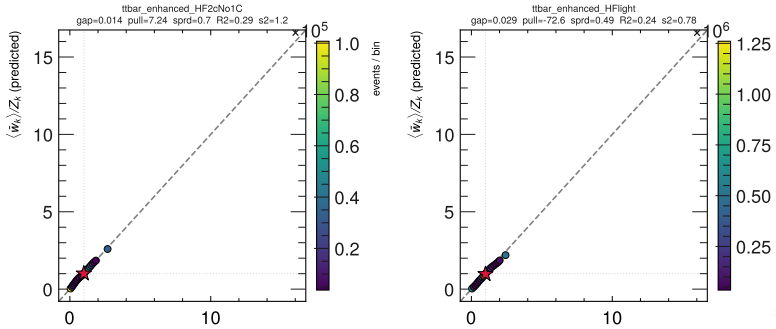
Weight-head closure, 0L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C



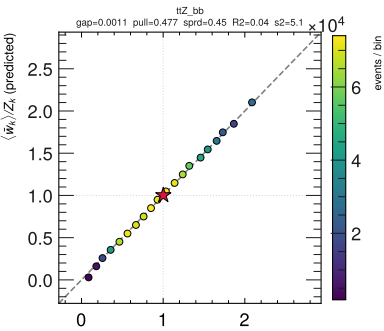


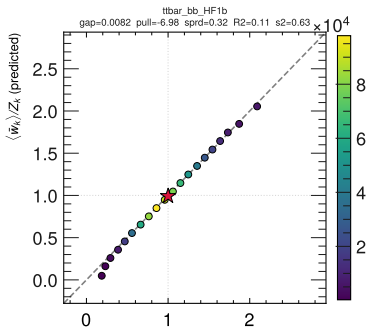


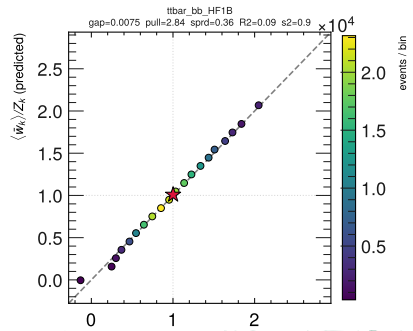
Weight-head closure, 0L — $t\bar{t}$ HF2c, $t\bar{t}$ light



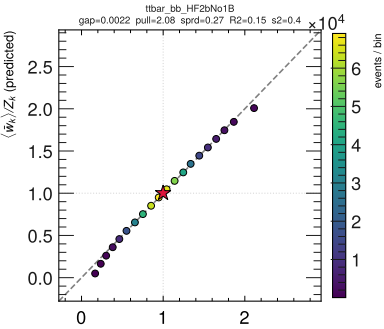
Weight-head closure, 1L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

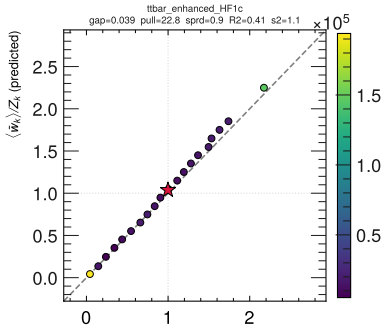


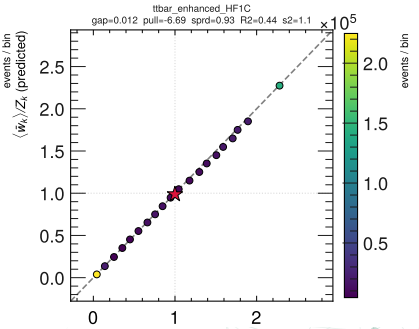




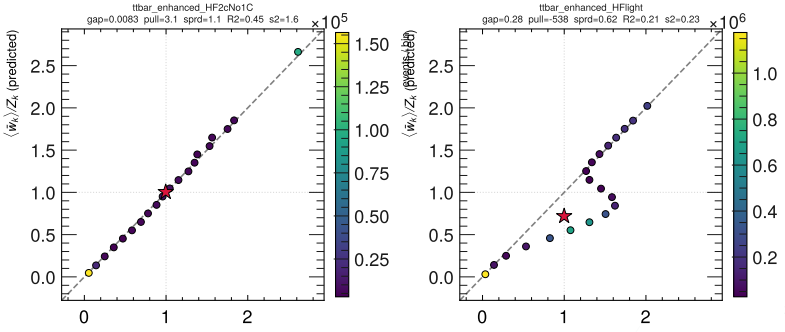
Weight-head closure, 1L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C



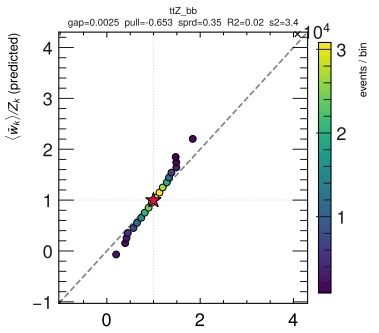


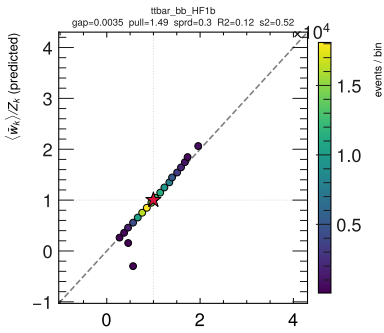


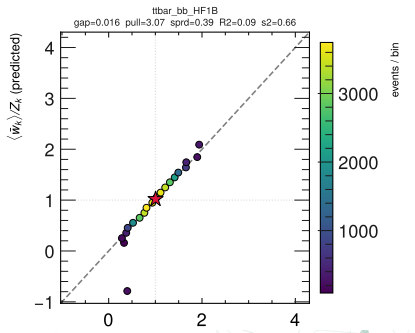
Weight-head closure, 1L — $t\bar{t}$ HF2c, $t\bar{t}$ light



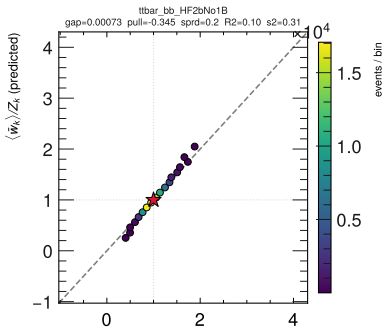
Weight-head closure, 2L — ttZ $b\bar{b}$, $t\bar{t}b\bar{b}$ HF1b/1B

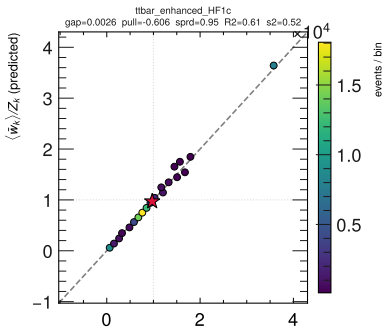


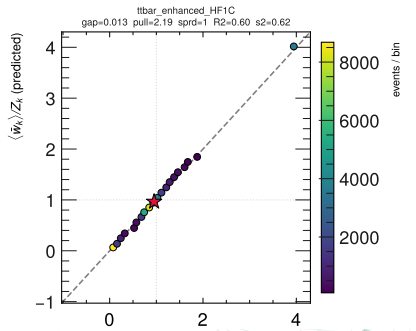




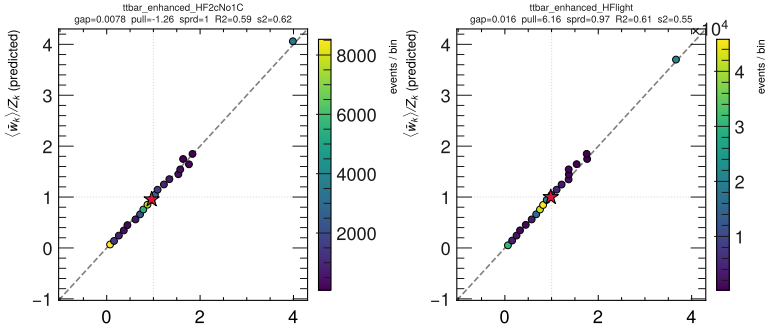
Weight-head closure, 2L — $t\bar{t}b\bar{b}$ HF2b, $t\bar{t}$ HF1c/1C







Weight-head closure, 2L — $t\bar{t}$ HF2c, $t\bar{t}$ light



Backup 6: References

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